

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\196-CChannel MOC=CSnew.pvdb

Date of Analysis : Sep 9, 2026 10:12am

PV Elite 28, January 2026



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FileName : 196-C_Channel MOC=CS_new

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Vessel Design Summary:

Step: 37 10:13am

Sep 9, 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2500.000 x 2506.000 mm ID

Vessel Design Length, Tangent to Tangent 9433.05 mm

Specified Datum Line Distance 0.00 mm

Shell Side Design Temperature 230 °C

Channel Side Design Temperature 82 °C

Shell Side Design Pressure 600.000 KPa.

Channel Side Design Pressure 1000.000 KPa.

Shop Shell Side Test Pressure 1483.657 KPa.

Field Shell Side Test Pressure 780.000 KPa.

Shop Channel Side Test Pressure 1361.025 KPa.

Field Channel Side Test Pressure 780.000 KPa.

Wind Design Code ASCE/SEI 7-16

Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F22	3	...	K21590	
Nozzle Flg	SA-266 2	...	...	K03506	
Rings	SA-516 70	...	...	K02700	
Tubes	SA-789	...	...	S32205	
Tubesheet	SA-182 F60	...	...	S32205	
Flo. Head	SA-240	...	...	S32205	
Bk. Bolting		...	...		
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Ch. cover	1028.426	101.13	1079.000	3.0000	N/A	No
Ch.flange-1	1024.172	101.13	1729.000	0.0000	N/A	No
Ch.shell	1024.534	101.13	1185.000	3.0000	N/A	No
ch, flange 2	1024.564	1.01	1728.000	0.0000	N/A	No
Shell flng-1	624.564	0.00	1013.000	3.0000	N/A	No
Shell-1	624.572	0.00	1064.000	0.8000	N/A	No
Shell-2	624.572	0.00	2168.000	0.8000	N/A	No



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Vessel Design Summary:

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Shell-3	624.572	0.00	1064.000	0.8000	N/A	No
Shell flng. 2	624.564	0.00	1013.000	3.0000	N/A	No
Shell girth fln	624.564	0.00	1013.000	3.0000	N/A	No
Shell -2	624.593	0.00	997.000	3.0000	N/A	No
head- front	624.593	0.00	968.000	3.0000	No	No

Liquid Level: 2900.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
3731.53	Bar 258.0 x 40.0	SR, 1000.0 Fr60
7032.70	Bar 258.0 x 40.0	SR, 4000.0 Fr60

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	191.3	185.0	193.0	185.0	179.6	...	1.00	1.00
Body Flg	486.4	295.0	255.0	247.0	242.2	...	1.00	1.00
Cylinder	1986.4	1500.0	16.0	16.0	14.0	...	0.85	0.85
Body Flg	2290.7	298.0	258.0	250.0	245.0	...	1.00	1.00
Body Flg	2732.7	317.0	268.0	260.0	258.4	...	1.00	1.00
Cylinder	4132.7	1400.0	14.0	12.5	7.5	...	0.85	0.85
Cylinder	6632.7	2500.0	26.0	24.5	7.5	...	0.85	0.85
Cylinder	8032.7	1400.0	14.0	12.5	7.5	...	0.85	0.85
Body Flg	8361.1	322.0	273.0	265.0	258.4	...	1.00	1.00
Body Flg	8683.1	322.0	273.0	265.0	258.4	...	1.00	1.00
Cylinder	9383.0	700.0	14.0	14.0	9.7	...	0.85	0.85
Ellipse	9433.0	50.0	16.0	13.6	9.7	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	185	147.193	101.13	No Calc
20	30	247	242.189	101.13	No Calc
30	40	16	9.48296	101.13	550.061
40	50	250	244.958	1.013	No Calc
50	60	260	258.369	...	No Calc
60	Ring	12.5	No Calc	...	548.69
Ring	70	12.5	No Calc	...	196.53
70	80	24.5	No Calc	...	1022.7
80	Ring	12.5	No Calc	...	196.53
Ring	90	12.5	No Calc	...	548.457
90	100	265	258.369	...	No Calc
100	110	265	258.369	...	No Calc
110	120	14	No Calc	...	506.692
120	130	13.6	No Calc	...	270.131

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	1500	7980.75	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc



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50	60	No Calc	No Calc	No Calc	No Calc
60	Ring	998.828	No Calc	No Calc	No Calc
Ring	70	3301.17	No Calc	No Calc	No Calc
70	80	3301.17	No Calc	No Calc	No Calc
80	Ring	3301.17	No Calc	No Calc	No Calc
Ring	90	1000	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	958.833	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Tube inlet	WRC-297	0.986	Shell	Passed
Tube outlet	WRC-297	0.993	Shell	Passed
Shell outlet	WRC-297	0.993	Shell	Passed
Shell inlet	WRC-297	0.993	Shell	Passed

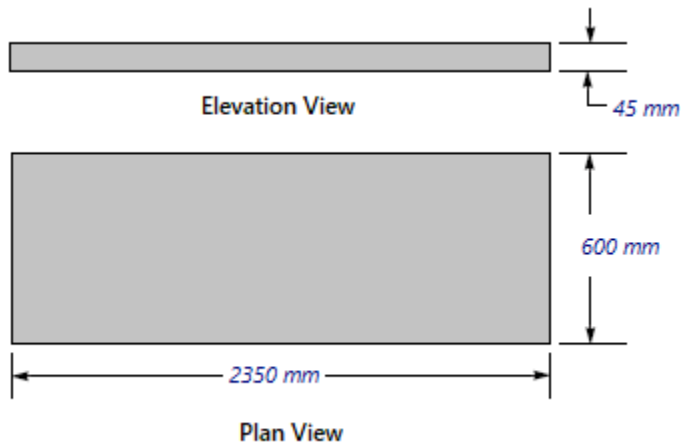
Foundation/Support Summary

Saddle Parameters:

Saddle Width	300.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1850.000 mm
Wear Pad Width	510.000 mm
Wear Pad Thickness	14.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1000.000 mm

Baseplate Length	2350.000 mm
Baseplate Thickness	45.000 mm
Baseplate Width	600.000 mm
Number of Ribs (including outside ribs)	5
Rib Thickness	20.000 mm
Web Thickness	20.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	8

Baseplate Sketch





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Vessel Design Summary:

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Maximum Tensile Bolt Load 21. Kgf**Summary of Maximum Saddle Loads, Operating Case, Un-Factored:**

Maximum Vertical Saddle Load	105211.31	Kgf
Maximum Transverse Saddle Shear Load	4787.19	Kgf
Maximum Longitudinal Saddle Shear Load	7750.70	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	92332.50	Kgf
Maximum Transverse Saddle Shear Load	318.45	Kgf
Maximum Longitudinal Saddle Shear Load	362.43	Kgf

Weights:

Fabricated - Bare w/o removable internals	86595.6	kg.
Shop Test - Fabricated + water (full)	131140.4	kg.
Shipping - Fab. + rem. intls.+ shipping app.	86595.6	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	86595.6	kg.
Empty - Fab. + intls. + details + wghts.	89470.3	kg.
Operating - Empty + operating liquid (no ca)	134219.9	kg.
Field Test - Empty weight + test medium (full)	133092.1	kg.

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Nozzle Summary:

Step: 35 10:13am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Tube inlet	1185	OK	...	OK	8.33	OK	Passed
Tube outlet	1185	OK	...	OK	8.33	OK	Passed
Shell outlet	2128	...	...	OK	6.49	OK	Passed
Shell inlet	2152	...	...	OK	6.27	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2128 Nozzle : Shell outlet [Shellside]
 Minimum MAWP Nozzles : 1185 Nozzle : Tube inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	Tube inlet	1236.350	0.000	688.950
30	Tube outlet	1236.350	180.000	688.950
70	Shell outlet	5453.230	180.000	2136.800
70	Shell inlet	5415.960	0.000	2136.800

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 34 10:13am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
Tube inlet	16.000 in	120	WNF	16.000	30.963	886.40	16.00	382.63	150
Tube outlet	16.000 in	120	WNF	16.000	30.963	886.40	16.00	382.63	150
Shell outlet	42.000 in Actual		WNF	43.890	24.000	...	...	577.27	150
Shell inlet	42.000 in Actual		WNF	43.969	25.000	...	...	577.27	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Tube inlet	SA-106 B	14.000	15.000	12.000	16.000	...
Tube outlet	SA-106 B	14.000	15.000	12.000	16.000	...
Shell outle	SA-266 2	12.500	20.000	...	...	...
Shell inlet	SA-266 2	12.500	20.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Tube inlet	1236.350	0.0	350.00	0.00	Ch.shell
Tube outlet	1236.350	180.0	350.00	0.00	Ch.shell
Shell outlet	5453.230	180.0	350.00	0.00	Shell-2
Shell inlet	5415.960	0.0	350.00	0.00	Shell-2

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Bill of Material:

Step: 33 10:13am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 193.0mm THK X 2900.0mm OD	SA-266 2
1	WELD NECK FLANGE: 255.0mm THK X 2900.0mm OD	SA-266 2
1	CYLINDER: 16.0mm THK X 2500.0mm ID X 1500.0mm	SA-516 70
1	WELD NECK FLANGE: 258.0mm THK X 2900.0mm OD	SA-266 2
1	WELD NECK FLANGE: 268.0mm THK X 2900.0mm OD	SA-266 2
2	WELD NECK FLANGE: 273.0mm THK X 2900.0mm OD	SA-266 2
1	CYLINDER: 14.0mm THK X 2506.0mm ID X 700.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm THK X 2506.0mm ID X 50.0mm	SA-516 70
1	INSULATION: 200mm X 100mm THK	Insulation
1	INSULATION: 303mm X 100mm THK	Insulation
1	INSULATION: 1500mm X 100mm THK	Hot Insulation
1	TUBESHEET: 2578.0mm X 125.0mm THK	SA-182 F60
4424	TUBES: 6100.0mm X 25.4mm DIA X 2.1mm THK	SA-789
1	FLOATING TUBESHEET: 2482.0mm X 125.0mm THK	SA-182 F60
44	FLOATING HEAD FLANGE BOLTS: 34.9mm DIA	SA-193 B7
88	NUTS FOR FLOATING HEAD FLANGE BOLTS: 34.9mm DIA	...
1	FLOATING HEAD FLANGE GASKET: 2360.0mm OD X 2320.0mm ID	...
1	LINING: 307.000mm X 6.000mm THK	...
2	SADDLE: 300.0mm X 120 DEG	SA-516 70
1	INSULATION: 1400mm X 100mm THK	Hot
2	LINING: 1400.000mm X 3.000mm THK	...
2	BAR RING STIFFENER: 2531.0mm ID X 3047.0mm OD X 40.0mm	SA-516 70
1	INSULATION: 2500mm X 100mm THK	Hot Insulation
1	LINING: 2500.000mm X 3.000mm THK	...
1	INSULATION: 1400mm X 100mm THK	
2	LINING: 314.000mm X 6.000mm THK	...
1	INSULATION: 700mm X 100mm THK	Hot Insulation
1	LINING: 700.000mm X 3.000mm THK	...
1	INSULATION: 619mm X 100mm THK	Hot Insulation
1	LINING: 676.500mm X 3.000mm THK	...



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Bill of Material:

Step: 33 10:13am Sep 9,2026

1	GASKET: 2576.0mm OD X 2532.0mm ID	...
108	BODY FLANGE BOLTS: 57.1mm DIA	SA-193 B7
216	NUTS FOR BODY FLANGE BOLTS: 57.1mm DIA	...
2	GASKET: 2570.0mm OD X 2526.0mm ID	...
4	CYLINDRICAL SEGMENT 14.0mm THK X 3048.0mm X 1400.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 14.0mm THK X 1776.8mm X 1400.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 26.0mm THK X 3105.3mm X 2500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 26.0mm THK X 3083.5mm X 2500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 26.0mm THK X 1683.9mm X 2500.0mm	SA-516 70
1	NAMEPLATE	...

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MDMT Summary:

Step: 36 10:13am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Shell flng-1 [11]	B	-22	-38	-29	0.702	12.500	1.00	No
Shell flng. 2[11]	B	-38	-48		0.702	12.500	1.00	No
Bolting [21]		-48						
Shell girth f[11]	B	-38	-48		0.741	12.500	1.00	No
Shell-1 [8]	B	-22	-33	-29	0.791	12.500	0.85	No
Shell-2 [8]	B	-2	-48	-29	0.390	24.500	0.85	No
Shell-3 [8]	B	-22	-33	-29	0.791	12.500	0.85	No
Left Saddle [24]	B	-22	-33	-29	0.791	12.500	0.85	No
Right saddle [24]	B	-22	-33	-29	0.791	12.500	0.85	No
Shell outlet [1]	B	-2	-104	-29	0.105	24.000	1.00	No
Nozzle Flg [5]	B	-18	-104					
Shell inlet [1]	B	-1	-104	-29	0.096	25.000	1.00	No
Nozzle Flg [5]	B	-18	-104					
Warmest MDMT:		-1	-33					
Ch. cover [11]	B	-2	-14		0.777	46.250	1.00	No
Ch.flange-1 [11]	B	-14	-41	-29	0.552	16.000	1.00	No
ch, flange 2 [11]	B	-14	-41	-29	0.552	16.000	1.00	No
Ch.shell [8]	B	-14	-32	-29	0.679	16.000	0.85	No
Shell -2 [8]	B	-18	-29	-29	0.805	14.000	0.85	No
head- front [10]	B	-19	-29	-29	0.829	13.600	0.85	No
head- front [7]	B	-14	-32	-29	0.681	16.000	0.85	No
Tube inlet [1]	B	-14	-31	-29	0.702	16.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
Tube outlet [1]	B	-14	-30	-29	0.719	16.000	1.00	No
Nozzle Flg [5]	B	-18	-47					
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		-2	-14					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-33.0	-12.0	Pass
Channel/Tube	-14.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material



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MDMT Summary:

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[22] - Nozzle Flange Bolting Material

[23] - Stiffening Ring to Shell Weld

[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: ch, flange 2 and Shell flng-1

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.



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Warnings and Errors:

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Input Echo:

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	600	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	230.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	9000	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	0	mm
Distance Upwind of Crest	0	mm
Distance from Crest to the Vessel	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor	1.000	
Importance Factor	1.250	
Table Value Fa	1.600	
Table Value Fv	2.400	
Max. Mapped Res. Acceleration	0.147	[Ss]
Max. Eff. Ground Acceleration	0.065	[S]
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	0.000	z/h
Amplification Factor	1.000	Ap
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Moment Reduction Factor Tau	1.000	

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

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Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element# 1/12
 Element From Node 10
 Element To Node 20
 Element Type Flange
 Description Ch. cover
 Distance "FROM" to "TO" 185 mm
 Flange Outside Diameter 2900 mm
 Element Thickness 185 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 193 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 1000 KPa.
 Design Temperature Internal Pressure 82 °C
 Design External Pressure 101.13 KPa.
 Design Temperature External Pressure 82 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-266 2
 Allowable Stress, Ambient 138 N./mm²
 Allowable Stress, Operating 138 N./mm²
 Allowable Stress, Hydrotest 225 N./mm²
 Material Density 0.00775 kg./cm³
 P Number Thickness 32.004 mm
 Yield Stress, Operating 230.08 N./mm²
 UCS-66 Chart Curve Designation B
 External Pressure Chart Name CS-2
 UNS Number K03506
 Product Form Forgings
 Perform Flange Stress Calculation (Y/N) Y
 Weld is pre-Heated No

Element From Node 10
 Detail Type Liquid
 Detail ID Liquid: 10
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 2900 mm
 Liquid Density 0.0009984 kg./cm³

Element From Node 10
 Detail Type Insulation
 Detail ID Insulation
 Dist. from "FROM" Node / Offset dist 0 mm



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Input Echo:

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Height/Length of Insulation	200	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element#	2/12
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Ch.flange-1
Distance "FROM" to "TO"	295 mm
Flange Inside Diameter	2500 mm
Element Thickness	247 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	255 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	101.13 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2466 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	303 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	3/12
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Ch.shell
Distance "FROM" to "TO"	1500 mm
Inside Diameter	2500 mm
Element Thickness	16 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	101.13 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm



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Yield Stress, Operating	242.6	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2500	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1500	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	Tube inlet	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	16	in.
Nozzle Schedule	120	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	232.68	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	Tube outlet	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	16	in.
Nozzle Schedule	120	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	232.68	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Weight	
Detail ID	Partition plate	
Dist. from "FROM" Node / Offset dist	750	mm
Miscellaneous Weight	1105	Kgf
Offset from Element Centerline	0	mm

Element#	4/12	
Element From Node	40	
Element To Node	50	
Element Type	Flange	
Description	ch, flange 2	
Distance "FROM" to "TO"	298	mm



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Flange Inside Diameter	2500	mm
Element Thickness	250	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	258	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	1.013	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2506	mm
Liquid Density	0.0009984	kg./cm ³

Element#	5/12	
Element From Node	50	
Element To Node	60	
Element Type	Flange	
Description	Shell flng-1	
Distance "FROM" to "TO"	317	mm
Flange Inside Diameter	2510	mm
Element Thickness	260	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	268	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	600	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2506	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	50	
Detail Type	Lining	
Detail ID	cladd	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	307	mm



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Thickness of Lining	6 mm
Density	0.008 kg./cm ³

Element#	6/12
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	1400 mm
Inside Diameter	2506 mm
Element Thickness	12.5 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	600 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	220 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	60
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1000 mm
Width of Saddle	300 mm
Height of Saddle at Bottom	1850 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	510 mm
Thickness of Wear Plate	14 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2506 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	60
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm



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Height/Length of Insulation	1400	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	60	
Detail Type	Lining	
Detail ID	Cladding	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1400	mm
Thickness of Lining	3	mm
Density	0.00785	kg./cm ³

Element From Node	60	
Detail Type	Ring	
Detail ID	SR, 1000.0 Fr60	
Dist. from "FROM" Node / Offset dist	998.83	mm
Inside Diameter of Ring	2531	mm
Thickness of Ring	40	mm
Outside Diameter of Ring	3047	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	60	
Detail Type	Weight	
Detail ID	Fireproofing we	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	520	Kgf
Offset from Element Centerline	1800	mm

Element#	7/12	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	2500	mm
Inside Diameter	2506	mm
Element Thickness	24.5	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	26	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	600	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2506	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2500	mm



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Input Echo:

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Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	70	
Detail Type	Lining	
Detail ID	SS clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	2500	mm
Thickness of Lining	3	mm
Density	0.00785	kg./cm ³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	Shell outlet	
Dist. from "FROM" Node / Offset dist	1320.5	mm
Nozzle Diameter	42	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	901.86	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	Shell inlet	
Dist. from "FROM" Node / Offset dist	1283.3	mm
Nozzle Diameter	42	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	856.65	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element#	8/12	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	1400	mm
Inside Diameter	2506	mm
Element Thickness	12.5	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	600	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	400	mm
Width of Saddle	300	mm



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Input Echo:

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Height of Saddle at Bottom	1850	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	510	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.40000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2506	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	80	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1400	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³
Element From Node	80	
Detail Type	Lining	
Detail ID	Cladding	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1400	mm
Thickness of Lining	3	mm
Density	0.00785	kg./cm ³
Element From Node	80	
Detail Type	Ring	
Detail ID	SR, 4000.0 Fr60	
Dist. from "FROM" Node / Offset dist	400	mm
Inside Diameter of Ring	2531	mm
Thickness of Ring	40	mm
Outside Diameter of Ring	3047	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	80	
Detail Type	Weight	
Detail ID	Fire proofing	
Dist. from "FROM" Node / Offset dist	400	mm
Miscellaneous Weight	520	Kgf
Offset from Element Centerline	1800	mm

Element#	9/12	
Element From Node	90	
Element To Node	100	
Element Type	Flange	
Description	Shell flng. 2	
Distance "FROM" to "TO"	322	mm
Flange Inside Diameter	2510	mm
Element Thickness	265	mm
Internal Corrosion Allowance	3	mm



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Nominal Thickness	273	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	600	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	136.8	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	208	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2506	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	90	
Detail Type	Lining	
Detail ID	cladd	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	314	mm
Thickness of Lining	6	mm
Density	0.008	kg./cm ³

Element#	10/12	
Element From Node	100	
Element To Node	110	
Element Type	Flange	
Description	Shell girth fln	
Distance "FROM" to "TO"	322	mm
Flange Inside Diameter	2510	mm
Element Thickness	265	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	273	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	600	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	100	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2506	mm
Liquid Density	0.0009984	kg./cm ³



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Input Echo:

Step: 1 10:12am Sep 9,2026

Element From Node	100
Detail Type	Lining
Detail ID	cladd
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	314 mm
Thickness of Lining	6 mm
Density	0.008 kg./cm ³

Element#	11/12
Element From Node	110
Element To Node	120
Element Type	Cylinder
Description	Shell -2
Distance "FROM" to "TO"	700 mm
Inside Diameter	2506 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	600 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	220 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	110
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2506 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	110
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	700 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	110
Detail Type	Lining
Detail ID	Ss clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	700 mm
Thickness of Lining	3 mm
Density	0.00785 kg./cm ³



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Input Echo:

Step: 1 10:12am Sep 9,2026

Element#	12/12
Element From Node	120
Element To Node	130
Element Type	Elliptical
Description	head- front
Distance "FROM" to "TO"	50 mm
Inside Diameter	2506 mm
Element Thickness	13.6 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	600 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	120
Detail Type	Liquid
Detail ID	Liquid: 80
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2506 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	120
Detail Type	Insulation
Detail ID	Ins: 80
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	620 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³
Element From Node	120
Detail Type	Lining
Detail ID	SS clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	676.5 mm
Thickness of Lining	3 mm
Density	0.00785 kg./cm ³



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XY Coordinate Calculations:

Step: 2 10:12am

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Ch. cover		191.35	...	185	...
Ch.flange-1		486.35	...	295	...
Ch.shell		1986.35	...	1500	...
ch, flange 2		2290.7	...	298	...
Shell flng-1		2732.7	...	317	...
Shell-1		4132.7	...	1400	...
Shell-2		6632.7	...	2500	...
Shell-3		8032.7	...	1400	...
Shell flng. 2		8361.05	...	322	...
Shell girth fln		8683.05	...	322	...
Shell -2		9383.05	...	700	...
head- front		9433.05	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: New Flange

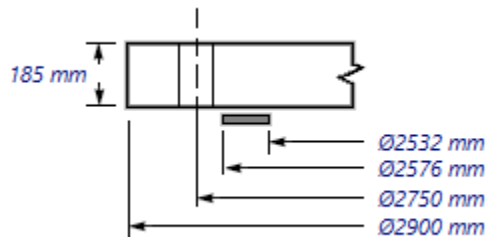
Flng: 31 10:12am Sep 9,2026

Flange Input Data Values

Description: New Flange :

Ch. cover

Flange Type	TEMA Channel Cover		
Design Pressure	P	1028.43	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2900.000	mm
Flange Thickness	t	185.0000	mm
Flange Material	SA-266 2		
Flange Material UNS number	K03506		
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	8639.380	mm
Youngs Modulus for Blind Flange Matl.	Sy	197846.66	N./mm ²
Allowed Channel Cover Deflection	dfAllow	3.1325	mm
Diameter of Bolt Circle	C	2750.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	36		
Flange Face Outside Diameter	Fod	2587.000	mm
Flange Face Inside Diameter	Fid	2500.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	2576.000	mm
Gasket Inside Diameter	Gi	2532.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	22.000 mm
Basic Gasket Width,	$bo = N/2$	11.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	8.358 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	2559.285 mm



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 Flg Calc [Int P]: New Flange Flng: 31 10:12am Sep 9,2026

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2559.2849^2 \cdot 1028.426$$

$$= 539506.500 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.3576 \cdot \pi \cdot 2559.2849 \cdot 3.0 \cdot 1028.43$$

$$= 42283.668 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(539507 + 42284 + 0, 0)$$

$$= 581790.188 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.3576 \cdot \pi \cdot 2559.285 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 472451.719 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((581790 + 0 + 4 \cdot 0 / 2559.28) / 172, 472452 / 172)$$

$$= 331.722 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	331.722	795.018	
Radial Distance between Bolts and Edge:	57.150	75.000	
Circ. Spacing between the Bolts:	120.650	239.678	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 795.018 \cdot 172.0 / (68.95 \cdot \pi (2576.0 + 2532.0))$$

$$= 12.358 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 795.0178$$

$$= 1394340.25 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 581790.19 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2750.0002 - 2559.2849) / 2$$

$$= 95.3577 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2$$

$$= 1.9 \cdot 581790 \cdot 95.3577 \cdot 1.0 / (182.0^2 \cdot 2559.2849) + 0.3 \cdot 1.0 \cdot 1028.43 \cdot 2559.2849^2 / 182.0^2$$

$$= 73.21 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:



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Flg Calc [Int P]: New Flange

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$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1394340 \cdot 95.3577 \cdot 1.0 / (182.0^2 \cdot 2559.2849)$$

$$= 29.22 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 581790 / 795.0178$$

$$= 71.77 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (472452 / 795.0178)$$

$$= 58.28 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	73.21	138.00	29.22	138.00
Bolting	71.77	172.00	58.28	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 3.0082 mm

Allowed deflection at center of Cover 3.1325 mm

Reqd. Blind Flange Thickness at Center 135.559 mm

Reqd. Blind Flange Thickness at Gasket 86.755 mm

Min. Required Flange Thickness per TEMA 179.559 mm

Estimated M.A.W.P. (Operating) 1108 KPa.

Estimated Finished Weight of Flange at given Thk. 9470.8 kg.

Estimated Unfinished Weight of Forging at given Thk 9470.8 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.777, Temperature Reduction per Fig. UCS 66.1 = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 15 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -14 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	70.85	138.00	28.28	138.00
Bolting	71.77	172.00	58.28	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 178.537 mm

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Flg Calc [Int P]: New Flange

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Flange Input Data Values

Description: New Flange :

Ch.flange-1

Flange Type	Integral Weld Neck		
Design Pressure	P	1024.17	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2500.000	mm
Flange Outside Diameter	A	2900.000	mm
Flange Thickness	t	247.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	23.0000	mm
Length of Hub	h	48.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2750.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	2587.000	mm
Flange Face Inside Diameter	Fid	2500.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2576.000	mm
Gasket Inside Diameter	Gi	2532.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2535.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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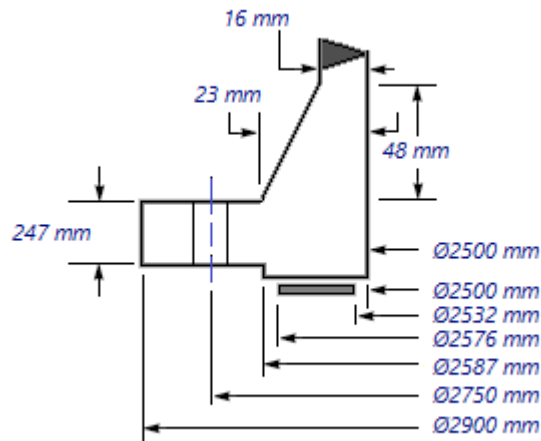
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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1024.17 \cdot (2500.0/2 + 0.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1024.17) + C_a$$

$$= 9.3191 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (1250.0 + 0.6 \cdot 16.0)$$

$$= 1752.804 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (1250.0 + 0.6 \cdot 16.0)$$

$$= 1752.804 \text{ KPa.}$$

Corroded Flange ID,	Bcor = B + 2 * ci	2500.000 mm
Corroded Large Hub,	g1Cor = g1 - ci	23.000 mm
Corroded Small Hub,	g0Cor = g0 - ci	16.000 mm
Code R Dimension,	R = (C - B) / 2 - g1	102.000 mm
Gasket Contact Width,	N = (Go - Gi) / 2	22.000 mm
Basic Gasket Width,	bo = N / 2	11.000 mm
Effective Gasket Width,	b = Cb * sqrt(bo)	8.358 mm
Gasket Reaction Diameter,	G = Go - 2 * b	2559.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 2559.2849^2 \cdot 1024.172$$

$$= 537274.812 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.3576 \cdot \pi \cdot 2559.2849 \cdot 3.0 \cdot 1024.17$$

$$+ 2 \cdot 2535.0 \cdot 5.0 \cdot 3.0 \cdot 1024.1718$$

$$= 50051.469 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 2500.0^2 \cdot 1024.1718$$

$$= 512671.625 \text{ Kgf}$$



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Flg Calc [Int P]: New Flange

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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 537275 - 512672$$

$$= 24603.207 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(537275 + 50051 + 0, 0)$$

$$= 587326.312 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.3576 \cdot \pi \cdot 2559.285 + 68.95 \cdot 5.0 \cdot 2535.0$$

$$= 561567.250 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((587326 + 0 + 4 \cdot 0 / 2559.28) / 172, 561567 / 172)$$

$$= 334.879 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 57.15 + 6 \cdot 247.0 / (3.0 + 0.5)$$

$$= 537.729 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2750.0 \cdot \sin(\pi / 36)$$

$$= 239.678 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 239.678 / (2 \cdot 57.15 + 247.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	334.879	795.018	
Radial Distance between Hub and Bolts:	69.850	102.000	
Radial Distance between Bolts and Edge:	57.150	75.000	
Circ. Spacing between the Bolts:	120.650	239.678	537.729

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 795.018 \cdot 172.0 / (68.95 \cdot \pi (2576.0 + 2532.0))$$

$$= 12.358 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 795.0178$$

$$= 1394340.25 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 587326 - 537275$$

$$= 50051.47 \text{ Kgf}$$

Moment Arm Calculations:



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Flg Calc [Int P]: New Flange

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Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2750.0002 - 2559.2849) / 2$$

$$= 95.3577 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2750.0002 - 2500.0) / 2 + 95.3577)$$

$$= 110.1788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2750.0 - 2500.0 - 23.0) / 2$$

$$= 113.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	512672.	113.5000	1.0000	58189.
Face Pressure, Mt	24603.	110.1788	1.0000	2711.
Gasket Load, Mg	50051.	95.3577	1.0000	4773.
Gasket Seating, Matm	1394340.	95.3577	1.0000	132964.

Total Moment for Operation, Mo 65673. Kg-m.

Total Moment for Gasket seating, Mg 132964. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 200.000 mm

Hub Ratio, h/h0 = HL / H0 0.240

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.438

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.895

Factor V per 4.16.5 0.405

Factor f per 4.16.5 1.256

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2900.0 / 2500.0$$

$$= 1.160$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.16^2 \cdot \log_{10}(1.16))) / (1.16^2 - 1)) / (1.16 - 1)$$

$$= 13.145$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.16^2 (1 + 8.55246 \cdot \log_{10}(1.16)) - 1) / ((1.0472 + 1.9448 \cdot 1.16^2) \cdot (1.16 - 1))$$

$$= 1.855$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.16^2 (1 + 8.55246 \cdot \log_{10}(1.16)) - 1) / (1.36136 (1.16^2 - 1) \cdot (1.16 - 1))$$

$$= 14.445$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.16^2 + 1) / (1.16^2 - 1)$$

$$= 6.787$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$

$$= 14.445 \cdot 16.0^2 \cdot 200.0 / 0.405$$



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Flg Calc [Int P]: New Flange

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$$= 1824986.750 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.895/200.0$$

$$= 0.004 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (247.0 \cdot 0.004 + 1) / 1.855 + 247.0^3/1824987$$

$$= 9.392$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$$

$$= (1.2556 \cdot 65673/2500.0) / (9.3921 \cdot 23.0^2)$$

$$= 65.10 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$$

$$= (1.2556 \cdot 132964/2500.0) / (9.3921 \cdot 23.0^2)$$

$$= 131.81 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.4734 \cdot 65673/2500.0) / (9.3921 \cdot 247.0^2)$$

$$= 1.11 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm/Bcor) / (L \cdot t^2)$$

$$= (2.4734 \cdot 132964/2500.0) / (9.3921 \cdot 247.0^2)$$

$$= 2.25 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (13.1451 \cdot 65673 / (247.0^2 \cdot 2500.0)) - 6.787 \cdot 1$$

$$= 47.96 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (13.1451 \cdot 132964 / (247.0^2 \cdot 2500.0)) - 6.787 \cdot 2$$

$$= 97.10 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (65 + \max(1, 48)) / 2$$

$$= 56.53 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (132 + \max(2, 97)) / 2$$

$$= 114.45 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 587326/795.0178$$

$$= 72.45 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (561567/795.0178)$$

$$= 69.27 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²



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Flg Calc [Int P]: New Flange

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	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	65.10	207.00	131.81	207.00
Radial Flange	1.11	138.00	2.25	138.00
Tangential Flange	47.96	138.00	97.10	138.00
Maximum Average	56.53	138.00	114.45	138.00
Bolting	72.45	172.00	69.27	172.00

Minimum Required Flange Thickness [Rigidity] 242.189 mm

Estimated M.A.W.P. (Operating) 1753 KPa.

Estimated Finished Weight of Flange at given Thk. 3305.1 kg.

Estimated Unfinished Weight of Forging at given Thk 3878.8 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	482.04	499.99	0.964	482.04	499.99	0.964
Tangential Flange	St	360.37	374.99	0.961	360.37	374.99	0.961
Tangential Flange	Sto	111.29	250.00	0.445	111.29	250.00	0.445
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	368.73	749.99	0.492	368.73	749.99	0.492
Combined	Sh + Sto	593.33	749.99	0.791	593.33	749.99	0.791
Radial Flange	Sr	8.36	499.99	0.017	8.36	499.99	0.017
Maximum Stress Ratio				1.000			1.000



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Flg Calc [Int P]: New Flange

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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.4218/1.1492)

= 1.2372

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.3026/1.0594) if $h/h0 < 0.35$, else max (0.25; 0.2521/0.7854)

= 0.2856

In this case, the hub ratio $h/h0$, 0.2400 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	638.4 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Flg Calc [Int P]: New Flange

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Root Area of a Single Bolt	Abs	22.0838 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	1.056 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (2576.0^2 - 2532.0^2) + (0.5 \cdot 2535.0 \cdot 5.0)$$

$$= 1828.6 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207.0 \cdot 1828.578 / (36 \cdot 22.084)$$

$$= 476.1 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(476.11; 507.5)$$

$$= 476.1 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(476.11; 290.0)$$

$$= 476.1 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(476.11; 638.354)$$

$$= 476.1 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 476.11 \geq 140.0(1828.578 / (36 \cdot 22.084))$$

$$:: 476.1 \geq 322.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 476.11 \geq (97.0 \cdot 1828.578 + \pi/4 \cdot 1024.172 \cdot 2532.0^2) / (0.7 \cdot 22.084 \cdot 36)$$

$$:: 476.1 \geq 411.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 476.11 \leq 276.0(1828.578 / (36 \cdot 22.084))$$

$$:: 476.1 \leq 634.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$$

$$:: 476.11 \leq 638.354(1.0/1.056)$$

$$:: 476.1 \leq 604.4 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 476.11 \cdot 0.2 \cdot 22.084 \cdot 57.15$$

$$= 1225.5 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 132963.6 / 1.0 \cdot 9806.637 \cdot 0.405 / (8.907 \cdot 201334 \cdot 16.0^2 \cdot 200.0 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 65673.1 / 1.0 \cdot 9806.637 \cdot 0.405 / (8.907 \cdot 197847 \cdot 16.0^2 \cdot 200.0 \cdot 0.3)$
 $= 0.502$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 132963.6 / 1.0 \cdot 9806.637 \cdot 0.405 / (9.392 \cdot 201334 \cdot 16.0^2 \cdot 200.0 \cdot 0.3)$
 $= 0.948$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 65673.1 / 1.0 \cdot 9806.637 \cdot 0.405 / (9.392 \cdot 197847 \cdot 16.0^2 \cdot 200.0 \cdot 0.3)$
 $= 0.476$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.552, Temperature Reduction per Fig. UCS 66.1 = 27 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-41 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	65.10	207.00	131.81	207.00
Radial Flange	1.11	138.00	2.25	138.00
Tangential Flange	47.96	138.00	97.10	138.00
Maximum Average	56.53	138.00	114.45	138.00
Bolting	72.45	172.00	69.27	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 242.189 mm

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Flg Calc [Int P]: New Flange

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Flange Input Data Values

Description: New Flange :

ch, flange 2

Flange Type	Integral Weld Neck, Integral type		
Design Pressure	P	1024.56	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2500.000	mm
Flange Outside Diameter	A	2900.000	mm
Flange Thickness	t	250.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	g1	23.0000	mm
Length of Hub	h	48.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2750.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	2581.000	mm
Flange Face Inside Diameter	Fid	2500.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2570.000	mm
Gasket Inside Diameter	Gi	2526.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2535.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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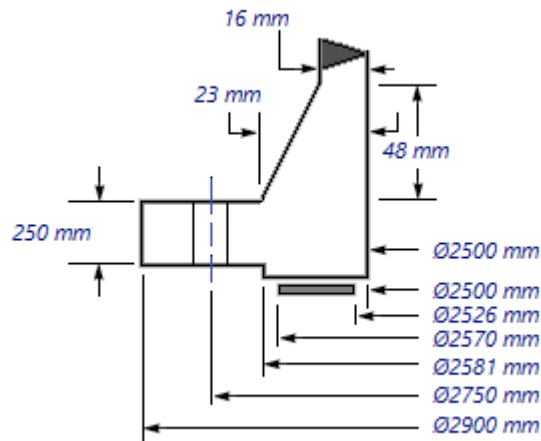
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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1024.56 \cdot (2500.0/2 + 0.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1024.56) + C_a$$

$$= 9.3227 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (1250.0 + 0.6 \cdot 16.0)$$

$$= 1752.804 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (1250.0 + 0.6 \cdot 16.0)$$

$$= 1752.804 \text{ KPa.}$$

Corroded Flange ID,	Bcor = B + 2 * ci	2500.000 mm
Corroded Large Hub,	g1Cor = g1 - ci	23.000 mm
Corroded Small Hub,	g0Cor = g0 - ci	16.000 mm
Code R Dimension,	R = (C - B) / 2 - g1	102.000 mm
Gasket Contact Width,	N = (Go - Gi) / 2	22.000 mm
Basic Gasket Width,	bo = N / 2	11.000 mm
Effective Gasket Width,	b = Cb * sqrt(bo)	8.358 mm
Gasket Reaction Diameter,	G = Go - 2 * b	2553.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 2553.2849^2 \cdot 1024.564$$

$$= 534963.312 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.3576 \cdot \pi \cdot 2553.2849 \cdot 3.0 \cdot 1024.56$$

$$+ 2 \cdot 2535.0 \cdot 5.0 \cdot 3.0 \cdot 1024.5638$$

$$= 49971.871 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 2500.0^2 \cdot 1024.5638$$

$$= 512867.875 \text{ Kgf}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 534963 - 512868$$

$$= 22095.422 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(534963 + 49972 + 0, 0)$$

$$= 584935.188 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.3576 \cdot \pi \cdot 2553.285 + 68.95 \cdot 5.0 \cdot 2535.0$$

$$= 560459.625 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((584935 + 0 + 4 \cdot 0 / 2553.28) / 172, 560460 / 172)$$

$$= 333.515 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 57.15 + 6 \cdot 250.0 / (3.0 + 0.5)$$

$$= 542.871 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2750.0 \cdot \sin(\pi / 36)$$

$$= 239.678 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 239.678 / (2 \cdot 57.15 + 250.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	333.515	795.018	
Radial Distance between Hub and Bolts:	69.850	102.000	
Radial Distance between Bolts and Edge:	57.150	75.000	
Circ. Spacing between the Bolts:	120.650	239.678	542.871

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 795.018 \cdot 172.0 / (68.95 \cdot \pi (2570.0 + 2526.0))$$

$$= 12.388 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 795.0178$$

$$= 1394340.25 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 584935 - 534963$$

$$= 49971.86 \text{ Kgf}$$

Moment Arm Calculations:



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Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2750.0002 - 2553.2849) / 2$$

$$= 98.3577 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2750.0002 - 2500.0) / 2 + 98.3577)$$

$$= 111.6788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2750.0 - 2500.0 - 23.0) / 2$$

$$= 113.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	512868.	113.5000	1.0000	58212.
Face Pressure, Mt	22095.	111.6788	1.0000	2468.
Gasket Load, Mg	49972.	98.3577	1.0000	4915.
Gasket Seating, Matm	1394340.	98.3577	1.0000	137147.

Total Moment for Operation, Mo 65595. Kg-m.

Total Moment for Gasket seating, Mg 137147. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 200.000 mm

Hub Ratio, h/h0 = HL / H0 0.240

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.438

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.895

Factor V per 4.16.5 0.405

Factor f per 4.16.5 1.256

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2900.0 / 2500.0$$

$$= 1.160$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.16^2 \cdot \log_{10}(1.16))) / (1.16^2 - 1)) / (1.16 - 1)$$

$$= 13.145$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.16^2 (1 + 8.55246 \cdot \log_{10}(1.16)) - 1) / ((1.0472 + 1.9448 \cdot 1.16^2) \cdot (1.16 - 1))$$

$$= 1.855$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.16^2 (1 + 8.55246 \cdot \log_{10}(1.16)) - 1) / (1.36136 (1.16^2 - 1) \cdot (1.16 - 1))$$

$$= 14.445$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.16^2 + 1) / (1.16^2 - 1)$$

$$= 6.787$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$

$$= 14.445 \cdot 16.0^2 \cdot 200.0 / 0.405$$



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Flg Calc [Int P]: New Flange

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$$= 1824986.750 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.895/200.0$$

$$= 0.004 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (250.0 \cdot 0.004 + 1) / 1.855 + 250.0^3/1824987$$

$$= 9.704$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$$

$$= (1.2556 \cdot 65595/2500.0) / (9.7039 \cdot 23.0^2)$$

$$= 62.93 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$$

$$= (1.2556 \cdot 137147/2500.0) / (9.7039 \cdot 23.0^2)$$

$$= 131.58 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.4913 \cdot 65595/2500.0) / (9.7039 \cdot 250.0^2)$$

$$= 1.06 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm/Bcor) / (L \cdot t^2)$$

$$= (2.4913 \cdot 137147/2500.0) / (9.7039 \cdot 250.0^2)$$

$$= 2.21 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (13.1451 \cdot 65595 / (250.0^2 \cdot 2500.0)) - 6.787 \cdot 1$$

$$= 46.94 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (13.1451 \cdot 137147 / (250.0^2 \cdot 2500.0)) - 6.787 \cdot 2$$

$$= 98.15 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (63 + \max(1, 47)) / 2$$

$$= 54.94 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (132 + \max(2, 98)) / 2$$

$$= 114.87 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 584935/795.0178$$

$$= 72.15 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (560460/795.0178)$$

$$= 69.13 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²



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Flg Calc [Int P]: New Flange

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	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	62.93	207.00	131.58	207.00
Radial Flange	1.06	138.00	2.21	138.00
Tangential Flange	46.94	138.00	98.15	138.00
Maximum Average	54.94	138.00	114.87	138.00
Bolting	72.15	172.00	69.13	172.00

Minimum Required Flange Thickness [Rigidity] 244.958 mm

Estimated M.A.W.P. (Operating) 1753 KPa.

Estimated Finished Weight of Flange at given Thk. 3344.5 kg.

Estimated Unfinished Weight of Forging at given Thk 3918.2 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	479.15	499.99	0.958	479.15	499.99	0.958
Tangential Flange	St	362.70	374.99	0.967	362.70	374.99	0.967
Tangential Flange	Sto	110.62	250.00	0.442	110.62	250.00	0.442
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	370.87	749.99	0.495	370.87	749.99	0.495
Combined	Sh + Sto	589.77	749.99	0.786	589.77	749.99	0.786
Radial Flange	Sr	8.17	499.99	0.016	8.17	499.99	0.016
Maximum Stress Ratio				1.000			1.000



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Flg Calc [Int P]: New Flange

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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.4218/1.1492)

= 1.2372

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.3026/1.0594) if $h/h0 < 0.35$, else max (0.25; 0.2521/0.7854)

= 0.2856

In this case, the hub ratio $h/h0$, 0.2400 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	635.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Flg Calc [Int P]: New Flange

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Root Area of a Single Bolt	Abs	22.0838 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetaamax	1.050 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (2570.0^2 - 2526.0^2) + (0.5 \cdot 2535.0 \cdot 5.0)$$

$$= 1824.4 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207.0 \cdot 1824.426 / (36 \cdot 22.084)$$

$$= 475.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(475.029; 507.5)$$

$$= 475.0 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(475.029; 290.0)$$

$$= 475.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(475.029; 635.599)$$

$$= 475.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 475.029 \geq 140.0(1824.426 / (36 \cdot 22.084))$$

$$:: 475.0 \geq 321.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 475.029 \geq (97.0 \cdot 1824.426 + \pi/4 \cdot 1024.564 \cdot 2526.0^2) / (0.7 \cdot 22.084 \cdot 36)$$

$$:: 475.0 \geq 410.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 475.029 \leq 276.0(1824.426 / (36 \cdot 22.084))$$

$$:: 475.0 \leq 633.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetaamax)$$

$$:: 475.029 \leq 635.599(1.0/1.05)$$

$$:: 475.0 \leq 605.4 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 475.029 \cdot 0.2 \cdot 22.084 \cdot 57.15$$

$$= 1222.7 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 137146.8 / 1.0 \cdot 9806.637 \cdot 0.405 / (9.183 \cdot 201334 \cdot 16.0^2 \cdot 200.0 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 65594.5 / 1.0 \cdot 9806.637 \cdot 0.405 / (9.183 \cdot 197847 \cdot 16.0^2 \cdot 200.0 \cdot 0.3)$
 $= 0.487$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 137146.8 / 1.0 \cdot 9806.637 \cdot 0.405 / (9.704 \cdot 201334 \cdot 16.0^2 \cdot 200.0 \cdot 0.3)$
 $= 0.947$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 65594.5 / 1.0 \cdot 9806.637 \cdot 0.405 / (9.704 \cdot 197847 \cdot 16.0^2 \cdot 200.0 \cdot 0.3)$
 $= 0.460$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.552, Temperature Reduction per Fig. UCS 66.1 = 27 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-41 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	62.93	207.00	131.58	207.00
Radial Flange	1.06	138.00	2.21	138.00
Tangential Flange	46.94	138.00	98.15	138.00
Maximum Average	54.94	138.00	114.87	138.00
Bolting	72.15	172.00	69.13	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 244.958 mm

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Flg Calc [Int P]: New Flange

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Flange Input Data Values

Description: New Flange :

Shell flng-1

Flange Type	Integral Weld Neck		
Design Pressure	P	624.56	KPa.
Design Temperature		230	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2510.000	mm
Flange Outside Diameter	A	2900.000	mm
Flange Thickness	t	260.0000	mm
Thickness of Hub at Small End	go	12.5000	mm
Thickness of Hub at Large End	gl	19.0000	mm
Length of Hub	h	57.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	136.80	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2750.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	2581.000	mm
Flange Face Inside Diameter	Fid	2510.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2570.000	mm
Gasket Inside Diameter	Gi	2526.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2535.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



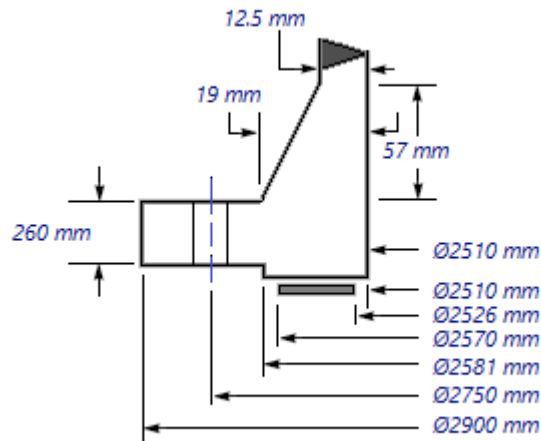
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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (624.56 \cdot (2510.0/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 624.56) + Ca$$

$$= 8.7094 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.5) / (1258.0 + 0.6 \cdot 9.5)$$

$$= 1037.351 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.5) / (1255.0 + 0.6 \cdot 12.5)$$

$$= 1366.232 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	257.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2516.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	16.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	9.500 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	101.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2553.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2553.2849^2 \cdot 624.564$$

$$= 326108.281 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.3576 \cdot \pi \cdot 2553.2849 \cdot 3.0 \cdot 624.56$$

$$+ 2 \cdot 2535.0 \cdot 5.0 \cdot 3.0 \cdot 624.5639$$

$$= 30462.352 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2516.0^2 \cdot 624.5639$$

$$= 316653.719 \text{ Kgf}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 326108 - 316654$$

$$= 9454.555 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(326108 + 30462 + 0, 0)$$

$$= 356570.656 \text{ Kg}$$

$$= 584935.188 \text{ Kg, Mating Flange Load Governs}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.3576 \cdot \pi \cdot 2553.285 + 68.95 \cdot 5.0 \cdot 2535.0$$

$$= 560459.625 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((584935 + 0 + 4 \cdot 0 / 2553.28) / 172, 560460 / 172)$$

$$= 333.515 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 57.15 + 6 \cdot 257.0 / (3.0 + 0.5)$$

$$= 554.871 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2750.0 \cdot \sin(\pi / 36)$$

$$= 239.678 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 239.678 / (2 \cdot 57.15 + 257.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	333.515	795.018	
Radial Distance between Hub and Bolts:	69.850	101.000	
Radial Distance between Bolts and Edge:	57.150	75.000	
Circ. Spacing between the Bolts:	120.650	239.678	554.871

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 795.018 \cdot 172.0 / (68.95 \cdot \pi (2570.0 + 2526.0))$$

$$= 12.388 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 795.0178$$

$$= 1394340.25 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 584935 - 326108$$

$$= 258826.88 \text{ Kg}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2750.0002 - 2553.2849) / 2$$

$$= 98.3577 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2750.0002 - 2516.0) / 2 + 98.3577)$$

$$= 107.6788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2750.0 - 2516.0 - 16.0) / 2$$

$$= 109.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	316654.	109.0000	1.0000	34516.
Face Pressure, Mt	9455.	107.6788	1.0000	1018.
Gasket Load, Mg	258827.	98.3577	1.0000	25458.
Gasket Seating, Matm	1394340.	98.3577	1.0000	137147.

Total Moment for Operation, Mo 60992. Kg-m.

Total Moment for Gasket seating, Mg 137147. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 154.603 mm

Hub Ratio, h/h0 = HL / H0 0.369

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.684

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.861

Factor V per 4.16.5 0.313

Factor f per 4.16.5 1.231

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2900.0 / 2516.0$$

$$= 1.153$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.153^2 \cdot \log_{10}(1.153))) / (1.153^2 - 1)) / (1.153 - 1)$$

$$= 13.724$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.153^2(1 + 8.55246 \cdot \log_{10}(1.153)) - 1) / ((1.0472 + 1.9448 \cdot 1.153^2) \cdot (1.153 - 1))$$

$$= 1.858$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.153^2(1 + 8.55246 \cdot \log_{10}(1.153)) - 1) / (1.36136(1.153^2 - 1) \cdot (1.153 - 1))$$

$$= 15.081$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.153^2 + 1) / (1.153^2 - 1)$$

$$= 7.088$$

Stress factor from Table 4.16.4 for integral flanges [d]:



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$$\begin{aligned}
 &= U \cdot g_0^2 \cdot h_0 / V \\
 &= 15.081 \cdot 9.5^2 \cdot 154.603 / 0.313 \\
 &= 671333.312 \text{ mm}^3
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned}
 &= F / h_0 \\
 &= 0.861 / 154.603 \\
 &= 0.006 \text{ 1/mm}
 \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned}
 &= (t \cdot e + 1) / T + t^3/d \\
 &= (257.0 \cdot 0.006 + 1) / 1.858 + 257.0^3 / 671333 \\
 &= 26.594
 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned}
 &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\
 &= (1.2314 \cdot 60992 / 2516.0) / (26.594 \cdot 16.0^2) \\
 &= 43.00 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned}
 &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\
 &= (1.2314 \cdot 137147 / 2516.0) / (26.594 \cdot 16.0^2) \\
 &= 96.69 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned}
 &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\
 &= (2.9091 \cdot 60992 / 2516.0) / (26.594 \cdot 257.0^2) \\
 &= 0.39 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned}
 &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\
 &= (2.9091 \cdot 137147 / 2516.0) / (26.594 \cdot 257.0^2) \\
 &= 0.89 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\
 &= (13.7235 \cdot 60992 / (257.0^2 \cdot 2516.0)) - 7.0875 \cdot 0.4 \\
 &= 46.60 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\
 &= (13.7235 \cdot 137147 / (257.0^2 \cdot 2516.0)) - 7.0875 \cdot 0.9 \\
 &= 104.79 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (SHo + \max(SRo, STo)) / 2 \\
 &= (43 + \max(0.4, 47)) / 2 \\
 &= 44.80 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned}
 &= (SHa + \max(SRa, STa)) / 2 \\
 &= (97 + \max(0.9, 105)) / 2 \\
 &= 100.74 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= Wm1 / Ab \\
 &= 584935 / 795.0178 \\
 &= 72.15 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (Wm2 / Ab) \\
 &= (560460 / 795.0178) \\
 &= 69.13 \text{ N./mm}^2
 \end{aligned}$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	43.00	205.20	96.69	207.00
Radial Flange	0.39	136.80	0.89	138.00
Tangential Flange	46.60	136.80	104.79	138.00
Maximum Average	44.80	136.80	100.74	138.00
Bolting	72.15	172.00	69.13	172.00

Minimum Required Flange Thickness [Rigidity] 258.369 mm

Estimated M.A.W.P. (Operating) 1037 KPa.

Estimated Finished Weight of Flange at given Thk. 3394.5 kg.

Estimated Unfinished Weight of Forging at given Thk 4071.4 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	330.13	475.43	0.694	330.13	499.99	0.660
Tangential Flange	St	356.57	356.57	1.000	356.57	374.99	0.951
Tangential Flange	Sto	111.51	237.71	0.469	111.51	250.00	0.446
Combined	Sh/f + St	623.74	713.14	0.875	623.74	749.99	0.832
Combined	Sr + St	359.58	713.14	0.504	359.58	749.99	0.479
Combined	Sh + Sto	441.64	713.14	0.619	441.64	749.99	0.589
Radial Flange	Sr	3.01	475.43	0.006	3.01	499.99	0.006
Maximum Stress Ratio				1.000			0.951



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Flg Calc [Int P]: New Flange

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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.6783/1.3582)

= 1.2357

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.5502/1.3211) if $h/h0 < 0.35$, else max (0.25; 0.3154/0.7556)

= 0.4174

In this case, the hub ratio $h/h0$, 0.3687 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	585.2 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	22.0838 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	208.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.505 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2570.0^2 - 2526.0^2) + (0.5 \cdot 2535.0 \cdot 5.0)$
 $= 1824.4 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207.0 \cdot 1824.426 / (36 \cdot 22.084)$
 $= 475.0 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(475.029; 507.5)$
 $= 475.0 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(475.029; 290.0)$
 $= 475.0 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(475.029; 585.233)$
 $= 475.0 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin - S(Ag / (nb \cdot Abs))$
 $:: 475.029 \geq 140.0 (1824.426 / (36 \cdot 22.084))$
 $:: 475.0 \geq 321.3 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 475.029 \geq (97.0 \cdot 1824.426 + \pi/4 \cdot 624.564 \cdot 2526.0^2) / (0.7 \cdot 22.084 \cdot 36)$
 $:: 475.0 \geq 374.2 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 475.029 \leq 276.0 (1824.426 / (36 \cdot 22.084))$
 $:: 475.0 \leq 633.4 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$
 $:: 475.029 \leq 585.233 (1.0/0.505)$
 $:: 475.0 \leq 1159.6 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 475.029 \cdot 0.2 \cdot 22.084 \cdot 57.15$
 $= 1222.7 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 92 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 137146.8 / 1.0 \cdot 9806.637 \cdot 0.313 / (26.106 \cdot 201334 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 60992.1 / 1.0 \cdot 9806.637 \cdot 0.313 / (26.106 \cdot 189090 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.473$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 137146.8 / 1.0 \cdot 9806.637 \cdot 0.313 / (26.594 \cdot 201334 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.980$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 60992.1 / 1.0 \cdot 9806.637 \cdot 0.313 / (26.594 \cdot 189090 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.464$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.702, Temperature Reduction per Fig. UCS 66.1 = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-38 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	45.54	207.00	101.58	207.00
Radial Flange	0.58	138.00	1.30	138.00
Tangential Flange	43.92	138.00	97.97	138.00
Maximum Average	44.73	138.00	99.78	138.00
Bolting	72.15	172.00	69.13	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 251.155 mm

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Flange Input Data Values

Description: New Flange :

Shell flng. 2

Flange Type	Integral Weld Neck		
Design Pressure	P	624.56	KPa.
Design Temperature		230	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2510.000	mm
Flange Outside Diameter	A	2900.000	mm
Flange Thickness	t	265.0000	mm
Thickness of Hub at Small End	go	12.5000	mm
Thickness of Hub at Large End	gl	19.0000	mm
Length of Hub	h	57.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	136.80	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2750.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	2581.000	mm
Flange Face Inside Diameter	Fid	2510.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2570.000	mm
Gasket Inside Diameter	Gi	2526.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2535.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



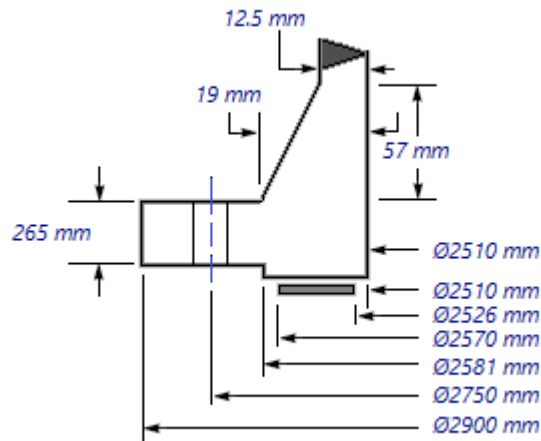
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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (624.56 \cdot (2510.0/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 624.56) + Ca$$

$$= 8.7094 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.5) / (1258.0 + 0.6 \cdot 9.5)$$

$$= 1037.351 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.5) / (1255.0 + 0.6 \cdot 12.5)$$

$$= 1366.232 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	262.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2516.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	16.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	9.500 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	101.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2553.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2553.2849^2 \cdot 624.564$$

$$= 326108.281 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.3576 \cdot \pi \cdot 2553.2849 \cdot 3.0 \cdot 624.56$$

$$+ 2 \cdot 2535.0 \cdot 5.0 \cdot 3.0 \cdot 624.5639$$

$$= 30462.352 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2516.0^2 \cdot 624.5639$$

$$= 316653.719 \text{ Kgf}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 326108 - 316654$$

$$= 9454.555 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(326108 + 30462 + 0, 0)$$

$$= 356570.656 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.3576 \cdot \pi \cdot 2553.285 + 68.95 \cdot 5.0 \cdot 2535.0$$

$$= 560459.625 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((356571 + 0 + 4 \cdot 0 / 2553.28) / 172, 560460 / 172)$$

$$= 319.560 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 57.15 + 6 \cdot 262.0 / (3.0 + 0.5)$$

$$= 563.443 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2750.0 \cdot \sin(\pi / 36)$$

$$= 239.678 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 239.678 / (2 \cdot 57.15 + 262.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	319.560	795.018	
Radial Distance between Hub and Bolts:	69.850	101.000	
Radial Distance between Bolts and Edge:	57.150	75.000	
Circ. Spacing between the Bolts:	120.650	239.678	563.443

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 795.018 \cdot 172.0 / (68.95 \cdot \pi (2570.0 + 2526.0))$$

$$= 12.388 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 795.0178$$

$$= 1394340.25 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 356571 - 326108$$

$$= 30462.36 \text{ Kgf}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2750.0002 - 2553.2849) / 2$$

$$= 98.3577 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2750.0002 - 2516.0) / 2 + 98.3577)$$

$$= 107.6788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2750.0 - 2516.0 - 16.0) / 2$$

$$= 109.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	316654.	109.0000	1.0000	34516.
Face Pressure, Mt	9455.	107.6788	1.0000	1018.
Gasket Load, Mg	30462.	98.3577	1.0000	2996.
Gasket Seating, Matm	1394340.	98.3577	1.0000	137147.

Total Moment for Operation, Mo 38530. Kg-m.

Total Moment for Gasket seating, Mg 137147. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 154.603 mm

Hub Ratio, h/h0 = HL / H0 0.369

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.684

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.861

Factor V per 4.16.5 0.313

Factor f per 4.16.5 1.231

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2900.0 / 2516.0$$

$$= 1.153$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.153^2 \cdot \log_{10}(1.153))) / (1.153^2 - 1)) / (1.153 - 1)$$

$$= 13.724$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.153^2 (1 + 8.55246 \cdot \log_{10}(1.153)) - 1) / ((1.0472 + 1.9448 \cdot 1.153^2) \cdot (1.153 - 1))$$

$$= 1.858$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.153^2 (1 + 8.55246 \cdot \log_{10}(1.153)) - 1) / (1.36136 (1.153^2 - 1) \cdot (1.153 - 1))$$

$$= 15.081$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.153^2 + 1) / (1.153^2 - 1)$$

$$= 7.088$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$



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$$= 15.081 \cdot 9.5^2 \cdot 154.603 / 0.313$$

$$= 671333.312 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.861 / 154.603$$

$$= 0.006 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (262.0 \cdot 0.006 + 1) / 1.858 + 262.0^3 / 671333$$

$$= 28.114$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$$

$$= (1.2314 \cdot 38530 / 2516.0) / (28.1136 \cdot 16.0^2)$$

$$= 25.70 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$$

$$= (1.2314 \cdot 137147 / 2516.0) / (28.1136 \cdot 16.0^2)$$

$$= 91.46 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.9462 \cdot 38530 / 2516.0) / (28.1136 \cdot 262.0^2)$$

$$= 0.23 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.9462 \cdot 137147 / 2516.0) / (28.1136 \cdot 262.0^2)$$

$$= 0.82 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (13.7235 \cdot 38530 / (262.0^2 \cdot 2516.0)) - 7.0875 \cdot 0.2$$

$$= 28.40 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (13.7235 \cdot 137147 / (262.0^2 \cdot 2516.0)) - 7.0875 \cdot 0.8$$

$$= 101.09 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (26 + \max(0.2, 28)) / 2$$

$$= 27.05 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (91 + \max(0.8, 101)) / 2$$

$$= 96.28 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 356571 / 795.0178$$

$$= 43.98 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (560460 / 795.0178)$$

$$= 69.13 \text{ N./mm}^2$$



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Flg Calc [Int P]: New Flange

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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	25.70	205.20	91.46	207.00
Radial Flange	0.23	136.80	0.82	138.00
Tangential Flange	28.40	136.80	101.09	138.00
Maximum Average	27.05	136.80	96.28	138.00
Bolting	43.98	172.00	69.13	172.00

Minimum Required Flange Thickness [Rigidity] 258.369 mm

Estimated M.A.W.P. (Operating) 1037 KPa.

Estimated Finished Weight of Flange at given Thk. 3458.7 kg.

Estimated Unfinished Weight of Forging at given Thk 4135.6 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	323.74	475.43	0.681	323.74	499.99	0.648
Tangential Flange	St	356.57	356.57	1.000	356.57	374.99	0.951
Tangential Flange	Sto	109.35	237.71	0.460	109.35	250.00	0.437
Combined	Sh/f + St	618.57	713.14	0.867	618.57	749.99	0.825
Combined	Sr + St	359.45	713.14	0.504	359.45	749.99	0.479
Combined	Sh + Sto	433.09	713.14	0.607	433.09	749.99	0.577
Radial Flange	Sr	2.88	475.43	0.006	2.88	499.99	0.006
Maximum Stress Ratio				1.000			0.951



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Flg Calc [Int P]: New Flange

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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.6783/1.3582)

= 1.2357

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.5502/1.3211) if $h/h0 < 0.35$, else max (0.25; 0.3154/0.7556)

= 0.4174

In this case, the hub ratio $h/h0$, 0.3687 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	606.7 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	22.0838 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	208.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.495 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2570.0^2 - 2526.0^2) + (0.5 \cdot 2535.0 \cdot 5.0)$
 $= 1824.4 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207.0 \cdot 1824.426 / (36 \cdot 22.084)$
 $= 475.0 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(475.029; 507.5)$
 $= 475.0 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(475.029; 290.0)$
 $= 475.0 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(475.029; 606.701)$
 $= 475.0 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin - S(Ag / (nb \cdot Abs))$
 $:: 475.029 \geq 140.0 (1824.426 / (36 \cdot 22.084))$
 $:: 475.0 \geq 321.3 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 475.029 \geq (97.0 \cdot 1824.426 + \pi/4 \cdot 624.564 \cdot 2526.0^2) / (0.7 \cdot 22.084 \cdot 36)$
 $:: 475.0 \geq 374.2 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 475.029 \leq 276.0 (1824.426 / (36 \cdot 22.084))$
 $:: 475.0 \leq 633.4 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$
 $:: 475.029 \leq 606.701 (1.0/0.495)$
 $:: 475.0 \leq 1225.9 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 475.029 \cdot 0.2 \cdot 22.084 \cdot 57.15$
 $= 1222.7 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 92 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 137146.8 / 1.0 \cdot 9806.637 \cdot 0.313 / (26.106 \cdot 201334 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38530.3 / 1.0 \cdot 9806.637 \cdot 0.313 / (26.106 \cdot 189090 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.298$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 137146.8 / 1.0 \cdot 9806.637 \cdot 0.313 / (28.114 \cdot 201334 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.927$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38530.3 / 1.0 \cdot 9806.637 \cdot 0.313 / (28.114 \cdot 189090 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.277$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.702, Temperature Reduction per Fig. UCS 66.1 = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-38 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	27.39	207.00	96.26	207.00
Radial Flange	0.34	138.00	1.20	138.00
Tangential Flange	26.93	138.00	94.67	138.00
Maximum Average	27.16	138.00	95.46	138.00
Bolting	43.98	172.00	69.13	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 251.155 mm

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Flg Calc [Int P]: New Flange

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Flange Input Data Values

Description: New Flange :

Shell girth fln

Flange Type	Integral Weld Neck		
Design Pressure	P	624.56	KPa.
Design Temperature		230	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2510.000	mm
Flange Outside Diameter	A	2900.000	mm
Flange Thickness	t	265.0000	mm
Thickness of Hub at Small End	go	12.5000	mm
Thickness of Hub at Large End	gl	19.0000	mm
Length of Hub	h	57.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	136.80	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2750.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	2581.000	mm
Flange Face Inside Diameter	Fid	2510.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2570.000	mm
Gasket Inside Diameter	Gi	2526.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2535.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



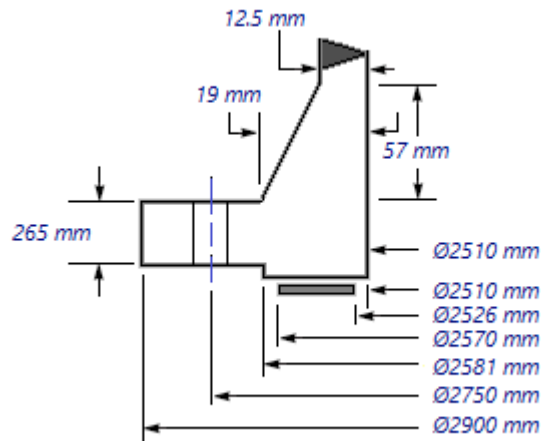
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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (624.56 \cdot (2510.0/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 624.56) + Ca$$

$$= 8.7094 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.5) / (1258.0 + 0.6 \cdot 9.5)$$

$$= 1037.351 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.5) / (1255.0 + 0.6 \cdot 12.5)$$

$$= 1366.232 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	262.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2516.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	16.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	9.500 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	101.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2553.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2553.2849^2 \cdot 624.564$$

$$= 326108.281 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.3576 \cdot \pi \cdot 2553.2849 \cdot 3.0 \cdot 624.56$$

$$+ 2 \cdot 2535.0 \cdot 5.0 \cdot 3.0 \cdot 624.5639$$

$$= 30462.352 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2516.0^2 \cdot 624.5639$$

$$= 316653.719 \text{ Kgf}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 326108 - 316654$$

$$= 9454.555 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(326108 + 30462 + 0, 0)$$

$$= 356570.656 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.3576 \cdot \pi \cdot 2553.285 + 68.95 \cdot 5.0 \cdot 2535.0$$

$$= 560459.625 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((356571 + 0 + 4 \cdot 0 / 2553.28) / 172, 560460 / 172)$$

$$= 319.560 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 57.15 + 6 \cdot 262.0 / (3.0 + 0.5)$$

$$= 563.443 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2750.0 \cdot \sin(\pi / 36)$$

$$= 239.678 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 239.678 / (2 \cdot 57.15 + 262.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	319.560	795.018	
Radial Distance between Hub and Bolts:	69.850	101.000	
Radial Distance between Bolts and Edge:	57.150	75.000	
Circ. Spacing between the Bolts:	120.650	239.678	563.443

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 795.018 \cdot 172.0 / (68.95 \cdot \pi (2570.0 + 2526.0))$$

$$= 12.388 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 795.0178$$

$$= 1394340.25 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 356571 - 326108$$

$$= 30462.36 \text{ Kg}$$



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Flg Calc [Int P]: New Flange

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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2750.0002 - 2553.2849) / 2$$

$$= 98.3577 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2750.0002 - 2516.0) / 2 + 98.3577)$$

$$= 107.6788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2750.0 - 2516.0 - 16.0) / 2$$

$$= 109.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	316654.	109.0000	1.0000	34516.
Face Pressure, Mt	9455.	107.6788	1.0000	1018.
Gasket Load, Mg	30462.	98.3577	1.0000	2996.
Gasket Seating, Matm	1394340.	98.3577	1.0000	137147.

Total Moment for Operation, Mo 38530. Kg-m.

Total Moment for Gasket seating, Mg 137147. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 154.603 mm

Hub Ratio, h/h0 = HL / H0 0.369

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.684

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.861

Factor V per 4.16.5 0.313

Factor f per 4.16.5 1.231

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2900.0 / 2516.0$$

$$= 1.153$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.153^2 \cdot \log_{10}(1.153))) / (1.153^2 - 1)) / (1.153 - 1)$$

$$= 13.724$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.153^2 (1 + 8.55246 \cdot \log_{10}(1.153)) - 1) / ((1.0472 + 1.9448 \cdot 1.153^2) \cdot (1.153 - 1))$$

$$= 1.858$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.153^2 (1 + 8.55246 \cdot \log_{10}(1.153)) - 1) / (1.36136 (1.153^2 - 1) \cdot (1.153 - 1))$$

$$= 15.081$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.153^2 + 1) / (1.153^2 - 1)$$

$$= 7.088$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$



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$$= 15.081 \cdot 9.5^2 \cdot 154.603/0.313$$

$$= 671333.312 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.861/154.603$$

$$= 0.006 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (262.0 \cdot 0.006 + 1) / 1.858 + 262.0^3/671333$$

$$= 28.114$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.2314 \cdot 38530/2516.0)/(28.1136 \cdot 16.0^2)$$

$$= 25.70 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.2314 \cdot 137147/2516.0)/(28.1136 \cdot 16.0^2)$$

$$= 91.46 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.9462 \cdot 38530/2516.0)/(28.1136 \cdot 262.0^2)$$

$$= 0.23 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm/Bcor) / (L \cdot t^2)$$

$$= (2.9462 \cdot 137147/2516.0)/(28.1136 \cdot 262.0^2)$$

$$= 0.82 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (13.7235 \cdot 38530/(262.0^2 \cdot 2516.0)) - 7.0875 \cdot 0.2$$

$$= 28.40 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (13.7235 \cdot 137147/(262.0^2 \cdot 2516.0)) - 7.0875 \cdot 0.8$$

$$= 101.09 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (26 + \max(0.2, 28))/2$$

$$= 27.05 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (91 + \max(0.8, 101))/2$$

$$= 96.28 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 356571/795.0178$$

$$= 43.98 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (560460/795.0178)$$

$$= 69.13 \text{ N./mm}^2$$



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Flg Calc [Int P]: New Flange

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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	25.70	205.20	91.46	207.00
Radial Flange	0.23	136.80	0.82	138.00
Tangential Flange	28.40	136.80	101.09	138.00
Maximum Average	27.05	136.80	96.28	138.00
Bolting	43.98	172.00	69.13	172.00

Minimum Required Flange Thickness [Rigidity] 258.369 mm

Estimated M.A.W.P. (Operating) 1037 KPa.

Estimated Finished Weight of Flange at given Thk. 3458.7 kg.

Estimated Unfinished Weight of Forging at given Thk 4135.6 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	323.74	475.43	0.681	323.74	499.99	0.648
Tangential Flange	St	356.57	356.57	1.000	356.57	374.99	0.951
Tangential Flange	Sto	109.35	237.71	0.460	109.35	250.00	0.437
Combined	Sh/f + St	618.57	713.14	0.867	618.57	749.99	0.825
Combined	Sr + St	359.45	713.14	0.504	359.45	749.99	0.479
Combined	Sh + Sto	433.09	713.14	0.607	433.09	749.99	0.577
Radial Flange	Sr	2.88	475.43	0.006	2.88	499.99	0.006
Maximum Stress Ratio				1.000			0.951



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Flg Calc [Int P]: New Flange

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Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.6783/1.3582)

= 1.2357

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.5502/1.3211) if $h/h_0 < 0.35$, else max (0.25; 0.3154/0.7556)

= 0.4174

In this case, the hub ratio h/h_0 , 0.3687 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	518.0 N./mm ²
Minimum permissible bolt stress	Sbmin	242.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	436.0 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	42.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	88.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Flg Calc [Int P]: New Flange

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Root Area of a Single Bolt	Abs	22.0838 cm ²
Single flange rotation at Sfmax	Thetamax	0.320 deg
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	208.0 N./mm ²

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (2570.0^2 - 2526.0^2) + (0.5 \cdot 2535.0 \cdot 5.0)$$

$$= 1824.4 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 1824.426 / (36 \cdot 22.084)$$

$$= 477.3 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(477.323; 518.0)$$

$$= 477.3 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(477.323; 242.0)$$

$$= 477.3 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(477.323; 436.0)$$

$$= 436.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 436.0 \geq 88.0(1824.426 / (36 \cdot 22.084))$$

$$:: 436.0 \geq 201.9 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 436.0 \geq (42.0 \cdot 1824.426 + \pi/4 \cdot 624.564 \cdot 2526.0^2) / (0.7 \cdot 22.084 \cdot 36)$$

$$:: 436.0 \geq 193.9 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 436.0 \leq 276.0(1824.426 / (36 \cdot 22.084))$$

$$:: 436.0 \leq 633.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetamax} / \text{thetamax})$$

$$:: 436.0 \leq 436.0(1.0/0.32)$$

$$:: 436.0 \leq 1362.5 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 436.0 \cdot 0.2 \cdot 22.084 \cdot 57.15$$

$$= 1122.2 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 92 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 137146.8 / 1.0 \cdot 9806.637 \cdot 0.313 / (26.106 \cdot 201334 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38530.3 / 1.0 \cdot 9806.637 \cdot 0.313 / (26.106 \cdot 189090 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.298$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 137146.8 / 1.0 \cdot 9806.637 \cdot 0.313 / (28.114 \cdot 201334 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.927$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38530.3 / 1.0 \cdot 9806.637 \cdot 0.313 / (28.114 \cdot 189090 \cdot 9.5^2 \cdot 154.603 \cdot 0.3)$
 $= 0.277$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.741, Temperature Reduction per Fig. UCS 66.1 = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -22 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -38 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	27.39	207.00	96.26	207.00
Radial Flange	0.34	138.00	1.20	138.00
Tangential Flange	26.93	138.00	94.67	138.00
Maximum Average	27.16	138.00	95.46	138.00
Bolting	43.98	172.00	69.13	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 251.155 mm

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Internal Pressure Calculations: Step: 9 10:12am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Ch. cover		1028.4	193	3	2900	138	1
Ch.flange-1		1024.2	255	...	2500	138	2
Ch.shell		1024.5	16	3	2500	117.3	3
ch, flange 2		1024.6	258	...	2500	138	4
Shell flng-1		624.56	268	3	2510	136.8	5
Shell-1		624.57	14	0.8	2506	117.3	6
Shell-2		624.57	26	0.8	2506	117.3	7
Shell-3		624.57	14	0.8	2506	117.3	8
Shell flng. 2		624.56	273	3	2510	136.8	9
Shell girth fln		624.56	273	3	2510	136.8	10
Shell -2		624.59	14	3	2506	117.3	11
head- front		624.59	16	3	2506	117.3	12

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Ch. cover		1000	1079.4	1245.71	185	179.559	1
Ch.flange-1		1000	1728.63	1752.8	247	242.189	2
Ch.shell		1000	1184.84	1489.88	16	14.0026	3
ch, flange 2		1000	1728.24	1752.8	250	244.958	4
Shell flng-1		600	1012.79	1366.23	260	258.369	5
Shell-1		600	1063.85	1163.14	12.5	7.49785	6
Shell-2		600	2167.66	2266.81	24.5	7.49785	7
Shell-3		600	1063.85	1163.14	12.5	7.49785	8
Shell flng. 2		600	1012.79	1366.23	265	258.369	9
Shell girth fln		600	1012.79	1366.23	265	258.369	10
Shell -2		600	997.268	1301.79	14	9.70984	11
head- front		600	967.595	1271.69	13.6	9.67072	12

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1013 KPa.
 Shell Side MAPnc = 1163 KPa.
 Channel Side MAWP = 968 KPa.
 Channel Side MAPnc = 1069 KPa.

Elements Suitable for Design Internal Pressure.

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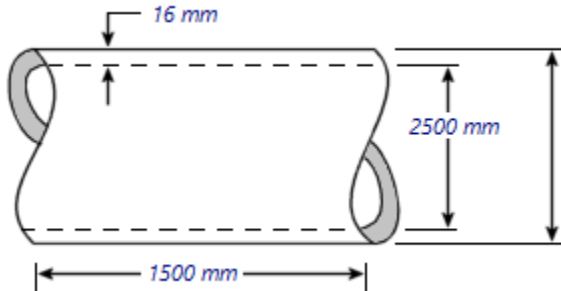


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 Element: Ch.shell

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Ch.shell

Cylindrical Shell Design Information:			
Design Pressure P	1024.534 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	1500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2500.000 mm	Actual Stress	116.9 N./mm ²
Spec. Min. Thk t _e , or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	119317.7 cm ²	SG of Contents	1.000
Element Volume	7364428.0 cm ³	Weight of Contents	7359.9 Kg
Empty Weight	1470.3 Kg	Operating Weight	8830.2 Kg
		Comp. All. Ope. Stress	111.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (1024.534 \cdot 1253.0) / (138.0 \cdot 0.85 - 0.6 \cdot 1024.534)$
 $= 11.0026 + 3.0000 = 14.0026 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.534 KPa.

$= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138.0 \cdot 0.85 \cdot 13.0) / (1253.0 + 0.6 \cdot 13.0)$
 $= 1209 - 25 = 1185 \text{ KPa.}$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138.0 \cdot 0.85 \cdot 16.0) / (1250.0 + 0.6 \cdot 16.0)$
 $= 1490 \text{ KPa.}$

Actual stress at given pressure and thickness, corroded [Sact]:

$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$
 $= (1024.534 \cdot (1253.0 + 0.6 \cdot 13.0)) / (0.85 \cdot 13.0)$
 $= 116.906 \text{ N./mm}^2$



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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.636 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 16.0$, $t_r = 10.388$, $c = 3.0$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.679$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	967.595 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.679	Temperature Difference	18 °C
UCS-66(c) red.	No	Computed Min. Temp.	-32 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Ch.shell

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	2532.00	1500.00	194.77	0.5924	0.0008555	80.36

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 80.3559) / (3 \cdot 194.7692) = 550.061 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.483	2532.00	1500.00	390.56	0.5924	0.0002963	29.63

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 29.6255) / (3 \cdot 390.5622) = 101.1322 \text{ KPa.}$$

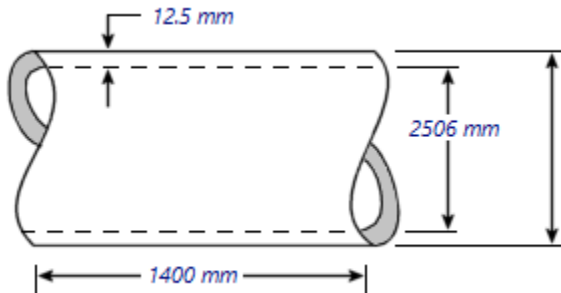
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	2532.00	7980.75	194.77	3.1520	0.0001478	14.78

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 14.7814) / (3 \cdot 194.7692) = 101.183 \text{ KPa.}$$



Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:			
Design Pressure P	624.572 KPa.	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	1400.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.8000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2506.000 mm	Actual Stress	79.2 N./mm ²
Spec. Min. Thk t,e, or sa	12.5000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	111451.1 cm ²	SG of Contents	1.000
Element Volume	3767591.8 cm ³	Weight of Contents	3765.3 Kg
Empty Weight	1202.6 Kg	Operating Weight	4967.9 Kg
		Comp. All. Ope. Stress	91.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (624.572 \cdot 1253.8) / (138.0 \cdot 0.85 - 0.6 \cdot 624.572) \\
 &= 6.6978 + 0.8000 = 7.4978 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.572 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 11.7) / (1253.8 + 0.6 \cdot 11.7) \\
 &= 1088 - 25 = 1064 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 12.5) / (1253.0 + 0.6 \cdot 12.5) \\
 &= 1163 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (624.572 \cdot (1253.8 + 0.6 \cdot 11.7)) / (0.85 \cdot 11.7) \\
 &= 79.187 \text{ N./mm}^2
 \end{aligned}$$



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Element: Shell-1

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.556 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 12.5$, $t_r = 10.883$, $c = 0.8$ mm, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.791$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -22 °C

Min Metal Temp. at Required thickness (UCS 66.1) -33 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

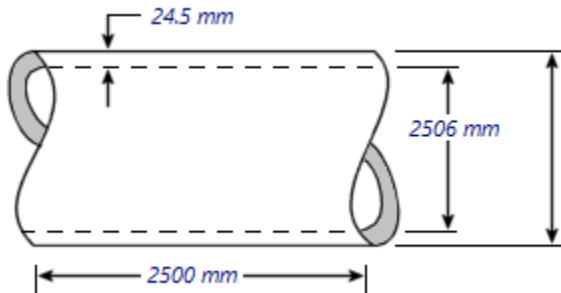
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1012.787 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.500 mm	Basic MDMT	-22 °C
Stress/Thk. Ratio	0.791	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-33 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:			
Design Pressure P	624.572 KPa.	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	2500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.8000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2506.000 mm	Actual Stress	39.3 N./mm ²
Spec. Min. Thk t,e, or sa	24.5000 mm	Spec. Nominal Thk.	26.000 mm
Surface Area	200904.9 cm ²	SG of Contents	1.000
Element Volume	6727841.5 cm ³	Weight of Contents	6723.7 Kg
Empty Weight	4007.3 Kg	Operating Weight	10731.1 Kg
		Comp. All. Ope. Stress	105.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (624.572 \cdot 1253.8) / (138.0 \cdot 0.85 - 0.6 \cdot 624.572) \\
 &= 6.6978 + 0.8000 = 7.4978 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.572 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 23.7) / (1253.8 + 0.6 \cdot 23.7) \\
 &= 2192 - 25 = 2168 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 24.5) / (1253.0 + 0.6 \cdot 24.5) \\
 &= 2267 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (624.572 \cdot (1253.8 + 0.6 \cdot 23.7)) / (0.85 \cdot 23.7) \\
 &= 39.316 \text{ N./mm}^2
 \end{aligned}$$



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FileName : 196-C_Channel MOC=CS_new

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Element: Shell-2

Step: 12 10:12am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.027 %**Minimum Design Metal Temperature Results:**Govrn. thk, $t_g = 24.5$, $t_r = 10.883$, $c = 0.8$ mm, $E^* = 0.85$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

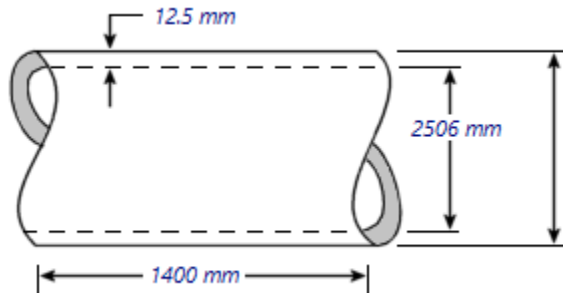
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1012.787 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	24.500 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.390	Temperature Difference	47 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:			
Design Pressure P	624.572 KPa.	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	1400.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.8000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2506.000 mm	Actual Stress	79.2 N./mm ²
Spec. Min. Thk t,e, or sa	12.5000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	111451.1 cm ²	SG of Contents	1.000
Element Volume	3767591.8 cm ³	Weight of Contents	3765.3 Kg
Empty Weight	1202.6 Kg	Operating Weight	4967.9 Kg
		Comp. All. Ope. Stress	91.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (624.572 \cdot 1253.8) / (138.0 \cdot 0.85 - 0.6 \cdot 624.572) \\
 &= 6.6978 + 0.8000 = 7.4978 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.572 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 11.7) / (1253.8 + 0.6 \cdot 11.7) \\
 &= 1088 - 25 = 1064 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 12.5) / (1253.0 + 0.6 \cdot 12.5) \\
 &= 1163 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (624.572 \cdot (1253.8 + 0.6 \cdot 11.7)) / (0.85 \cdot 11.7) \\
 &= 79.187 \text{ N./mm}^2
 \end{aligned}$$



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FileName : 196-C_Channel MOC=CS_new

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Element: Shell-3

Step: 13 10:12am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.556 %

Minimum Design Metal Temperature Results:Govrn. thk, $t_g = 12.5$, $t_r = 10.883$, $c = 0.8$ mm, $E^* = 0.85$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.791$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -22 °C

Min Metal Temp. at Required thickness (UCS 66.1) -33 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

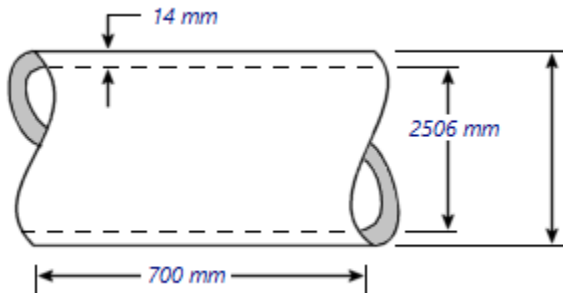
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1012.787 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.500 mm	Basic MDMT	-22 °C
Stress/Thk. Ratio	0.791	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-33 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell -2

Cylindrical Shell Design Information:					
Design Pressure	P	624.593 KPa.	Int. Design Temperature		230 °C
Material		SA-516 70	Ext. Design Temperature		25 °C
Length		700.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2506.000 mm	Actual Stress		84.3 N./mm ²
Spec. Min. Thk t,e, or sa		14.0000 mm	Spec. Nominal Thk.		14.000 mm
Surface Area		55725.6 cm ²	SG of Contents		1.000
Element Volume		3453248.8 cm ³	Weight of Contents		3451.1 Kg
Empty Weight		601.3 Kg	Operating Weight		4052.5 Kg
			Comp. All. Ope. Stress		89.7 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (624.593 \cdot 1256.0) / (138.0 \cdot 0.85 - 0.6 \cdot 624.593) \\
 &= 6.7098 + 3.0000 = 9.7098 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.593 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 11.0) / (1256.0 + 0.6 \cdot 11.0) \\
 &= 1022 - 25 = 997 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 14.0) / (1253.0 + 0.6 \cdot 14.0) \\
 &= 1302 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (624.593 \cdot (1256.0 + 0.6 \cdot 11.0)) / (0.85 \cdot 11.0) \\
 &= 84.348 \text{ N./mm}^2
 \end{aligned}$$



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FileName : 196-C_Channel MOC=CS_new

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Element: Shell -2

Step: 14 10:12am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.556 %

Minimum Design Metal Temperature Results:Govrn. thk, $t_g = 14.0$, $t_r = 10.413$, $c = 3.0$ mm, $E^* = 0.85$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.805$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -29 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

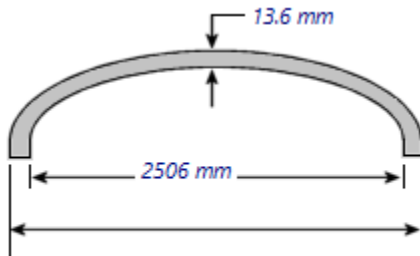
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	967.595 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.805	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-29 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: head- front

Elliptical Head Design Information:			
Design Pressure P	624.593 KPa.	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2506.000 mm	Actual Stress	86.9 N./mm ²
Spec. Min. Thk t,e, or sa	13.6000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	73811.0 cm ²	SG of Contents	1.000
Element Volume	2307099.0 cm ³	Weight of Contents	2305.7 Kg
Empty Weight	993.1 Kg	Operating Weight	3298.8 Kg
		Comp. All. Ope. Stress	89.7 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.325 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.662 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (624.593 \cdot 2512.0 \cdot 0.997) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 624.593)$$

$$= 6.6707 + 3.0000 = 9.6707 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.593 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 10.6) / (0.997 \cdot 2512.0 + 0.2 \cdot 10.6)$$

$$= 992 - 25 = 968 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 13.6) / (1.0 \cdot 2506.0 + 0.2 \cdot 13.6)$$

$$= 1272 \text{ KPa.}$$



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 Element: head- front Step: 15 10:12am Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (624.593 \cdot (0.997 \cdot 2512.0 + 0.2 \cdot 10.6)) / (2 \cdot 0.85 \cdot 10.6)$$

$$= 86.871 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (624.593 \cdot 1256.0) / (138.0 \cdot 0.85 - 0.6 \cdot 624.593) + 3.0$$

$$= 9.710 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 24.593 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 13.0) / (1256.0 + 0.6 \cdot 13.0)$$

$$= 1207 - 25 = 1182 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (2512.0 / (2 \cdot 629.5)))^2 / 6$$

$$= 0.996827$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.765 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 13.6, tr = 10.337, c = 3.0 mm, E* = 0.85
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.829$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -19 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -29 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16.0, tr = 10.413, c = 3.0 mm, E* = 0.85
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.681$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	967.595 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.681	Temperature Difference	18 °C
UCS-66(c) red.	No	Computed Min. Temp.	-32 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.325 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.662 mm

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 Shop Pressure Test Results: Step: 16 10:12am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1316.623 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	780.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1483.657 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1114.066 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	858.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	858.000 KPa.
App 27-4	= M.A.W.P.	1012.787 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1257.873 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	780.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1361.025 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1064.354 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	858.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	858.000 KPa.
App 27-4	= M.A.W.P.	967.595 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch.shell	158.1	234.0	0.676	1385.53
Shell-1	191.2	234.0	0.817	1508.22
Shell-2	94.9	234.0	0.406	1508.22
Shell-3	191.2	234.0	0.817	1508.22
Shell -2	187.1	234.0	0.800	1385.59
head- front	192.7	234.0	0.824	1385.59

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube inlet	Nozzle	118.00	118.00	1.000
Tube inlet	Pad	138.00	138.00	1.000
Tube outlet	Nozzle	118.00	118.00	1.000
Tube outlet	Pad	138.00	138.00	1.000
Shell outlet	Nozzle	138.00	136.80	1.009
Shell inlet	Nozzle	138.00	136.80	1.009

Minimum 1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
SR, 1000.0 Fr60	138.00	138.00	1.000
SR, 4000.0 Fr60	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):



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Shop Pressure Test Results:

Step: 16 10:12am Sep 9, 2026

Description	Ambient	Operating	Ratio
Ch. cover	138.00	138.00	1.000
Ch.flange-1	138.00	138.00	1.000
Ch.shell	138.00	138.00	1.000
ch, flange 2	138.00	138.00	1.000
Shell flng-1	138.00	136.80	1.009
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell flng. 2	138.00	136.80	1.009
Shell girth fln	138.00	136.80	1.009
Shell -2	138.00	138.00	1.000
head- front	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Floating Head Matl.	187.00	171.60	1.090
Backing Ring Matl.	137.90	137.90	1.000
Flange Material	187.00	171.60	1.090
Tube Material	187.00	187.00	1.000
Tubesheet Material	187.00	187.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube inlet	8.54	216.00	0.040
Tube outlet	8.54	216.00	0.040
Shell outlet	35.64	225.00	0.158
Shell inlet	34.20	225.00	0.152

Elements Suitable for Test Pressure.

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 FileName : 196-C_Channel MOC=CS_new Page 91 of 263
 Field Pressure Test Results: Step: 17 10:12am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1316.623 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	780.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1483.657 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1114.066 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	858.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	858.000 KPa.
App 27-4	= M.A.W.P.	1012.787 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1257.873 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	780.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1361.025 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1064.354 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	858.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	858.000 KPa.
App 27-4	= M.A.W.P.	967.595 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 600.0 • 1.0
 = 780.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 600.0 • 1.0
 = 780.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch.shell	91.8	234.0	0.392	804.51
Shell-1	102.0	234.0	0.436	804.56
Shell-2	50.6	234.0	0.216	804.56
Shell-3	102.0	234.0	0.436	804.56
Shell -2	108.7	234.0	0.464	804.56
head- front	111.9	234.0	0.478	804.56

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube inlet	Nozzle	118.00	118.00	1.000
Tube inlet	Pad	138.00	138.00	1.000
Tube outlet	Nozzle	118.00	118.00	1.000
Tube outlet	Pad	138.00	138.00	1.000
Shell outlet	Nozzle	138.00	136.80	1.009
Shell inlet	Nozzle	138.00	136.80	1.009

Minimum 1.000

Stress ratios for Stiffening Ring Materials (N./mm²):



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Field Pressure Test Results:

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Description	Ambient	Operating	Ratio
SR, 1000.0 Fr60	138.00	138.00	1.000
SR, 4000.0 Fr60	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Ch. cover	138.00	138.00	1.000
Ch.flange-1	138.00	138.00	1.000
Ch.shell	138.00	138.00	1.000
ch, flange 2	138.00	138.00	1.000
Shell flng-1	138.00	136.80	1.009
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell flng. 2	138.00	136.80	1.009
Shell girth fln	138.00	136.80	1.009
Shell -2	138.00	138.00	1.000
head- front	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Floating Head Matl.	187.00	171.60	1.090
Backing Ring Matl.	137.90	137.90	1.000
Flange Material	187.00	171.60	1.090
Tube Material	187.00	187.00	1.000
Tubesheet Material	187.00	187.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube inlet	4.96	216.00	0.023
Tube outlet	4.96	216.00	0.023
Shell outlet	19.01	225.00	0.084
Shell inlet	18.24	225.00	0.081

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell-1

Shell Test Pressure		780.00 KPa.
Shell Thickness	ts	14.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2506.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Test Pressure		780.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2500.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4424
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6100.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5850.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		187.00 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	479.99 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1124.300 mm
Largest Center-to-Center Tube Distance	U1	41.2750 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	38519.9 cm ²
Perimeter of Tube Layout	Cp	7243.0 mm



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Tubesheet Data:

Tubesheet TYPE: Stationary, Gasketed both Sides, Conf. d
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-182 F60
Tubesheet Material UNS Number		S32205
Tubesheet Allowable Stress at Temperature	S	187.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	125.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2578.000 mm
Dimension G for the Channel Side	Gc	2553.285 mm
Area of the Untubed Lanes	AL	595.3 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	2900.000 mm
Flange Inside Diameter	B	2510.000 mm
Flange Face Outside Diameter	Fod	2581.000 mm
Flange Face Inside Diameter	Fid	2510.000 mm
Gasket Outside Diameter	Go	2570.000 mm
Gasket Inside Diameter	Gi	2526.000 mm
Small end Hub thk.	g0	12.5000 mm
Large end Hub thk.	g1	19.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	2535.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:

Diameter of Bolt Circle	C	2750.000 mm
Nominal Bolt Diameter	dB	57.1500 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	36
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm
Alternate Flange Operating Bolt Load, Wm1		584935.19 KgF
Alternate Flange Seating Bolt Load, Wm2		560459.62 KgF
Alternate Flange Design Bolt Load, W		1394340.25 KgF

Note: The full bolt load Sa*Ab, was used for all load cases.

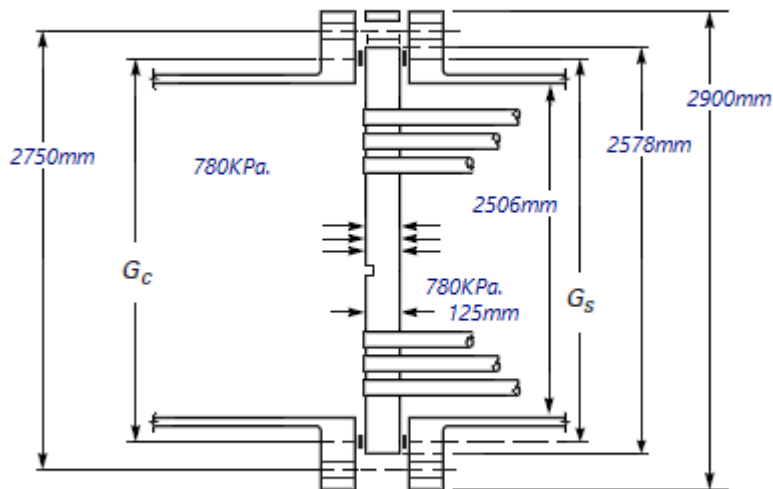


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Tubesheet Integral with Tubesheet Extended as Flange None
Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1
Tubes - TM-1 Material Group H

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19994E+09 KPa.



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Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1
TubeSheet - TM-1 Material Group H

Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (780.0 \cdot (25.4/2 - 0)) / (187.0 \cdot 1.0 + 0.4 \cdot 780.0)$$

$$= 0.0529 + 0.0000 = 0.0529 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-5	at	21.11 °C
Elastic Modulus for Material			195117184.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5850.00	12.04	50.0000	0.0075908	200.17

$$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 200.1697) / (3 \cdot 12.0379) = 22169.7383 \text{ KPa.}$$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4470	25.40	5850.00	56.82	50.0000	0.0003407	33.24

$$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 33.2419) / (3 \cdot 56.8197) = 780.0105 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4470 mm
Allowable Internal Pressure at Corroded thickness	33277.66 KPa.
Required Internal Design Pressure	780.00 KPa.
Allowable External Pressure at Corroded thickness	22169.74 KPa.
Required External Design Pressure	780.00 KPa.
Required Thickness due to Shell Side pressure	0.4470 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	22.000 mm
Basic Gasket Width,	$b0 = N / 2$	11.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	8.355 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2553.291 mm

Results for ASME Stationary Tubesheet Calculations for Configuration d:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio $[\rho]$:



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$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/125.0$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 2.11 (0.19994E+09/.19994E+09) \cdot (186/186) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1124.3 + 25.4$$

$$= 2274.000 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 2274.0/2$$

$$= 1137.0000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$

$$= 31.75 / \sqrt{ 1 - 4 \cdot \min(595.31 \cdot 100.0 ; 4 \cdot 2274.0 \cdot 31.75) / (\pi \cdot 2274.0^2) }$$

$$= 31.9853 \text{ mm}$$

Effective ligament efficiency for bending [μ *]:

$$= (p^* - d^*) / p^*$$

$$= (31.9853 - 25.4) / 31.9853$$

$$= 0.2059$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 1276.6425/1137.0$$

$$= 1.1228$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 1276.6425/1137.0$$

$$= 1.1228$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 4424 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1137.0))^2$$

$$= 0.6162$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 4424 (25.4 / (2 \cdot 1137.0))^2$$

$$= 0.4480$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0



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Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_c , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 3.937008$; $\mu^* = 0.205885$; $E^* = 32454492$ KPa.

$E^*/E = 0.162318$; $\nu^* = 0.450106$

Note: As h/p (3.937) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:

$$= \left(\left(24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.45^2)4424 \cdot 0.19994E+09 \cdot 2.11 \right. \right. \\ \left. \left. (25.4 - 2.11)1137.0^2 \right) / (32454492 \cdot 5850.0 \cdot 125.0^3) \right)^{1/4}$$

$$= 7.3385$$

Values from Table 4.18.3

$Z_d = 0.003738$; $Z_v = 0.018464$; $Z_m = 0.193274$

$Z_a = 0.457704E+03$; $Z_w = 0.018464$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio $[K]$:

$= A / Do$

$= 2578.0/2274.0$

$= 1.1337$

Compute coefficient $[F]$:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.45) / 32454492 \cdot (0 + 0 + 0.19994E+09 \cdot \ln(1.13))$$

$$= 0.4251$$

Compute parameter $[\phi]$:

$= (1 + \nu^*)F$

$= (1 + 0.4501) 0.4251$

$= 0.6164$

Compute parameter $[Q1]$:

$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$

$= (1.1228 - 1 - 0.6164 \cdot 0.0185) / (1 + 0.6164 \cdot 0.1933)$

$= 0.099573$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor $[\gamma_b]$:

$= (G_c - G_s) / Do$ (config d)

$= (2553.2849 - 2553.2849) / 2274.0 = 0$

Calculate parameter $[\omega_s]$:

$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$

$= 1.1228 \cdot 0 \cdot 0 \cdot 0 (1 + 125.0 \cdot 0)$

$= 0.0000 \text{ mm}^2$

Calculate parameter $[\omega^*s]$:

$= ao^2(\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$

$= 1137.0^2(1.123^2 - 1) \cdot (1.123 - 1) / 4 - 0$

$= 10348.7168 \text{ mm}^2$

Parameter $[\omega_c]$:

$= \rho_c \cdot k_c [\delta_C (C_m \Theta + \beta_C + C_q \Delta \cdot h \cdot \beta_C^2) -$

$\zeta_c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta_C / 2)]$

$= 1.123 \cdot 0 [0 (0 + 0 \cdot 0 \cdot 125.0) -$

$0 (0/125.0 + 0 \cdot 0/2)]$

$= 0.0000 \text{ mm}^2$



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Parameter [$\omega \cdot c$]:

$$= a_o^2 [(p_c^2 + 1) \cdot (p_c - 1)/4 - (p_s - 1)/2] - \omega c$$

$$= 1137.2 [(1.12282^2 + 1) \cdot (1.12282 - 1)/4 - (1.12282 - 1)/2] - 0$$

$$= 10348.7236 \text{ mm}^2$$

Pressure [$P \cdot S$]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [$P \cdot C$]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 780.0 - 780.0 = 0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q_2]:

$$= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) +$$

$$W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((10348.717 \cdot 780.0 - 10348.724 \cdot 780.0) -$$

$$(0 \cdot 0 - 0 \cdot 0)) \cdot 0. +$$

$$1394340 \cdot 0 / (2 \cdot \pi)] / (1 + 0.6164 \cdot 0.19327)$$

$$= -0.0004 \text{ Kg}$$

Tubesheet Bending Stress - original thickness [σ]:

$$= 6 \cdot Q_2 / (\mu \cdot (H - h'g)^2)$$

$$= 6 \cdot -0.0004 / (0.206 \cdot (125.0 - 0)^2)$$

$$= -0.0000 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 479.99 = 683.99 \text{ N./mm}^2, 4.18.233$$

Could not converge on a min. required tubesheet thk. to satisfy the Bending Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

Tubesheet thickness = 11.5133 mm
Tubesheet Bending Stress = 0.0011 N./mm²

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$$

$$= (2 \cdot 0.2 \cdot 125.0 / 1137.0) \cdot 364.793$$

$$= 16.0419 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (38519.9 \cdot 0) / (0.2 \cdot 11.513 \cdot 7243.0)$$

$$= 0.0000 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [H_{reqS}]:

$$= H + C_{ats} + C_{atc}$$



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$$= 11.5133 + 0 + 0$$

$$= 11.5133 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(11.5133 ; 11.5133)$$

$$= 11.5133 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = -7.4998, F_{tmax} = 26.7744$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - 2 \cdot Q_2 \cdot F_{tmin} / a_o^2) / (X_t - X_s)$$

$$= ((780.0 \cdot 0.448 - 780.0 \cdot 0.6162) - 2 \cdot -0.001 \cdot -7.5 / 1137.0^2) / (0.6162 - 0.448)$$

$$= -0.7800 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - 2 \cdot Q_2 \cdot F_{tmax} / a_o^2) / (X_t - X_s)$$

$$= ((780.0 \cdot 0.448 - 780.0 \cdot 0.6162) - 2 \cdot -0.001 \cdot 26.774 / 1137.0^2) / (0.6162 - 0.448)$$

$$= -0.7800 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(| \sigma_{t,1} | ; | \sigma_{t,2} |) = 0.78 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta_T \cdot S_{y,t} = 455.9918 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= 1.2500 \text{ (as } P_e \text{ is 0.0)}$$

Determine the factor [rt]:

$$= (\sqrt{ (\frac{d_t^2}{4} + (\frac{d_t}{2} - t_t)^2) }) / 4$$

$$= (\sqrt{ (\frac{25.4^2}{4} + (\frac{25.4}{2} - 2.11)^2) }) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (\frac{2 \cdot \pi^2 \cdot E_t / S_{yt}}{L^2}) }$$

$$= \sqrt{ (\frac{2 \cdot \pi^2 \cdot 0.19994E+09 / 479}{L^2}) }$$

$$= 93.8884$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 0 \cdot 0 / 8.268$$

$$= 0.0000$$

Buckling Allowable Stress [Stb]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (479 / 1.25) \cdot (1 - 0 / (2 \cdot 93.888))$$

$$= 455.992 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 0.78 \cdot 1.5438$$

$$= 12.28 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$



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$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 455.992$$

$$= 7178.441 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 455.99$$

$$= 7024.8394 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 455.992 / 455.992$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 455.99$$

$$= 4545.4844 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1.0 \cdot 1.0)} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm
 Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 7178.44 is $\geq$ tube strength 7178.44 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Computations completed for ASME tubesheet configuration: d

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	0.0 $\leq$	684.0 N./mm ²	Pass
Tubesheet Shear Stress	0.0 $\leq$	364.8 N./mm ²	Pass
Maximum Stress in the Tubes	0.8 $\leq$	456.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-0.8 $\leq$	-456.0 N./mm ²	Pass
Maximum Force on any one Tube	12.3 $\leq$	7178.4 Kgf	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	11.513	125.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025



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Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd			
D1uc	55.499	3.293	146	684	17	365	Ps+Pt-Th	D1	Ok
D2uc	55.499	3.293	146	684	17	365	Ps+Pt-Th	D2	Ok
D3uc	11.513	...	...	684	...	365	Ps+Pt-Th	D3	Ok
Max:	55.4995	3.293 mm		0.214		0.045	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	36	455	-8	-455	573	7178	...	...	...	...	Ok
D2uc	37	455	-37	-455	586	7178	...	...	...	...	Ok
D3uc	1	455	-1	-455	12	7178	...	...	...	...	Ok
Max RATIO	0.082			0.082		0.082	...			...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	55.499	125.000	Ok
Tube thickness :	0.447	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Note: This is a full strength Tube to Tubesheet Joint.

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Input Echo, Tubesheet Number 1, Description: Floating tubesh

Shell Data:

Main Shell Description: Shell-1

Shell Test Pressure		780.00 KPa.
Shell Thickness	ts	14.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2506.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Test Pressure		780.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2500.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4424
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6100.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5850.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		187.00 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	479.99 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1124.300 mm
Largest Center-to-Center Tube Distance	U1	41.2750 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	38519.9 cm ²
Perimeter of Tube Layout	Cp	7243.0 mm



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Tubesheet Data:

Tubesheet TYPE: Floating, Tubesheet Gasketed, Conf. B
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-182 F60
Tubesheet Material UNS Number		S32205
Tubesheet Allowable Stress at Temperature	S	187.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	125.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2482.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		No
Area of the Untubed Lanes	AL	595.3 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	2482.000 mm
Flange Inside Diameter	B	2288.000 mm
Flange Face Outside Diameter	Fod	2363.000 mm
Flange Face Inside Diameter	Fid	2288.000 mm
Gasket Outside Diameter	Go	2360.000 mm
Gasket Inside Diameter	Gi	2320.000 mm
Gasket Factor,	m	4.25
Gasket Design Seating Stress	y	69.64 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	4.0000 mm
Full face Gasket Flange Option		Program Selects

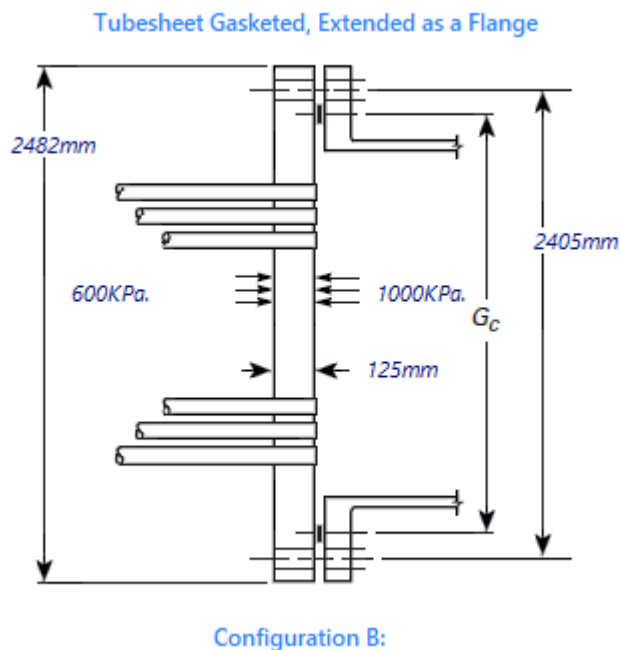
Bolting Information:

Diameter of Bolt Circle	C	2405.000 mm
Nominal Bolt Diameter	dB	34.9250 mm
Thread Series		UNC Thread Series
Number of Bolts	n	44
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	109.0000 mm
Is Bolt Load Transferred to the Tubesheet		Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No
---	--	----



Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1
Tubes - TM-1 Material Group H

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1
TubeSheet - TM-1 Material Group H



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Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

$= (780.0 \cdot (25.4/2 - 0)) / (187.0 \cdot 1.0 + 0.4 \cdot 780.0)$

$= 0.0529 + 0.0000 = 0.0529 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-5	at	21.11 °C
Elastic Modulus for Material			195117184.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5850.00	12.04	50.0000	0.0075908	200.17

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 200.1697) / (3 \cdot 12.0379) = 22169.7383 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4470	25.40	5850.00	56.82	50.0000	0.0003407	33.24

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 33.2419) / (3 \cdot 56.8197) = 780.0105 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4470 mm
Allowable Internal Pressure at Corroded thickness	33277.66 KPa.
Required Internal Design Pressure	780.00 KPa.
Allowable External Pressure at Corroded thickness	22169.74 KPa.
Required External Design Pressure	780.00 KPa.
Required Thickness due to Shell Side pressure	0.4470 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	20.000 mm
Basic Gasket Width,	$b0 = N / 2$	10.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	7.966 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2344.068 mm

Results for ASME Floating Tubesheet Calculations for Configuration B:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [ρ]:

$= ltx / h$ (modified for corrosion if present)

$= 0/125.0$

$= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d^*]:

$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$



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$$= \max(25.4 - 2 \cdot 2.11 (0.19994E+09 / 0.19994E+09) \cdot (186/186) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1124.3 + 25.4$$

$$= 2274.000 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 2274.0 / 2$$

$$= 1137.0000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)}$$

$$= 31.75 / \sqrt{1 - 4 \cdot \min(595.31 \cdot 100.0; 4 \cdot 2274.0 \cdot 31.75) / (\pi \cdot 2274.0^2)}$$

$$= 31.9853 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.9853 - 25.4) / 31.9853$$

$$= 0.2059$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 1172.0342 / 1137.0$$

$$= 1.0308$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 1172.0342 / 1137.0$$

$$= 1.0308$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 4424 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1137.0))^2$$

$$= 0.6162$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 4424 (25.4 / (2 \cdot 1137.0))^2$$

$$= 0.4480$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 3.937008$; $\mu^* = 0.205885$; $E^* = 32454492 \text{ KPa}$.



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$$E^*/E = 0.162318 ; v^* = 0.450106$$

Note: As h/p (3.937) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$\begin{aligned} &= \left(\left(24(1 - v^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4} \\ &= \left(\left(24(1 - 0.45^2)4424 \cdot 0.19994E+09 \cdot 2.11 \right. \right. \\ &\quad \left. \left. (25.4 - 2.11)1137.0^2 \right) / (32454492 \cdot 5850.0 \cdot 125.0^3) \right)^{1/4} \\ &= 7.3385 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned} Z_d &= 0.003738 ; Z_v = 0.018464 ; Z_m = 0.193274 \\ Z_a &= 0.457704E+03 ; Z_w = 0.018464 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / Do \\ &= 2482.0/2274.0 \\ &= 1.0915 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.45) / 32454492 \cdot (0 + 0 + \\ &\quad 0.19994E+09 \cdot \ln(1.09)) \\ &= 0.2965 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + v^*)F \\ &= (1 + 0.4501) 0.2965 \\ &= 0.4300 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.0308 - 1 - 0.43 \cdot 0.0185) / (1 + 0.43 \cdot 0.1933) \\ &= 0.021119 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [γ_b]:

$$\begin{aligned} &= (G_c - C) / Do \text{ (config b)} \\ &= (2344.0684 - 2405.0) / 2274.0 = -0.02679 \end{aligned}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= ps \cdot ks \cdot \beta_s \cdot \delta S(1 + h \cdot \beta_s) \\ &= 1.0308 \cdot 0 \cdot 0 \cdot 0(1 + 125.0 \cdot 0) \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ω^*s]:

$$\begin{aligned} &= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega_s \\ &= 1137.0^2(1.031^2 - 1) \cdot (1.031 - 1) / 4 - 0 \\ &= 623.1480 \text{ mm}^2 \end{aligned}$$

Parameter [ω_c]:

$$\begin{aligned} &= pc \cdot kc[\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\ &\quad \zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)) \\ &= 1.031 \cdot 0[0(0 + 0 \cdot 0 \cdot 125.0) - \\ &\quad 0(0/125.0 + 0 \cdot 0/2)] \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$

Parameter [ω^*c]:

$$\begin{aligned} &= ao^2[(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega_c \\ &= 1137.0^2[(1.03081^2 + 1) \cdot (1.03081 - 1) / 4 - (1.03081 - 1) / 2] - 0 \\ &= 623.1490 \text{ mm}^2 \end{aligned}$$



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Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 780.0 - 780.0 = 0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s * P_s - \omega * c * P_t) - (\omega_s * P * s - \omega_c * P * c)) + W * \gamma_b / (2 * \pi)] / (1 + \phi * Z_m)$$

$$= [((623.148 * 780.0 - 623.149 * 780.0) -$$

$$(0 * 0 - 0 * 0)) * 0. +$$

$$524750 * -0.027 / (2 * \pi)] / (1 + 0.42997 * 0.19327)$$

$$= -2066.1240 \text{ Kgf}$$

Tube sheet Bending Stress - original thickness [σ]:

$$= 6 * Q2 / (\mu * (H - h'g)^2)$$

$$= 6 * -2066.124 / (0.206 * (125.0 - 0)^2)$$

$$= -37.7913 \text{ N./mm}^2$$

Allowable Tube sheet Bending Stress [σ allowed]:

$$= 1.5 * \beta T * S_y = 1.5 * 0.95 * 479.99 = 683.99 \text{ N./mm}^2, 4.18.233$$

Tube sheet Bending Stress - final thickness [σ_f]:

$$= 6 * Q2 / (\mu * (H - h'g)^2)$$

$$= 6 * -2180.3623 / (0.206 * (30.183 - 0)^2)$$

$$= 683.9876 \text{ N./mm}^2$$

Reqd Tube sheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 30.1833 + 0 + 0$$

$$= 30.1833 \text{ mm}$$

$$H_{reqd.} = H_{ReqB} = 30.1833 \text{ mm}$$

Computations completed for ASME tube sheet configuration: B

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tube sheet bend. stress	37.8 <=	684.0 N./mm ²	Pass

Floating Tube sheet results per ASME Div. 2 Part 4.18.9 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tube sheet Bend	Stresses Allwd	Case Type	Pass/ Fail
D1uc	43.310	28.740	106	684	Ps+Pt-Th	D1 Ok
D2uc	31.512	...	76	684	Ps+Pt-Th	D2 Ok
D3uc	30.183	...	38	684	Ps+Pt-Th	D3 Ok
Max:	43.3096	28.740 mm	0.154	...	(Str. Ratio)	

Load Case Definitions:



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ASME Fl-TS Calc: Pressure Test

Case: 1 10:12a Sep 9, 2026

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(-)Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	43.310	125.000	Ok
Tubesheet thickness flanged extension :	28.740	109.000	Ok
Tube thickness :	0.447	2.110	Ok
Tube-tubesheet fillet weld leg :	0.000	2.110	Ok
Tube-tubesheet groove weld leg :	0.000	2.110	Ok

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Floating Head: Operating

Case: 1 10:13a Sep 9,2026

Input Echo, Floating Head Item 1, Description: Operating

Floating Head Type		D	
Tube Side (Internal) Design Pressure	Pts	1000.00	KPa.
Shell Side (External) Design Pressure	Pss	600.00	KPa.
Design Temperature for Spherical Head	Temp	230.00	°C
Head Material		SA-240	
Head Material UNS Number		S32205	
Head Allowable Stress at Temperature	Sho	187.00	N./mm ²
Head Allowable Stress at Ambient	Sha	187.00	N./mm ²
Crown Radius for Spherical Head	L	2113.200	mm
Head Thickness	Th	25.000	mm
Tube Side (Internal) Corrosion Allowance	Cats	0.0000	mm
Shell Side (External) Corrosion Allowance	Cass	0.0000	mm
Flange Material		SA-182 F60	
Flange Material UNS Number		S32205	
Flange Allowable Stress at Temperature	Sfo	187.00	N./mm ²
Flange Allowable Stress at Ambient	Sfa	187.00	N./mm ²
Flange Outside Diameter	Fod	2482.000	mm
Flange Inside Diameter	Fid	2288.000	mm
Flange Thickness	Tf	195.0000	mm
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2405.000	mm
Nominal Bolt Diameter	dB	34.9250	mm
Type of Threads	UNC Thread Series		
Number of Bolts		44	
Full face Gasket Flange Option	Program Selects		
Flange Face Outside Diameter	Ffod	2363.000	mm
Flange Face Inside Diameter	Ffid	2288.000	mm
Gasket Outside Diameter	Go	2360.000	mm
Gasket Inside Diameter	Gi	2320.000	mm
Gasket Factor,	m	4.2500	
Gasket Design Seating Stress	y	69.64	N./mm ²
Flange Facing Sketch	Code Sketch 1a		
Column for Gasket Seating	Code Column I		
Gasket Thickness		4.0000	mm
Width of raised face		0.0000	mm
Dist. from Head Centerline to Flange Centroid hr		-14.9751	mm
The Flange is not Slotted.			
This unit does not have a Backing Ring.			
Use Full Bolt Load in Flange Design		Yes	

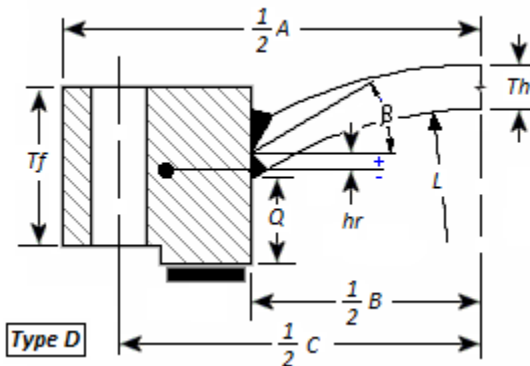


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Floating Head: Operating

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Internal Pressure results for Spherical Heads:

ASME Code, Section VIII, Division 1, 2025, App. 1-6

Thickness Due to Internal Pressure[thr]:
 $= 5PL / 6Sho$
 $= (5 \cdot 1000.0 \cdot 2113.2) / (6 \cdot 186)$
 $= 9.4178 \text{ mm}$

Maximum Allowable Working Pressure at Given Thickness[Pa]:
 $= 6Sho \cdot (Th - Cass - Cats) / 5L$
 $= (6 \cdot 186 \cdot 25.0) / (5 \cdot 2113.2)$
 $= 2654.54 \text{ KPa.}$

Maximum Allowable Pressure, New and Cold[Pnc]:
 $= 6Sha / 5L$
 $= (6 \cdot 186 \cdot 25.0) / (5 \cdot 2113.2)$
 $= 2654.54 \text{ KPa.}$

Actual Stress at given Pressure and Thickness[Sact]:
 $= 5PL / 6(Th - Cass - Cats)$
 $= (5 \cdot 1000.0 \cdot 2113.2) / (6 \cdot 25.0)$
 $= 70. \text{ N./mm}^2$

Spherical head Percent Elongation $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$: 0.882 %

External Pressure results, Spherical Head

ASME Code, Section VIII Division 1, 2025

External Pressure Chart HA-5 at 230.00 °C
 Elastic Modulus for Material 180805456.00 KPa.

Results for Maximum Allowable External Pressure:

Corroded Thickness of Shell	TCA	25.0000	mm
Outside Crown Radius of Shell	OD/2	2138.2000	mm
Crown Radius / Thickness Ratio	$(OD/2)/T$	85.5280	
Geometry Factor, A	f(D/T, LD)	0.0014615	
Materials Factor, B	f(A, Chart)	108.2847	N./mm ²
Maximum Allowable Working Pressure		1266.00	KPa.
$EMAWP = B / ((OD/T)/2) = 108.2847 / (171.056/2.0) = 1266.0$			

Results for Req'd Thickness for Ext. Pressure (Tca):

Corroded Thickness of Head	TCA	15.5809	mm
Outside Diameter of Shell	ODCA	4276.400	mm
Diameter / Thickness Ratio	(D/T)	274.4641	
Geometry Factor, A	f(DT, LD)	0.0009109	
Materials Factor, B	f(A, Chart)	82.3495	N./mm ²
Maximum Allowable Working Pressure		600.04	KPa.
$EMAWP = B / ((D/T)/2) = 82.3495 / (274.4641/2) = 600.0402$			



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Floating Head: Operating

Case: 1 10:13a Sep 9, 2026

Intermediate Calculations for Flanged portion:

ASME Code, Section VIII, Div. 1, 2025 App. 2

Gasket Contact Width,	$N = (Goc - Gic) / 2$	20.000 mm
Basic Gasket Width,	$b0 = N / 2$	10.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	7.966 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	2344.068 mm

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$= 2a + 6t / (m + 0.5)$
 $= 2 * 34.925 + 6 * 195.0 / (4.25 + 0.5)$
 $= 316.166 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:

$= C * \sin(\pi / n)$
 $= 2405.0 * \sin(\pi / 44)$
 $= 171.571 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$= \max(\sqrt{Bs / (2a + t)}, 1)$
 $= \max(\sqrt{171.571 / (2 * 34.925 + 195.0)}, 1)$
 $= 1.0000$

Bolting Information for UNC Thread Series (Non Mandatory):

Total Area of Bolts 299.199 cm²

	Minimum	Actual	Maximum
Radial Distance between Bolts and Edge:	34.925	38.500	
Circ. Spacing between the Bolts:	77.800	171.571	316.166

Results for the Internal Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	440075.3	Kgf
Contact Load on Gasket Surfaces	Hp	50846.7	Kgf
Hydrostatic End Load at Flange ID	Hd	419274.6	Kgf
Pressure Force on Flange Face	Ht	20800.7	Kgf
Radial Component of Head Membrane Force	Hr	654573.6	Kgf
Operating Bolt Load:	Wm1	490922.0	Kgf
Gasket Seating Bolt Load	Wm2	416557.2	Kgf
Required Bolt Area	Am	279.911	cm ²

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:

$= Ab * Sb / (y * PI * (Go + Gi))$
 $= 299.199 * 172.0 / (69.64 * 3.14 * (2360.0 + 2320.0))$
 $= 5.026 \text{ mm}$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

= 38.100 mm [Note: Exceeds actual gasket width, 20.000]

Flange Design Bolt Load (Seating)	W	524750.2	Kgf
Gasket Seating Force (Operating)	Hg	50846.7	Kgf
Distance to Gasket Load Reaction	hg	30.4659	mm
Distance to Face Pressure Reaction	ht	44.4829	mm
Distance to End Pressure Reaction	hd	58.5000	mm

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	419275.	58.5000	1.0000
				24528. Kg-m.



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Floating Head: Operating

Case: 1 10:13a Sep 9, 2026

Face Pressure,	Mt	20801.	44.4829	1.0000	925. Kg-m.
Gasket Load,	Mg	50847.	30.4659	1.0000	1549. Kg-m.
Floating Hd. Load,	Mh	654574.	-14.9751	1.0000	9802. Kg-m.
Gasket Seating,	Ma	524750.	30.4659	1.0000	15987. Kg-m.

Total Moment for Operation (Internal Pressure) 36804.965 Kg-m.

Total Moment for Gasket Seating (Int. Pressure) 15987.292 Kg-m.

Results for the External Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	264045.2	Kgf
Hydrostatic End Load at Flange ID	Hd	251564.7	Kgf
Pressure Force on Flange Face	Ht	12480.4	Kgf
Radial Component of Head Membrane Force	Hr	392744.2	Kgf
Gasket Seating Bolt Load	Wm2	416557.2	Kgf
Required Bolt Area	Am	237.510	cm ²

Flange Design Bolt Load (Seating)	W	524750.2	Kgf
Distance to Gasket Load Reaction	hg	30.4659	mm
Distance to Face Pressure Reaction	ht	44.4829	mm
Distance to End Pressure Reaction	hd	58.5000	mm

Summary of Moments for External Pressure:

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	251565.	28.0341	1.0000	7053. Kg-m.
Face Pressure,	Mt	12480.	14.0171	1.0000	175. Kg-m.
Floating Hd. Load,	Mh	392744.	-14.9751	1.0000	5882. Kg-m.
Gasket Seating,	Ma	524750.	30.4659	1.0000	15987. Kg-m.

Total Moment for Operation (External Pressure) 13108.983 Kg-m.

Total Moment for Gasket Seating (Ext. Pressure) 15987.292 Kg-m.

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 1000.0 \cdot 2288.0 \cdot \sqrt{(4 \cdot 2113.2 \cdot 2113.2 - 2288.0 \cdot 2288.0) / (8 \cdot 186.997(2482.0 - 2288.0))}$$

$$= 28.0166 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 36804(2482.0 + 2288.0) / (186.997 \cdot 2288.0)(2482.0 - 2288.0)$$

$$= 20742.1348 \text{ mm}^2$$

Required thickness for Main Flange, Internal Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 28.017 + \sqrt{(28.017 \cdot 28.017 + 20742.135)}$$

$$= 174.7376 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 15987(2482.0 + 2288.0) / (186.997 \cdot 2288.0)(2482.0 - 2288.0)$$

$$= 9009.9404 \text{ mm}^2$$

Required thickness for Main Flange, Internal Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 0 + \sqrt{(0 \cdot 0 + 9009.94)}$$

$$= 94.9207 \text{ mm}$$

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 600.0 \cdot 2288.0 \cdot \sqrt{(4 \cdot 2113.2 \cdot 2113.2 - 2288.0 \cdot 2288.0) / (8 \cdot 186.997(2482.0 - 2288.0))}$$



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Floating Head: Operating

Case: 1 10:13a Sep 9, 2026

= 16.8099 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $13108(2482.0+2288.0)/(186.997 \cdot 2288.0)(2482.0-2288.0)$

= 7387.8154 mm²

Required thickness for Main Flange, External Operating Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $16.81 + \sqrt{(16.81 \cdot 16.81 + 7387.815)}$

= 104.3907 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $15987(2482.0+2288.0)/(186.997 \cdot 2288.0)(2482.0-2288.0)$

= 9009.9404 mm²

Required thickness for Main Flange, External Bolt-up Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $0 + \sqrt{(0 \cdot 0 + 9009.94)}$

= 94.9207 mm

Summary of Required Thicknesses(mm):

	Head	Flange
Tubeside (Internal) Pressure	9.4178	174.7376
Shellside (External) Pressure	15.5809	104.3907
Tubeside Gasket Seating Load		94.9207
Shellside Gasket Seating Load		94.9207
Maximum + Corrosion Allowance	15.5809	174.7376
Actual Thickness as Given	25.0000	195.0000

Floating Head MAWP and MAPnc (KPa.):

Stress Condition	Tubeside		Shellside	
	MAWP	MAPnc	MAWP	MAPnc
Flange Stress	1199.50	1199.75	1634.12	1634.70
Floating head Stress	2654.44	2654.44	1265.90	1534.35
Bolt Stress	1068.81	1068.81	...	...
Minimum MAWP & MAPnc	1068.81	1068.81	1265.90	1534.35

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	182	No Calc	No Calc
20	30	No Calc	...	247	No Calc	No Calc
30	40	1500	2532	13	0.00085547	80.3559
40	50	No Calc	...	250	No Calc	No Calc
50	60	No Calc	...	257	No Calc	No Calc
60	Ring	998.828	2531	11.7	0.0011322	89.0266
Ring	70	3301.17	2531	11.7	0.00031895	31.8876
70	80	3301.17	2555	23.7	0.00091999	82.6946
80	Ring	3301.17	2531	11.7	0.00031895	31.8876
Ring	90	1000	2531	11.7	0.0011308	88.9887
90	100	No Calc	...	262	No Calc	No Calc
100	110	No Calc	...	262	No Calc	No Calc
110	120	958.833	2534	11	0.001077	87.5476
120	130	No Calc	2533.2	10.6	0.00058117	58.104

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	185	147.193	101.13	No Calc
20	30	247	242.189	101.13	No Calc
30	40	16	9.48296	101.13	550.061
40	50	250	244.958	1.013	No Calc
50	60	260	258.369	...	No Calc
60	Ring	12.5	No Calc	...	548.69
Ring	70	12.5	No Calc	...	196.53
70	80	24.5	No Calc	...	1022.7
80	Ring	12.5	No Calc	...	196.53
Ring	90	12.5	No Calc	...	548.457
90	100	265	258.369	...	No Calc
100	110	265	258.369	...	No Calc
110	120	14	No Calc	...	506.692
120	130	13.6	No Calc	...	270.131

Minimum

197

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	1500	7980.75	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	Ring	998.828	No Calc	No Calc	No Calc
Ring	70	3301.17	No Calc	No Calc	No Calc
70	80	3301.17	No Calc	No Calc	No Calc
80	Ring	3301.17	No Calc	No Calc	No Calc



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External Pressure Calculations: Step: 18 10:13am Sep 9,2026

Ring	90	1000	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	958.833	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 19 10:13am Sep 9,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	9880.37	...	9880.37	...	...
20	30	3410.27	1487614	3410.27	1487614	...
30	40	1470.28	7364428	1196.03	7399819	...
40	50	3449.71	1502343	3449.71	1502343	...
50	60	3497.25	1614765	3437.6	1616115	...
60	70	1202.64	3767591	1134.27	3776414	...
70	80	4007.34	6727841	3885.25	6743596	...
80	90	1202.64	3767591	1134.27	3776414	...
90	100	3561.47	1639628	3500.9	1640978	...
100	110	3561.47	1639628	3500.9	1640978	...
110	120	601.319	3453248	473.027	3469804	...
120	130	993.145	2307099	806.931	2323116	...
Total		36837	35271780.00	35809	35377196.00	0



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	...	92.5	...	...	Liquid: 10
10	Insl	265.275	100	...	...	Insulation
20	Liqd	1482.72	147.5	17	...	Liquid: 20
20	Insl	70.7089	151.5	...	...	Insulation
30	Liqd	7359.93	750	...	...	Liquid: 30
30	Insl	437.097	750	...	...	Ins: 40
30	Nozl	232.679	750	1422.24	...	Tube inlet
30	Nozl	232.679	750	-1422.24	...	Tube outlet
30	Wght	1105	750	...	...	Partition plate
40	Liqd	1501.43	149	-3.00011	...	Liquid: 40
50	Liqd	1613.61	158.5	1.99988	...	Liquid: 50
50	Lini	116.058	153.5	...	...	cladd
60	Sadl	1202.64	1000	1538.25	...	Left Saddle
60	Liqd	3765.29	700	...	...	Liquid: 60
60	Insl	407.802	700	...	...	Insulation
60	Lini	259.553	700	...	...	Cladding
60	Ring	700.822	998.828	...	...	SR, 1000.0 Fr60
60	Wght	520	1000	-1800	...	Fireproofing we
70	Liqd	6723.74	1250	...	...	Liquid: 40
70	Insl	734.861	1250	...	...	Ins: 40
70	Lini	463.487	1250	...	...	SS clad
70	Nozl	901.857	1320.53	-1786.4	...	Shell outlet
70	Nozl	856.652	1283.26	1786.4	...	Shell inlet
80	Sadl	1202.64	400	1538.25	...	Right saddle
80	Liqd	3765.29	700	...	...	Liquid: 80
80	Insl	407.802	700	...	...	Insulation
80	Lini	259.553	700	...	...	Cladding
80	Ring	700.823	400	...	...	SR, 4000.0 Fr60
80	Wght	520	400	-1800	...	Fire proofing
90	Liqd	1638.45	161	1.99988	...	Liquid: 70
90	Lini	118.704	157	...	...	cladd
100	Liqd	1638.45	161	1.99988	...	Liquid: 80



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Element and Detail Weights:

Step: 19 10:13am Sep 9, 2026

100 Lini	118.704	157	...	...	cladd
110 Liqd	3451.14	350	...	...	Liquid: 70
110 Insl	204.134	350	...	...	Ins: 70
110 Lini	129.776	350	...	...	Ss clad
120 Liqd	2305.69	258.833	-0.00014534	...	Liquid: 80
120 Insl	347.014	310	...	...	Ins: 80
120 Lini	186.062	338.25	...	...	SS clad
40 FTsh	2905.88	366.85	...	...	Tubesheet
40 Tube	32521.3	3354.35	...	...	Tubes
40 FLTs	2533.62	6404.35	...	...	
40 FlHd	1969.21	6501.85	...	...	

Total Weight of Each Detail Type:

Saddles	2405.3
Liquid	35245.7
Insulation	2874.7
Lining	1651.9
Stiffeners	1401.6
Nozzles	2223.9
Weights	2145.0
Exchanger Components	39930.0
Liquid in Tubes	9503.8

Sum of the Detail Weights 97381.9 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	36837.9	36837.9	36837.9	36837.9	36837.9	36837.9
Saddles	2405.3	2405.3	2405.3	2405.3	2405.3	2405.3
Stif. Rings	1401.6	1401.6	1401.6	1401.6	1401.6	1401.6
Nozzles	2223.9	2223.9	2223.9	2223.9	2223.9	2223.9
Wld Weights	2145.0	2145.0	2145.0	2145.0	2145.0	2145.0
Exchanger	39930.0	39930.0	39930.0	39930.0	39930.0	39930.0
Insulation	...	...	...	...	2874.7	2874.7 *
Lining	1651.9	1651.9	1651.9	1651.9	1651.9	1651.9
Ope. Liquid	...	...	...	...	...	35245.7
Tube Ope Lqd	...	...	...	...	...	9503.8
Test Liquid	...	35041.0	...	...	...	...
Tube Tst Lqd	...	9503.8	...	...	...	...
Totals	86595.6	131140.4	86595.6	86595.6	89470.3	134219.9

Field Installation Options:

* Insulation installed after lifting.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	86595.6 kg.
Shop Test Wt.	- Fabricated weight + water (full)	131140.4 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	86595.6 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	86595.6 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	89470.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	134219.9 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	133191.5 kg.
Field Test Wt.	- Empty weight + test medium (full)	133092.1 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	9509651.0 cm ³
Weight of Ope Liq in tubes :	9503.8 kg.
Weight of Water in tubes :	9503.8 kg.



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Element and Detail Weights:

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Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	83635.5	...
20	30	44036	23797.6
30	40	119318	117810
40	50	44309.3	24033.2
50	60	45550.3	25627.5
60	70	111451	110220
70	80	200905	196821
80	90	111451	110220
90	100	46005.8	26021.8
100	110	46005.8	26021.8
110	120	55725.6	55109.8
120	130	73811	72011.1

Total 982204.1 787692.6 cm²

Total 1057.2 847.9 ft²

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Nozzle Flange MAWP:

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Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Tube inlet	1820.4	1960.0	82	150	GR 1.1	...	...	...	...
Tube outlet	1820.4	1960.0	82	150	GR 1.1	...	...	...	...
Shell outlet	1278.0	1960.0	230	150	GR 1.1	...	...	...	...
Shell inlet	1278.0	1960.0	230	150	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1278.040 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1960.006 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1820.434 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1960.006 KPa.

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation:

Step: 21 10:13am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	9000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.222 \cdot 0.939) / (1 + 1.7 \cdot 3.4 \cdot 0.222)))$$

$$= \min(0.85, 0.893)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.548
Structure Height to Diameter ratio	3.862

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Z_g = 274320.0$ mm

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 1850.0 + 9000.0 = 10850.0 \text{ mm}$$

$$= 35.597 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (35.597 \text{ ft.}) > 15 \text{ ft.}$$

$$= 2.01 (z / Z_g)^{2/\alpha}$$

$$= 2.01 (35.597/900.0)^{2/9.5}$$

$$= 1.018$$

Type of Hill: No Hill



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Wind Load Calculation:

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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0.0 • 0.0 • 0.0)²

= 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 1.018 • 1.0 • 1.0 • 1.0 • 60.399²)

= 16.0 psf [78.12] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16.0 • 0.85 • 0.548 • 7.408

= 55.2 lbs. [25.0] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Ch. cover	10850.0	0.000	0.000	0.000	1.018	1.000	78.120
Ch.flange-1	10850.0	0.000	0.000	0.000	1.018	1.000	78.120
Ch.shell	10850.0	0.000	0.000	0.000	1.018	1.000	78.120
ch, flange 2	10850.0	0.000	0.000	0.000	1.018	1.000	78.120
Shell flng-1	10850.0	0.000	0.000	0.000	1.018	1.000	78.120
Shell-1	10850.0	0.000	0.000	0.000	1.018	1.000	78.120
Shell-2	10850.0	0.000	0.000	0.000	1.018	1.000	78.120
Shell-3	10850.0	0.000	0.000	0.000	1.018	1.000	78.120
Shell flng. 2	10850.0	0.000	0.000	0.000	1.018	1.000	78.120
Shell girth fln	10850.0	0.000	0.000	0.000	1.018	1.000	78.120
Shell -2	10850.0	0.000	0.000	0.000	1.018	1.000	78.120
head- front	10850.0	0.000	0.000	0.000	1.018	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Partition pla	0.00	9000.00	78.12	0.00
Fireproofing	0.00	9000.00	78.12	0.00
Fire proofing	0.00	9000.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	10850	3720	6882	78.12	27.5315
20	30	10850	3240	9558	78.12	38.2369
30	40	10850	3278.4	49176	78.12	196.729
40	50	10850	3000	8940	78.12	35.7646
50	60	10850	3012	9548.04	78.12	38.1971
60	70	10850	3277.2	45880.8	78.12	183.547
70	80	10850	3306	82650	78.12	330.643
80	90	10850	3277.2	45880.8	78.12	183.547
90	100	10850	3012	9698.64	78.12	38.7996
100	110	10850	3012	9698.64	78.12	38.7996
110	120	10850	3280.8	22965.6	78.12	91.8742
120	130	10850	3279.84	17527.3	78.12	70.1181



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Wind Load Calculation:

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The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(10.82680.75)$
 = 0.119 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.119$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.157 \cdot 131815 \cdot 1.25$
 = 7750.698 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 131815$
 = 4133.7 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1450	9415.33	633.655



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Earthquake Load Calculation:

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20	30	1250	9415.33	546.255
30	40	1250	9415.33	546.255
40	50	1250	9415.33	546.255
50	60	1255	9415.33	548.44
60	70	1253	9415.33	547.566
60	70	1253	9415.33	547.566
70	80	1253	9415.33	547.566
80	0	1253	9415.33	547.566
80	90	1253	9415.33	547.566
90	100	1255	9415.33	548.44
100	110	1255	9415.33	548.44
110	120	1253	9415.33	547.566
120	130	1253	9415.33	547.566

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 23 10:13am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5382.700 mm
Center of Gravity of the Liquid	4951.004 mm
Center of Gravity of the Insulation	4920.328 mm
Center of Gravity of the Lining	6387.958 mm
Center of Gravity of the Stiffening Rings	5382.114 mm
Center of Gravity of the Nozzles	4556.465 mm
Center of Gravity of the Added Weights (Operating)	3246.702 mm
Center of Gravity of the Added Weights (Empty)	3246.702 mm
Center of Gravity of the Tubesheet(s)	5165.358 mm
Center of Gravity of the Tubes	5340.700 mm
Center of Gravity of the Floating Head	8494.550 mm
Center of Gravity of Bare Shell New and Cold	3516.047 mm
Center of Gravity of Bare Shell Corroded	3453.033 mm
Vessel CG in the Operating Condition	4727.824 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4575.000 mm
Vessel CG in the Test Condition	4732.995 mm

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Saddle Calcs: Bundle Pull New:

Step: 24 10:13am

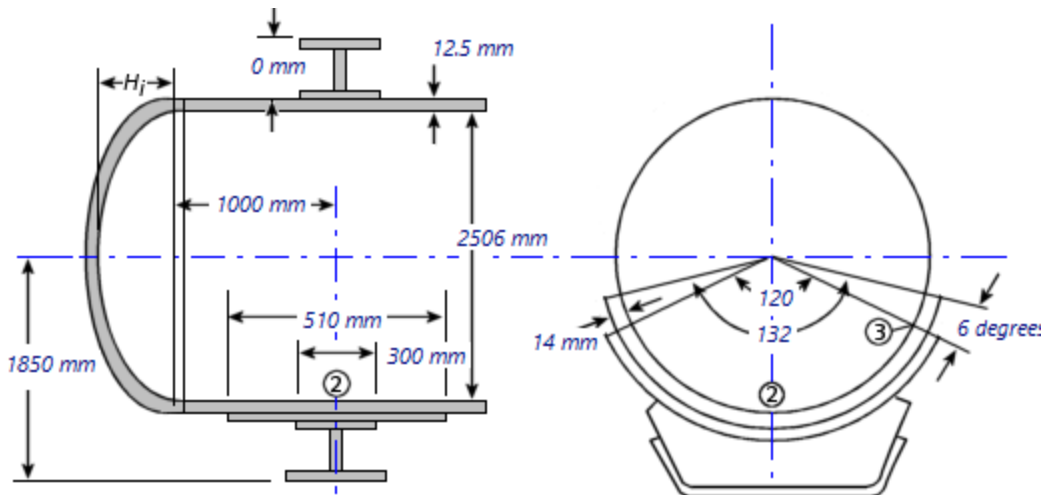
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	1259.25	mm
Shell Thickness used in this Case	t	12.500	mm
Stiffened Vessel Length per 4.15.6	L	6261.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		42764.10	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force	63290.07	Kgf
Pressure used in Analysis	P	0.000 KPa.

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	-6.90	-234.00
Long. Stress at Top	of Midspan	6.90	138.00
Long. Stress at Bottom	of Midspan	6.90	138.00
Long. Stress at Top	of Saddles	0.33	138.00



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Saddle Calcs: Bundle Pull New: Step: 24 10:13am Sep 9, 2026

Long. Stress at Bottom of Saddles	-0.33	-234.00
Long. Stress at Bottom of Saddles	0.33	138.00
Tangential Shear in Shell	8.54	110.40
Circ. Compressive Stress in Shell	3.56	138.00
Stiffener Circ. Stress at Shell	30.48	138.00
Stiffener Circ. Stress at Tip	82.15	234.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

= Bundle Pull Out Force • Impact Factor • B / Ls

= 42764 • 1.5 • 1850.0/3300.0007

= 35960.7 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 27329 + max(35961, 0, 0, 0)

= 63290.1 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 64492.70 Kgf

Transverse Shear Load 0.00 Kgf

Longitudinal Shear Load 64146.16 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(2506.0/2 + 12.5 + 14.0)sin(120.0/2), 2•1259.25)

= 2216.159 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0365 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0300

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -63290•1000.0[1-(1-1000.0/6261.0+(1259.25²-0.0²)/

(2•1000.0•6261.0))/(1+(4•0.0)/(3•6261.0))]

= -2094.0 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 63290•6261/4((1+2(1259²-0²)/(6261²))/(1+(4•0)/

(3•6261))-4•1000/6261)

= 43790.2 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0.0 • 1259.25/(2•12.5) - 43790.2/(π•1259.2²•12.5)

= -6.90 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0.0 • 1259.25/(2 • 12.5) + 43790.2/(π • 1259.2² • 12.5)

= 6.90 N./mm²



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Saddle Calcs: Bundle Pull New: Step: 24 10:13am Sep 9,2026

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1259.25 / (2 \cdot 12.5) - 2094.0 / (\pi \cdot 1259.2^2 \cdot 12.5)$$

$$= 0.33 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1259.25 / (2 \cdot 12.5) + 2094.0 / (\pi \cdot 1259.2^2 \cdot 12.5)$$

$$= -0.33 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 63290(6261.0 - 2 \cdot 1000.0) / (6261.0 + (4 \cdot 0.0 / 3))$$

$$= 43072.8 \text{ Kgf}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 43072.83 / (\pi \cdot 1259.25 \cdot 12.5)$$

$$= 8.54 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1259.25 \cdot 12.5)}$$

$$= 97.860 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.76 \cdot 63290 \cdot 0.1 / 132.66$$

$$= -3.56 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/I$$

$$= -0.3405 \cdot 63290 / 132.66 + 0.0529 \cdot 63290 \cdot 1259.25 \cdot 111.46 / 9921.12$$

$$= 30.48 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/I$$

$$= -0.3405 \cdot 63290 / 132.66 - 0.0529 \cdot 63290 \cdot 1259.25 \cdot 159.04 / 9921.12$$

$$= -82.15 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	600.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1253.0 \cdot 12.5)} + 510.0)12.5^3) / 12$$

$$= 11.478 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 11.478 + 88.154(350.616 - 6.25)^2$$

$$= 104551.812 \text{ cm}^4$$



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Saddle Calcs: Bundle Pull New:

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	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	705.2	12.5	6.2	88.2	55096.4	0.105E+06
Wearplate	510.0	14.0	19.5	71.4	139230.0	0.783E+05
Web	20.0	525.5	289.3	105.1	3040019.0	0.281E+05
BasePlate	600.0	45.0	574.5	270.0	15511501.0	0.136E+06
Totals	...	...	...	534.7	18745846.0	0.347E+06

Distance to Centroid [C1]:

= AY / A

= 18745846/534.654

= 350.616 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 120.0/2

= 120.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(120.0) - 0.5•sin(120.0)²)/(π - 2.094 + sin(120.0)cos(120.0))

= 0.2035

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.204 • 63290.07

= 12880.9023 Kgf

Tension Stress, St = (Fh/As) = 2.8291 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 1850.0 - 1253.0 • sin(120.0/2)/1.0472

= 813.777 mm

Bending Moment, M = Fh * d = 10482.3926 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 10.3934 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(63290 + 1203)600.0/(4 • 2350.0 • 173.33))

= 26.434 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 120.0/2)(1253.0 + 12.5 + 14.0)

= 2216.159 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 2216.1589/(5 - 1)

= 554.040 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 554.0397 • 600.0/2

= 1662.119 cm²



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Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 64492.703 / (2350.0 \cdot 600.0)$$

$$= 4.574 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1662.1 \cdot 4.57$$

$$= 7602.452 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 56.0 + 55.404$$

$$= 111.404 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 7602.5 / 111.404$$

$$= 6.692 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	300.0	...	60.0	...	0.450E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 20.0$$

$$= 280.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= A_Y / A$$

$$= 0.0 / 111.404$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0 / 2$$

$$= 150.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{ (4500.0 / 111.404)}$$

$$= 63.556 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1850.0 - \cos(120.0 / 2) (1253.0 + 12.5 + 14.0) - 45.0$$

$$= 1165.250 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= K L / r$$

$$= 1 \cdot 1165.25 / 63.556$$

$$= 18.334$$

Bending Moment [Rm]:

$$= F l / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$



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$$= 64146.2 / (2 \cdot 2350.0) \cdot 554.04 \cdot 1165.25 / 2$$

$$= 4405.664 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (18.3343 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.33)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (18.33) / (8 \cdot 123.21) - (18.33^3) / (8 \cdot 123.21^3))$$

$$= 149.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 6.69 / 149.31 + (4405.66 \cdot 150.0 / 44999992) / 173.33$$

$$= 0.876$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	280.0	0.0	56.0	0.0	0.450E+04
Web	554.0	20.0	0.0	110.8	0.0	36.9
Totals	...	...	...	166.8	...	0.454E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 166.808$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0 / 2$$

$$= 150.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1850.0 - \cos(1279.5/2) (45.0 + 0.0 + 0.0) - 0.0$$

$$= 525.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{4535.6 / 166.808}$$

$$= 52.145 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 525.5 / 52.145$$

$$= 10.078$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 64146.156 / (2 \cdot 2350.0)$$

$$= 13.648 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 13.648 \cdot 554.04 \cdot 525.5$$

$$= 3973.699 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3973.699 / 302373.438$$

$$= 128.876 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (10.0778 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$



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$$= (1 - (10.08)^2 / (2 \cdot 123.21^2)) \cdot 260 / (5/3 + 3 \cdot (10.08) / (8 \cdot 123.21) - (10.08^3) / (8 \cdot 123.21^3))$$

$$= 152.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 6.72/152.67 + (3973.7 \cdot 150.0/4535.598) / 173.33$$

$$= 0.788$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load QO (Weight)	QO	64492.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	64146.2 Kg
Maximum Transverse Force	Ft	0.0 Kg
Maximum Longitudinal Force	Fl	64146.2 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (BoltArea \cdot Number\ of\ Bolts)$$

$$= 64146.2 / (9.797 \cdot 8)$$

$$= 80.3 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (BoltArea \cdot Number\ of\ Bolts)$$

$$= 0.0 / (9.797 \cdot 8)$$

$$= 0.0 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (BoltShearAllowable \cdot Nbolts)$$

$$= 64146.16 / (103.2 \cdot 8.0)$$

$$= 7.6197 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1850.0 \cdot 0.0 + 0.0$$

$$= 0.00 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 0.0/64492.7$$

$$= 0.00 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 7.6197, 0.0]$$

$$= 7.6197 \text{ cm}^2$$



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Saddle Calcs: Bundle Pull New:

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ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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Saddle Calcs: Bundle Pull Cor:

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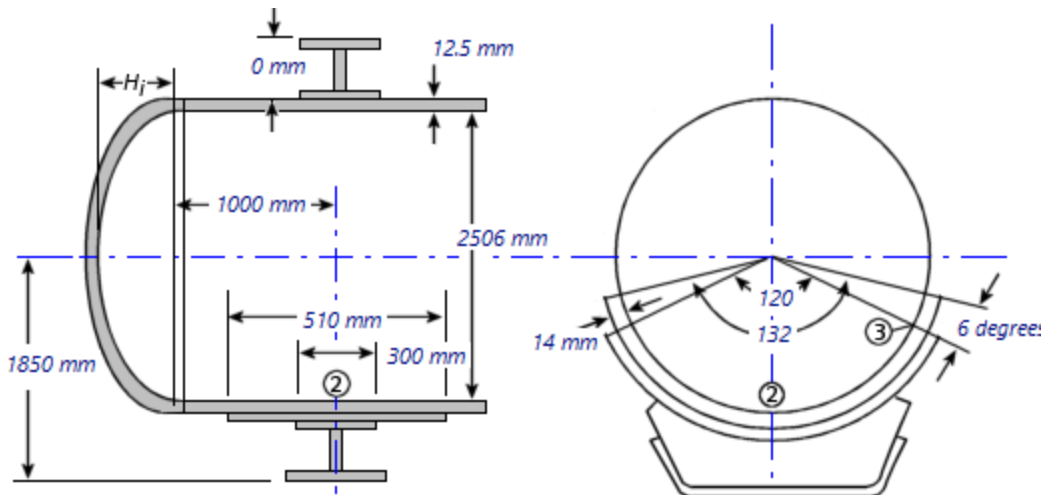
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	1259.65	mm
Shell Thickness used in this Case	t	12.500	mm
Stiffened Vessel Length per 4.15.6	L	6261.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		42764.10	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force	62775.87	Kgf
Pressure used in Analysis	P	0.000 KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	-7.30	-234.00
Long. Stress at Top	of Midspan	7.30	138.00
Long. Stress at Bottom	of Midspan	7.30	138.00
Long. Stress at Top	of Saddles	0.35	138.00



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Saddle Calcs: Bundle Pull Cor: Step: 25 10:13am Sep 9, 2026

Long. Stress at Bottom of Saddles	-0.35	-234.00
Long. Stress at Bottom of Saddles	0.35	138.00
Tangential Shear in Shell	9.05	110.40
Circ. Compressive Stress in Shell	3.60	138.00
Stiffener Circ. Stress at Shell	32.07	138.00
Stiffener Circ. Stress at Tip	83.07	234.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

= Bundle Pull Out Force • Impact Factor • B / Ls

= 42764 • 1.5 • 1850.0/3300.0007

= 35960.7 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 26815 + max(35961, 0, 0, 0)

= 62775.9 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 63978.51 Kgf

Transverse Shear Load 0.00 Kgf

Longitudinal Shear Load 64146.16 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(2506.0/2 + 12.5 + 14.0)sin(120.0/2), 2•1259.65)

= 2216.159 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0365 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0300

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -62776•1000.0[1-(1-1000.0/6261.0+(1259.65²-0.0²)/

(2•1000.0•6261.0))/(1+(4•0.0)/(3•6261.0))]

= -2071.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 62776•6261/4((1+2(1260²-0²)/(6261²))/(1+(4•0)/

(3•6261))-4•1000/6261)

= 43439.5 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0.0 • 1259.65/(2•11.7) - 43439.5/(π•1259.7²•11.7)

= -7.30 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0.0 • 1259.65/(2 • 11.7) + 43439.5/(π • 1259.7² • 11.7)

= 7.30 N./mm²



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Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1259.65 / (2 \cdot 11.7) - 2071.9 / (\pi \cdot 1259.7^2 \cdot 11.7)$$

$$= 0.35 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1259.65 / (2 \cdot 11.7) + 2071.9 / (\pi \cdot 1259.7^2 \cdot 11.7)$$

$$= -0.35 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 62776(6261.0 - 2 \cdot 1000.0) / (6261.0 + (4 \cdot 0.0 / 3))$$

$$= 42722.9 \text{ Kgf}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 42722.89 / (\pi \cdot 1259.65 \cdot 11.7)$$

$$= 9.05 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1259.65 \cdot 11.7)}$$

$$= 94.692 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.76 \cdot 62776 \cdot 0.1 / 130.04$$

$$= -3.60 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.3405 \cdot 62776 / 130.04 + 0.0529 \cdot 62776 \cdot 1259.65 \cdot 112.87 / 9600.66$$

$$= 32.07 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.3405 \cdot 62776 / 130.04 - 0.0529 \cdot 62776 \cdot 1259.65 \cdot 156.83 / 9600.66$$

$$= -83.07 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	600.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1253.8 \cdot 11.7)} + 510.0)11.7^3) / 12$$

$$= 9.329 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 9.329 + 81.776(354.036 - 5.85)^2$$

$$= 99149.375 \text{ cm}^4$$



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	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	698.9	11.7	5.9	81.8	47839.2	0.991E+05
Wearplate	510.0	14.0	18.7	71.4	133518.0	0.803E+05
Web	20.0	525.5	288.5	105.1	3031609.5	0.287E+05
BasePlate	600.0	45.0	573.7	270.0	15489899.0	0.131E+06
Totals	...	...	...	528.3	18702866.0	0.339E+06

Distance to Centroid [C1]:

= AY / A

= 18702866/528.276

= 354.036 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 120.0/2

= 120.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(120.0) - 0.5•sin(120.0)²)/(π - 2.094 + sin(120.0)cos(120.0))

= 0.2035

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.204 • 62775.871

= 12776.2510 Kgf

Tension Stress, St = (Fh/As) = 2.8061 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 1850.0 - 1253.8 • sin(120.0/2)/1.0472

= 813.116 mm

Bending Moment, M = Fh * d = 10388.7773 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 10.6431 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(62776 + 1203)600.0/(4 • 2350.0 • 173.33))

= 26.328 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 120.0/2)(1253.0 + 12.5 + 14.0)

= 2216.159 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 2216.1589/(5 - 1)

= 554.040 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 554.0397 • 600.0/2

= 1662.119 cm²



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Saddle Calcs: Bundle Pull Cor: Step: 25 10:13am Sep 9,2026

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 63978.5 / (2350.0 \cdot 600.0)$$

$$= 4.537 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1662.1 \cdot 4.54$$

$$= 7541.837 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 56.0 + 55.404$$

$$= 111.404 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 7541.8 / 111.404$$

$$= 6.639 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	300.0	...	60.0	...	0.450E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 20.0$$

$$= 280.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= A_Y / A$$

$$= 0.0 / 111.404$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0 / 2$$

$$= 150.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4500.0 / 111.404)}$$

$$= 63.556 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1850.0 - \cos(120.0/2) (1253.0 + 12.5 + 14.0) - 45.0$$

$$= 1165.250 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1165.25 / 63.556$$

$$= 18.334$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$



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$$= 64146.2 / (2 \cdot 2350.0) \cdot 554.04 \cdot 1165.25 / 2$$

$$= 4405.664 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (18.3343 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.33)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (18.33) / (8 \cdot 123.21) - (18.33^3) / (8 \cdot 123.21^3))$$

$$= 149.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 6.64 / 149.31 + (4405.66 \cdot 150.0 / 44999992) / 173.33$$

$$= 0.875$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	280.0	0.0	56.0	0.0	0.450E+04
Web	554.0	20.0	0.0	110.8	0.0	36.9
Totals	...	...	...	166.8	...	0.454E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 166.808$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0 / 2$$

$$= 150.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1850.0 - \cos(1279.5/2) (45.0 + 0.0 + 0.0) - 0.0$$

$$= 525.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{4535.6 / 166.808}$$

$$= 52.145 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 525.5 / 52.145$$

$$= 10.078$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 64146.156 / (2 \cdot 2350.0)$$

$$= 13.648 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 13.648 \cdot 554.04 \cdot 525.5$$

$$= 3973.699 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3973.699 / 302373.438$$

$$= 128.876 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (10.0778 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$



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$$= (1 - (10.08)^2 / (2 \cdot 123.21^2)) \cdot 260 / (5/3 + 3 \cdot (10.08) / (8 \cdot 123.21) - (10.08^3) / (8 \cdot 123.21^3))$$

$$= 152.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 6.67/152.67 + (3973.7 \cdot 150.0/4535.598) / 173.33$$

$$= 0.787$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	63978.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	64146.2 Kg
Maximum Transverse Force	Ft	0.0 Kg
Maximum Longitudinal Force	F1	64146.2 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 64146.2 / (9.797 \cdot 8)$$

$$= 80.3 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0.0 / (9.797 \cdot 8)$$

$$= 0.0 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (QL - Q0) / (Stba \cdot Rnbolts)$$

$$= (64146.16 - 63978.51) / (172.0 \cdot 8.0)$$

$$= 0.0119 \text{ cm}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 64146.16 / (103.2 \cdot 8.0)$$

$$= 7.6197 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1850.0 \cdot 0.0 + 0.0$$

$$= 0.00 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 0.0/63978.51$$

$$= 0.00 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$



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= max[0.0119, 7.6197, 0.0]
= 7.6197 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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Saddle Calcs: Road Transport:

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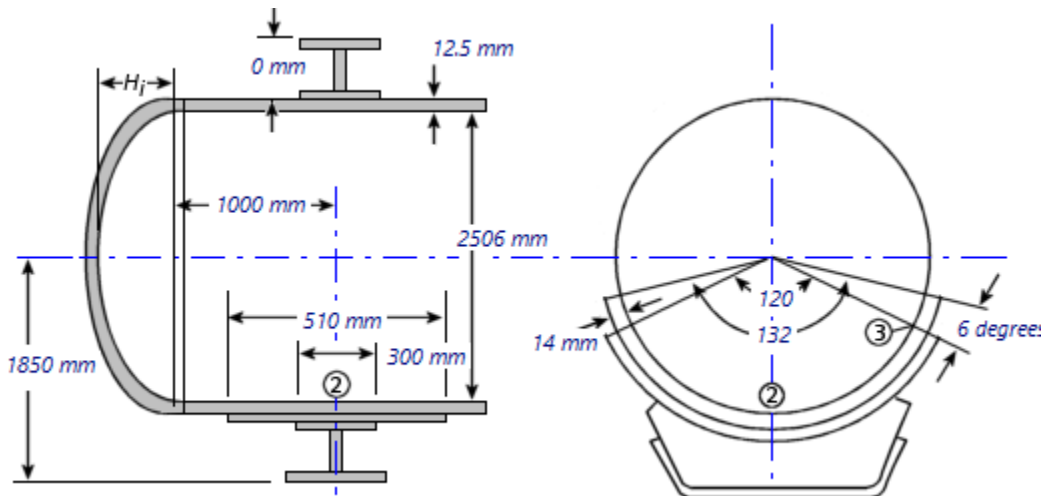
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	1259.25	mm
Shell Thickness used in this Case	t	12.500	mm
Stiffened Vessel Length per 4.15.6	L	6261.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		71245.16	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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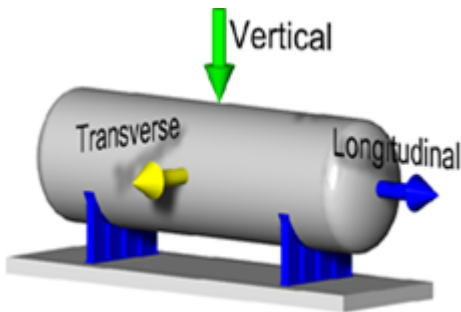
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Saddle Calcs: Road Transport:

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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-7.76	-234.00
Long. Stress at Top of Midspan	7.76	138.00
Long. Stress at Bottom of Midspan	7.76	138.00
Long. Stress at Top of Saddles	0.37	138.00
Long. Stress at Bottom of Saddles	-0.37	-234.00
Long. Stress at Bottom of Saddles	0.37	138.00
Tangential Shear in Shell	9.62	110.40
Circ. Compressive Stress in Shell	4.00	138.00
Stiffener Circ. Stress at Shell	34.31	138.00
Stiffener Circ. Stress at Tip	92.47	234.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf$$

$$= 342.73 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0$$

$$= 2911.2 \text{ lbf}$$

$$= 1320.534 \text{ Kg}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$$

$$= \pi \cdot 8.96^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2$$

$$= 642.7 \text{ lbf}$$

$$= 291.5 \text{ Kg}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw}$$

$$= 0.1 \cdot 82538.43 \cdot 1.0 + 1320.53$$

$$= 9574.4 \text{ Kg}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$$

$$= 0.1 \cdot 82538.43 \cdot 1.0 + 291.53$$

$$= 8545.4 \text{ Kg}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$= F_t / \text{Num of Saddles} \cdot B / E$$

$$= 9574.4/2 \cdot 1850.0/2216.1592$$

$$= 3996.2 \text{ Kg}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$= F_l \cdot B / L_s$$

$$= 8545 \cdot 1850.0/3300.0007$$

$$= 4790.6 \text{ Kg}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$= \text{Saddle Load} \cdot g_y$$



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$$= 63290.1 \cdot 0.05$$

$$= 3164.504 \text{ Kgf}$$

Load Combination Results due to Transport Loading [Q]:

$$= \text{Saddle Load} + \text{Frv} + \max(\text{Fwl}, \text{Fwt})$$

$$= 63290 + 3165 + \max(4791, 3996)$$

$$= 71245.2 \text{ Kgf}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	72447.80	Kgf
Transverse Shear Load	4787.19	Kgf
Longitudinal Shear Load	8545.37	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2506.0/2 + 12.5 + 14.0)\sin(120.0/2), 2 \cdot 1259.25)$$

$$= 2216.159 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0365	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0300			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -71245 \cdot 1000.0 [1 - (1 - 1000.0/6261.0 + (1259.25^2 - 0.0^2)/(2 \cdot 1000.0 \cdot 6261.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6261.0))]$$

$$= -2357.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 71245 \cdot 6261/4 [(1 + 2(1259^2 - 0^2)/(6261^2))/(1 + (4 \cdot 0)/(3 \cdot 6261)) - 4 \cdot 1000/6261]$$

$$= 49294.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 1259.25/(2 \cdot 12.5) - 49294.4/(\pi \cdot 1259.2^2 \cdot 12.5)$$

$$= -7.76 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1259.25/(2 \cdot 12.5) + 49294.4/(\pi \cdot 1259.2^2 \cdot 12.5)$$

$$= 7.76 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m/(2t) - M1/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1259.25/(2 \cdot 12.5) - 2357.2/(\pi \cdot 1259.2^2 \cdot 12.5)$$

$$= 0.37 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m/(2t) + M1/(\pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 1259.25/(2 \cdot 12.5) + 2357.2/(\pi \cdot 1259.2^2 \cdot 12.5)$$

$$= -0.37 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 71245(6261.0 - 2 \cdot 1000.0)/(6261.0 + (4 \cdot 0.0/3))$$



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= 48486.8 Kgf

Shear Stress in the shell with a single ring (4.15.13) [T1]:

= $T / (\pi \cdot R_m \cdot t)$

= $48486.77 / (\pi \cdot 1259.25 \cdot 12.5)$

= 9.62 N./mm²

Decay Length (4.15.20) [x1,x2]:

= $0.78 \cdot \sqrt{(R_m \cdot t)}$

= $0.78 \cdot \sqrt{(1259.25 \cdot 12.5)}$

= 97.860 mm

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ6*]:

= $-K5 \cdot Q \cdot k/A$

= $-0.76 \cdot 71245 \cdot 0.1/132.66$

= -4.00 N./mm²

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ8*]:

= $-K8 \cdot Q/A + K6 \cdot Q \cdot R_m \cdot c1/l$

= $-0.3405 \cdot 71245/132.66 + 0.0529 \cdot 71245 \cdot 1259.25 \cdot 111.46/9921.12$

= 34.31 N./mm²

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ9*]:

= $-K8 \cdot Q/A - K6 \cdot Q \cdot R_m \cdot c2/l$

= $-0.3405 \cdot 71245/132.66 - 0.0529 \cdot 71245 \cdot 1259.25 \cdot 159.04/9921.12$

= -92.47 N./mm²

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	600.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

= $((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3)/12$

= $((1.56 \cdot \sqrt{(1253.0 \cdot 12.5)} + 510.0)^3)/12$

= 11.478 cm⁴

Resolved Inertia for the Shell [I,shell]:

= $\text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$

= $11.478 + 88.154(350.616 - 6.25)^2$

= 104551.812 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	705.2	12.5	6.2	88.2	55096.4	0.105E+06
Wearplate	510.0	14.0	19.5	71.4	139230.0	0.783E+05
Web	20.0	525.5	289.3	105.1	3040019.0	0.281E+05
BasePlate	600.0	45.0	574.5	270.0	15511501.0	0.136E+06
Totals	...	...	...	534.7	18745846.0	0.347E+06

Distance to Centroid [C1]:

= AY / A

= 18745846/534.654



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$$= 350.616 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 71245.164$$

$$= 14499.9355 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.1847 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1850.0 - 1253.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 813.777 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 11799.9521 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 11.6998 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3 \cdot (71245 + 1203) \cdot 600.0) / (4 \cdot 2350.0 \cdot 173.33)}$$

$$= 28.016 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) \cdot (1253.0 + 12.5 + 14.0)$$

$$= 2216.159 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2216.1589 / (5 - 1)$$

$$= 554.040 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 554.0397 \cdot 600.0 / 2$$

$$= 1662.119 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 72447.797 / (2350.0 \cdot 600.0)$$

$$= 5.138 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1662.1 \cdot 5.14$$

$$= 8540.204 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 56.0 + 55.404$$



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$$= 111.404 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 8540.2/111.404$$

$$= 7.518 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	300.0	...	60.0	...	0.450E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 20.0$$

$$= 280.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/111.404$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0/2$$

$$= 150.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4500.0/111.404)}$$

$$= 63.556 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1850.0 - \cos(120.0/2) (1253.0 + 12.5 + 14.0) - 45.0$$

$$= 1165.250 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1165.25/63.556$$

$$= 18.334$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 8545.4 / (2 \cdot 2350.0) \cdot 554.04 \cdot 1165.25/2$$

$$= 586.910 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($18.3343 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.33)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (18.33) / (8 \cdot 123.21) - (18.33^3) / (8 \cdot 123.21^3))$$

$$= 149.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 7.52/149.31 + (586.91 \cdot 150.0/44999992) / 173.33$$

$$= 0.161$$



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Saddle Calcs: Road Transport:

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Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	280.0	0.0	56.0	0.0	0.450E+04
Web	554.0	20.0	0.0	110.8	0.0	36.9
Totals	...	...	...	166.8	...	0.454E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/166.808$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0/2$$

$$= 150.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1850.0 - \cos(1279.5/2)(45.0 + 0.0 + 0.0) - 0.0$$

$$= 525.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4535.6/166.808)}$$

$$= 52.145 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 525.5/52.145$$

$$= 10.078$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8545.373 / (2 \cdot 2350.0)$$

$$= 1.818 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.818 \cdot 554.04 \cdot 525.5$$

$$= 529.365 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 529.365/302373.438$$

$$= 17.168 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($10.0778 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.08)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (10.08) / (8 \cdot 123.21) - (10.08^3) / (8 \cdot 123.21^3))$$

$$= 152.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 7.55/152.67 + (529.37 \cdot 150.0/4535.598) / 173.33$$

$$= 0.149$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Saddle Calcs: Road Transport: Step: 26 10:13am Sep 9,2026

Vessel Mean Radius	Rm	1259.25	mm
Shell Thickness used in this Case	t	12.500	mm
Stiffened Vessel Length per 4.15.6	L	6261.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		26735.85	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-2.91	-234.00
Long. Stress at Top of Midspan		2.91	138.00
Long. Stress at Bottom of Midspan		2.91	138.00
Long. Stress at Top of Saddles		0.14	138.00
Long. Stress at Bottom of Saddles		-0.14	-234.00
Long. Stress at Bottom of Saddles		0.14	138.00
Tangential Shear in Shell		3.61	110.40
Circ. Compressive Stress in Shell		1.50	138.00
Stiffener Circ. Stress at Shell		12.88	138.00
Stiffener Circ. Stress at Tip		34.70	234.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf$$

$$= 342.73 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0$$

$$= 2911.2 \text{ lbf}$$

$$= 1320.534 \text{ Kgf}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$$

$$= \pi \cdot 8.96^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2$$

$$= 642.7 \text{ lbf}$$

$$= 291.5 \text{ Kgf}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$= gz \cdot \text{Shipping Weight} \cdot gw + Ftw$$

$$= 0.1 \cdot 82538.43 \cdot 1.0 + 1320.53$$



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Saddle Calcs: Road Transport:

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= 9574.4 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$

= $0.1 \cdot 82538.43 \cdot 1.0 + 291.53$

= 8545.4 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$

= $9574.4/2 \cdot 1850.0/2216.1592$

= 3996.2 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$

= $8545 \cdot 1850.0/3300.0007$

= 4790.6 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$

= $20900.3 \cdot 0.05$

= 1045.013 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= $20900 + 1045 + \max(4791, 3996)$

= 26735.9 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)

27938.49 Kgf

Transverse Shear Load

4787.19 Kgf

Longitudinal Shear Load

8545.37 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$

= $\min(2(2506.0/2 + 12.5 + 14.0) \sin(120.0/2), 2 \cdot 1259.25)$

= 2216.159 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0365 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0300

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$

= $-26736 \cdot 1000.0 [1 - (1 - 1000.0/6261.0 + (1259.25^2 - 0.0^2) / (2 \cdot 1000.0 \cdot 6261.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 6261.0))]$

= -884.6 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$

= $26736 \cdot 6261/4 ((1 + 2(1259^2 - 0^2)/(6261^2)) / (1 + (4 \cdot 0) / (3 \cdot 6261)) - 4 \cdot 1000/6261)$

= 18498.5 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$

= $0.0 \cdot 1259.25 / (2 \cdot 12.5) - 18498.5 / (\pi \cdot 1259.2^2 \cdot 12.5)$

= -2.91 N./mm²



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Saddle Calcs: Road Transport:

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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1259.25 / (2 \cdot 12.5) + 18498.5 / (\pi \cdot 1259.2^2 \cdot 12.5)$$

$$= 2.91 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1259.25 / (2 \cdot 12.5) - 884.6 / (\pi \cdot 1259.2^2 \cdot 12.5)$$

$$= 0.14 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1259.25 / (2 \cdot 12.5) + 884.6 / (\pi \cdot 1259.2^2 \cdot 12.5)$$

$$= -0.14 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 26736(6261.0 - 2 \cdot 1000.0) / (6261.0 + (4 \cdot 0.0 / 3))$$

$$= 18195.4 \text{ Kgf}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 18195.41 / (\pi \cdot 1259.25 \cdot 12.5)$$

$$= 3.61 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1259.25 \cdot 12.5)}$$

$$= 97.860 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k / A$$

$$= -0.76 \cdot 26736 \cdot 0.1 / 132.67$$

$$= -1.50 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q / A + K_6 \cdot Q \cdot R_m \cdot c_1 / I$$

$$= -0.3405 \cdot 26736 / 132.67 + 0.0529 \cdot 26736 \cdot 1259.25 \cdot 111.46 / 9921.14$$

$$= 12.88 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q / A - K_6 \cdot Q \cdot R_m \cdot c_2 / I$$

$$= -0.3405 \cdot 26736 / 132.67 - 0.0529 \cdot 26736 \cdot 1259.25 \cdot 159.04 / 9921.14$$

$$= -34.70 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	600.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$



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Saddle Calcs: Road Transport: Step: 26 10:13am Sep 9,2026

$$= ((1.56 \cdot \sqrt{(1253.0 \cdot 12.5)} + 510.0) 12.5^3) / 12$$

$$= 11.478 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 11.478 + 88.154 (350.616 - 6.25)^2$$

$$= 104551.812 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	705.2	12.5	6.2	88.2	55096.4	0.105E+06
Wearplate	510.0	14.0	19.5	71.4	139230.0	0.783E+05
Web	20.0	525.5	289.3	105.1	3040019.0	0.281E+05
BasePlate	600.0	45.0	574.5	270.0	15511501.0	0.136E+06
Totals	...	...	...	534.7	18745846.0	0.347E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 18745846 / 534.654$$

$$= 350.616 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 26735.852$$

$$= 5441.3257 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.1951 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1850.0 - 1253.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 813.777 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4428.1152 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.3905 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(26736 + 1203)600.0) / (4 \cdot 2350.0 \cdot 173.33)}$$

$$= 17.398 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1253.0 + 12.5 + 14.0)$$

$$= 2216.159 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2216.1589 / (5 - 1)$$



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Saddle Calcs: Road Transport:

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= 554.040 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 554.0397 • 600.0/2

= 1662.119 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)

= 27938.484/(2350.0 • 600.0)

= 1.981 Kg/cm²

Axial Load [P]:

= Ap • Bp

= 1662.1 • 1.98

= 3293.411 Kg

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 56.0 + 55.404

= 111.404 cm²

Compressive Stress [Sc]:

= P/Ar

= 3293.4/111.404

= 2.899 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	300.0	...	60.0	...	0.450E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 300.0 - 20.0

= 280.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/111.404

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 300.0/2

= 150.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{4500.0/111.404}$

= 63.556 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1850.0 - cos(120.0/2)(1253.0 + 12.5 + 14.0) - 45.0

= 1165.250 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0}$

= 123.211



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Saddle Calcs: Road Transport:

Step: 26 10:13am

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Slenderness ratio [KL/r]:

= KL/r

= 1 • 1165.25/63.556

= 18.334

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 8545.4 / (2 • 2350.0) • 554.04 • 1165.25/2

= 586.910 Kg-m.

Compressive Allowable, KL/r < Cc (18.3343 < 123.2105) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))

= (1 - (18.33)² / (2 • 123.21²)) 260 /

(5/3 + 3 • (18.33) / (8 • 123.21) - (18.33³) / (8 • 123.21³))

= 149.3 N/mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 2.9/149.31 + (586.91 • 150.0/44999992) / 173.33

= 0.130

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	280.0	0.0	56.0	0.0	0.450E+04
Web	554.0	20.0	0.0	110.8	0.0	36.9
Totals	...	...	...	166.8	...	0.454E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/166.808

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 300.0/2

= 150.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1850.0 - cos(1279.5/2) (45.0 + 0.0 + 0.0) - 0.0

= 525.500 mm

Radius of Gyration [r]:

= √(Total Inertia / Total Area)

= √(4535.6/166.808)

= 52.145 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 525.5/52.145

= 10.078

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 8545.373 / (2 • 2350.0)

= 1.818 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 1.818 • 554.04 • 525.5

= 529.365 Kg-m.



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Saddle Calcs: Road Transport: Step: 26 10:13am Sep 9, 2026

Bending Stress due to Transverse Force and Weight Load [σ_B , base, c]:

= Bending Moment / Section Modulus

= 529.365/302373.438

= 17.168 N./mm²

Compressive Allowable, $KL/r < C_c$ (10.0778 < 123.2105) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(10.08)²/(2 • 123.21²))260/

(5/3+3•(10.08)/(8• 123.21)-(10.08³)/(8•123.21³)

= 152.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$

= 2.91/152.67 + (529.37 • 150.0/4535.598)/173.33

= 0.118

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Operating Case:

Step: 27 10:13am

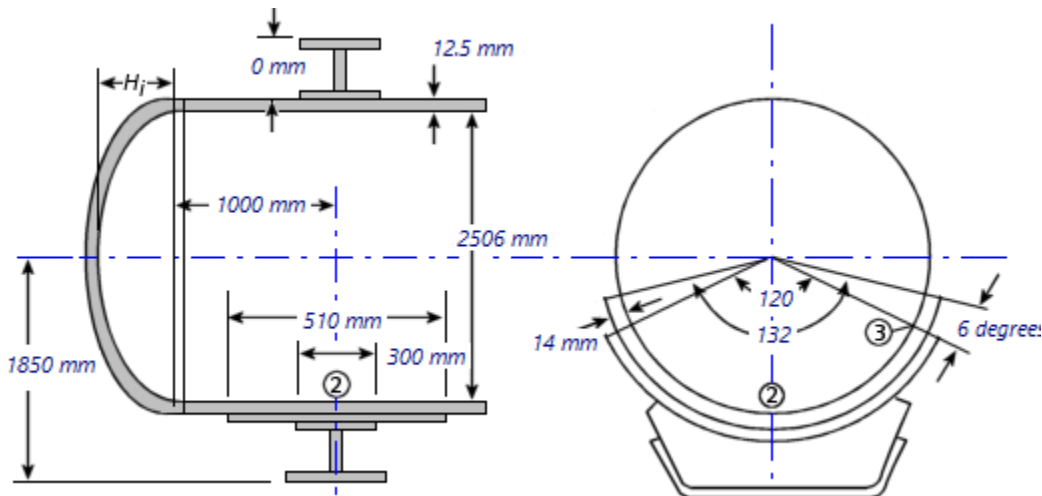
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1259.65	mm
Shell Thickness used in this Case	t	12.500	mm
Stiffened Vessel Length per 4.15.6	L	6261.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	104008.68	Kgf
Pressure used in Analysis	P	612.282 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	20.86	138.00
Long. Stress at Bottom of Midspan	45.06	138.00
Long. Stress at Top of Saddles	33.54	138.00
Long. Stress at Bottom of Saddles	32.38	138.00
Tangential Shear in Shell	14.99	110.40
Circ. Compressive Stress in Shell	5.96	138.00
Stiffener Circ. Stress at Shell	53.13	172.50



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Stiffener Circ. Stress at Tip	137.63	172.50
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(1273.8/2 + 0) \cdot 1850.0/2216.1592$
 $= 1595.0 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(724.86, 0) \cdot 1850.0/3300.0007$
 $= 406.4 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(7750.7, 0) \cdot 1850.0/3300.0007$
 $= 4345.1 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(7751/2 + 0) \cdot 1850.0/2216.1592$
 $= 9705.2 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 94304 + \max(406, 1595, 4345, 9705)$
 $= 104008.7 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08(0.0 + \pi/4(2.731 \cdot 1.2)^2)$
 $= 7108.300 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	105211.31	Kgf
Transverse Shear Load	3875.35	Kgf
Longitudinal Shear Load	7750.70	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2506.0/2 + 12.5 + 14.0)\sin(120.0/2), 2 \cdot 1259.65)$
 $= 2216.159 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0365	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0300			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 $= -104009 \cdot 1000.0 [1 - (1 - 1000.0/6261.0 + (1259.65^2 - 0.0^2)/(2 \cdot 1000.0 \cdot 6261.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6261.0))]$
 $= -3432.8 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$



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$$= 104009 \cdot 6261 / 4 \cdot ((1 + 2(1260^2 - 0^2) / (6261^2)) / (1 + (4 \cdot 0) / (3 \cdot 6261))) - 4 \cdot 1000 / 6261$$

$$= 71971.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 612.282 \cdot 1259.65 / (2 \cdot 11.7) - 71971.7 / (\pi \cdot 1259.7^2 \cdot 11.7)$$

$$= 20.86 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 612.282 \cdot 1259.65 / (2 \cdot 11.7) + 71971.7 / (\pi \cdot 1259.7^2 \cdot 11.7)$$

$$= 45.06 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 612.282 \cdot 1259.65 / (2 \cdot 11.7) - 3432.8 / (\pi \cdot 1259.7^2 \cdot 11.7)$$

$$= 33.54 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 612.282 \cdot 1259.65 / (2 \cdot 11.7) + 3432.8 / (\pi \cdot 1259.7^2 \cdot 11.7)$$

$$= 32.38 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 104009(6261.0 - 2 \cdot 1000.0) / (6261.0 + (4 \cdot 0.0 / 3))$$

$$= 70784.4 \text{ Kg}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 70784.38 / (\pi \cdot 1259.65 \cdot 11.7)$$

$$= 14.99 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1259.65 \cdot 11.7)}$$

$$= 94.692 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k / A$$

$$= -0.76 \cdot 104009 \cdot 0.1 / 130.04$$

$$= -5.96 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q / A + K_6 \cdot Q \cdot R_m \cdot c_1 / I$$

$$= -0.3405 \cdot 104009 / 130.04 + 0.0529 \cdot 104009 \cdot 1259.65 \cdot 112.87 / 9600.66$$

$$= 53.13 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q / A - K_6 \cdot Q \cdot R_m \cdot c_2 / I$$

$$= -0.3405 \cdot 104009 / 130.04 - 0.0529 \cdot 104009 \cdot 1259.65 \cdot 156.83 / 9600.66$$

$$= -137.63 \text{ N./mm}^2$$

Distance between Saddle Supports [L_s]:

$$= 3300.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000013 \cdot 3300.001 (230.0 - 21.1)$$

$$= 8.923 \text{ mm}$$

Results for Vessel Ribs, Web and Base:



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Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	600.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1253.8 \cdot 11.7)} + 510.0)11.7^3)/12$$

$$= 9.329 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 9.329 + 81.776(354.036 - 5.85)^2$$

$$= 99149.375 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	698.9	11.7	5.9	81.8	47839.2	0.991E+05
Wearplate	510.0	14.0	18.7	71.4	133518.0	0.803E+05
Web	20.0	525.5	288.5	105.1	3031609.5	0.287E+05
BasePlate	600.0	45.0	573.7	270.0	15489899.0	0.131E+06
Totals	...	...	...	528.3	18702866.0	0.339E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 18702866/528.276$$

$$= 354.036 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 104008.68$$

$$= 21168.0215 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 4.6493 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 138.2389 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1850.0 - 1253.8 \cdot \sin(120.0/2) / 1.0472$$

$$= 813.116 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 17212.3906 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 17.6337 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 153.5988 \text{ N./mm}^2$$



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Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(104009 + 1203)600.0}{4 \cdot 2350.0 \cdot 153.599}}$$

$$= 35.865 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1253.0 + 12.5 + 14.0)$$

$$= 2216.159 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 2216.1589 / (5 - 1)$$

$$= 554.040 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 554.0397 \cdot 600.0 / 2$$

$$= 1662.119 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 105211.305 / (2350.0 \cdot 600.0)$$

$$= 7.462 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 1662.1 \cdot 7.46$$

$$= 12402.393 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 56.0 + 55.404$$

$$= 111.404 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 12402.4 / 111.404$$

$$= 10.918 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	300.0	...	60.0	...	0.450E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 20.0$$

$$= 280.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 111.404$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0 / 2$$

$$= 150.000 \text{ mm}$$



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Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{4500.0 / 111.404}$
 $= 63.556 \text{ mm}$

Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 1850.0 - \cos(120.0/2) (1253.0 + 12.5 + 14.0) - 45.0$
 $= 1165.250 \text{ mm}$

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 230.4}$
 $= 130.885$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 1165.25 / 63.556$
 $= 18.334$

Bending Moment [Rm]:
 $= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
 $= 7750.7 / (2 \cdot 2350.0) \cdot 554.04 \cdot 1165.25 / 2$
 $= 532.331 \text{ Kg-m.}$

Compressive Allowable, $KL/r < Cc$ ($18.3343 < 130.8854$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (18.33)^2 / (2 \cdot 130.89^2)) 230 /$
 $(5/3 + 3 \cdot (18.33) / (8 \cdot 130.89) - (18.33^3) / (8 \cdot 130.89^3))$
 $= 132.7 \text{ N./mm}^2$

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 10.92 / 132.73 + (532.33 \cdot 150.0 / 449999992) / 153.6$
 $= 0.196$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	280.0	0.0	56.0	0.0	0.450E+04
Web	554.0	20.0	0.0	110.8	0.0	36.9
Totals	...	...	...	166.8	...	0.454E+04

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 166.808$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 300.0 / 2$
 $= 150.000 \text{ mm}$

Length of Inner Rib [L]:
 $= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$
 $= 1850.0 - \cos(1279.5/2) (45.0 + 0.0 + 0.0) - 0.0$
 $= 525.500 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{4535.6 / 166.808}$
 $= 52.145 \text{ mm}$



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Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 525.5/52.145$$

$$= 10.078$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 7750.698 / (2 \cdot 2350.0)$$

$$= 1.649 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.649 \cdot 554.04 \cdot 525.5$$

$$= 480.137 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 480.137/302373.438$$

$$= 15.572 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($10.0778 < 130.8854$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.08)^2 / (2 \cdot 130.89^2)) 230 /$$

$$(5/3 + 3 \cdot (10.08) / (8 \cdot 130.89) - (10.08^3) / (8 \cdot 130.89^3))$$

$$= 135.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_1/I) / S_{ba}$$

$$= 10.97/135.49 + (480.14 \cdot 150.0/4535.598) / 153.6$$

$$= 0.182$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	95506.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	4345.1 Kg
Maximum Transverse Force	Ft	3875.3 Kg
Maximum Longitudinal Force	Fl	7750.7 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 17139.0 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 17139.0 / (9.797 \cdot 8)$$

$$= 21.4 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3875.3 / (9.797 \cdot 8)$$

$$= 4.8 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (BoltShearAllowable \cdot Nbolts)$
 = $7750.7 / (82.8 \cdot 8.0)$
 = 1.1475 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1850.0 \cdot 3875.35 + 0.0$
 = 7169.54 Kg-m.

Eccentricity (e):
 = $Rmom / QO$
 = $7169.54 / 95506.15$
 = 75.07 mm < $Bplen / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 1.1475, 0.0]$
 = 1.1475 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.5, tr = 10.883, c = 0.8 mm, E* = 0.85
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.791$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-33 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1259.65	mm
Shell Thickness used in this Case	t	12.500	mm
Stiffened Vessel Length per 4.15.6	L	6261.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:



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Saddle Force Q, Operating Case 51349.93 Kgf
Pressure used in Analysis P 612.282 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	26.99	138.00
Long. Stress at Bottom of Midspan	38.94	138.00
Long. Stress at Top of Saddles	33.25	138.00
Long. Stress at Bottom of Saddles	32.68	138.00
Tangential Shear in Shell	7.40	110.40
Circ. Compressive Stress in Shell	2.94	138.00
Stiffener Circ. Stress at Shell	26.23	172.50
Stiffener Circ. Stress at Tip	67.95	172.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3.0(1273.8/2 + 0) • 1850.0/2216.1592
= 1595.0 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= max(Fl,wind; Sum of X Forces) • B / Ls
= max(724.86, 0) • 1850.0/3300.0007
= 406.4 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= max(Fl,eq; Sum of X Forces) • B / Ls
= max(7750.7, 0) • 1850.0/3300.0007
= 4345.1 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3.0(7751/2 + 0) • 1850.0/2216.1592
= 9705.2 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
= 41645 + max(406, 1595, 4345, 9705)
= 51349.9 Kgf

Longitudinal Wind Force [Fl]:
= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
= 1.1 • 766.08(0.0 + $\pi/4$ (2.731 • 1.2)²)
= 7108.300 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 52552.57 Kgf
Transverse Shear Load 3875.35 Kgf
Longitudinal Shear Load 7750.70 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
= min(2(2506.0/2 + 12.5 + 14.0)sin(120.0/2), 2•1259.65)
= 2216.159 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
K5 = 0.7603 K6 = 0.0529 K7 = 0.0365 K8 = 0.3405
K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434



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$$K7p = 0.0300$$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned} &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\ &= -51350 \cdot 1000.0 [1 - (1 - 1000.0/6261.0 + (1259.65^2 - 0.0^2)/(2 \cdot 1000.0 \cdot 6261.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6261.0))] \\ &= -1694.8 \text{ Kg-m.} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} &= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L] \\ &= 51350 \cdot 6261/4 [(1 + 2(1260^2 - 0^2)/(6261^2))/(1 + (4 \cdot 0)/(3 \cdot 6261)) - 4 \cdot 1000/6261] \\ &= 35533.0 \text{ Kg-m.} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t) \\ &= 612.282 \cdot 1259.65/(2 \cdot 11.7) - 35533.0/(\pi \cdot 1259.7^2 \cdot 11.7) \\ &= 26.99 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned} &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\ &= 612.282 \cdot 1259.65/(2 \cdot 11.7) + 35533.0/(\pi \cdot 1259.7^2 \cdot 11.7) \\ &= 38.94 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$\begin{aligned} &= P \cdot Rm/(2t) - M1/(\pi \cdot Rm^2 \cdot t) \\ &= 612.282 \cdot 1259.65/(2 \cdot 11.7) - 1694.8/(\pi \cdot 1259.7^2 \cdot 11.7) \\ &= 33.25 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$\begin{aligned} &= P \cdot Rm/(2t) + M1/(\pi \cdot Rm^2 \cdot t) \\ &= 612.282 \cdot 1259.65/(2 \cdot 11.7) + 1694.8/(\pi \cdot 1259.7^2 \cdot 11.7) \\ &= 32.68 \text{ N./mm}^2 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned} &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\ &= 51350(6261.0 - 2 \cdot 1000.0)/(6261.0 + (4 \cdot 0.0/3)) \\ &= 34946.8 \text{ Kg} \end{aligned}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$\begin{aligned} &= T / (\pi \cdot Rm \cdot t) \\ &= 34946.82/(\pi \cdot 1259.65 \cdot 11.7) \\ &= 7.40 \text{ N./mm}^2 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned} &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ &= 0.78 \cdot \sqrt{(1259.65 \cdot 11.7)} \\ &= 94.692 \text{ mm} \end{aligned}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$\begin{aligned} &= -K5 \cdot Q \cdot k/A \\ &= -0.76 \cdot 51350 \cdot 0.1/130.04 \\ &= -2.94 \text{ N./mm}^2 \end{aligned}$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$\begin{aligned} &= -K8 \cdot Q/A + K6 \cdot Q \cdot Rm \cdot c1/l \\ &= -0.3405 \cdot 51350/130.04 + 0.0529 \cdot 51350 \cdot 1259.65 \cdot 112.87/9600.67 \\ &= 26.23 \text{ N./mm}^2 \end{aligned}$$



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Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/I$$

$$= -0.3405 \cdot 51350/130.04 - 0.0529 \cdot 51350 \cdot 1259.65 \cdot 156.83/9600.67$$

$$= -67.95 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	600.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1253.8 \cdot 11.7)} + 510.0)11.7^3)/12$$

$$= 9.329 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 9.329 + 81.776(354.036 - 5.85)^2$$

$$= 99149.375 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	698.9	11.7	5.9	81.8	47839.2	0.991E+05
Wearplate	510.0	14.0	18.7	71.4	133518.0	0.803E+05
Web	20.0	525.5	288.5	105.1	3031609.5	0.287E+05
BasePlate	600.0	45.0	573.7	270.0	15489899.0	0.131E+06
Totals	...	...	...	528.3	18702866.0	0.339E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 18702866/528.276$$

$$= 354.036 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2)/(\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2)/(\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.204 \cdot 51349.934$$

$$= 10450.8252 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 2.2954 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 138.2389 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2)/(\text{theta}/2 \text{ in radians})$$



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$$= 1850.0 - 1253.8 \cdot \sin(120.0/2)/1.0472$$

$$= 813.116 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 8497.8984 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 8.7059 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 153.5988 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(51350 + 1203)600.0}{4 \cdot 2350.0 \cdot 153.599}}$$

$$= 25.348 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1253.0 + 12.5 + 14.0)$$

$$= 2216.159 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2216.1589 / (5 - 1)$$

$$= 554.040 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 554.0397 \cdot 600.0 / 2$$

$$= 1662.119 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 52552.57 / (2350.0 \cdot 600.0)$$

$$= 3.727 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1662.1 \cdot 3.73$$

$$= 6194.938 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 56.0 + 55.404$$

$$= 111.404 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 6194.9 / 111.404$$

$$= 5.453 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	300.0	...	60.0	...	0.450E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 20.0$$

$$= 280.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$



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= 0.0/111.404
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 300.0/2
= 150.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{4500.0/111.404}$
= 63.556 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 1850.0 - cos(120.0/2)(1253.0 + 12.5 + 14.0) - 45.0
= 1165.250 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/230.4}$
= 130.885

Slenderness ratio [KL/r]:

= KL/r
= 1 • 1165.25/63.556
= 18.334

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
= $7750.7 / (2 \cdot 2350.0) \cdot 554.04 \cdot 1165.25/2$
= 532.331 Kg-m.

Compressive Allowable, KL/r < Cc (18.3343 < 130.8854) per AISC E2-1 [Sca]:

= $(1 - (Kl/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Kl/r) / (8 \cdot Cc) - (Kl/r)^3 / (8 \cdot Cc^3))$
= $(1 - (18.33)^2 / (2 \cdot 130.89^2)) 230 /$
 $(5/3 + 3 \cdot (18.33) / (8 \cdot 130.89) - (18.33^3) / (8 \cdot 130.89^3))$
= 132.7 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $Sc/Sca + (Rm \cdot C1 / I) / Sba$
= $5.45/132.73 + (532.33 \cdot 150.0/44999992) / 153.6$
= 0.154

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	280.0	0.0	56.0	0.0	0.450E+04
Web	554.0	20.0	0.0	110.8	0.0	36.9
Totals	...	...	...	166.8	...	0.454E+04

Distance to Centroid from Datum [ytot]:

= AY / A
= 0.0/166.808
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 300.0/2
= 150.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk



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$$= 1850.0 - \cos(1279.5/2)(45.0 + 0.0 + 0.0) - 0.0$$

$$= 525.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{4535.6/166.808}$$

$$= 52.145 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 525.5/52.145$$

$$= 10.078$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 7750.698 / (2 \cdot 2350.0)$$

$$= 1.649 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.649 \cdot 554.04 \cdot 525.5$$

$$= 480.137 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 480.137/302373.438$$

$$= 15.572 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($10.0778 < 130.8854$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.08)^2 / (2 \cdot 130.89^2)) 230 / (5/3 + 3 \cdot (10.08) / (8 \cdot 130.89) - (10.08^3) / (8 \cdot 130.89^3))$$

$$= 135.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 5.48/135.49 + (480.14 \cdot 150.0/4535.598) / 153.6$$

$$= 0.142$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	42847.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	4345.1 Kg
Maximum Transverse Force	Ft	3875.3 Kg
Maximum Longitudinal Force	F1	7750.7 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 7750.7 / (9.797 \cdot 8)$$

$$= 9.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3875.3 / (9.797 \cdot 8)$$



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= 4.8 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = FI / (BoltShearAllowable • Nbolts)
 = 7750.7/(82.8 • 8.0)
 = 1.1475 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 1850.0 • 3875.35 + 0.0
 = 7169.54 Kg-m.

Eccentricity (e):
 = Rmom / QO
 = 7169.54/42847.4
 = 167.32 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 1.1475, 0.0]
 = 1.1475 cm²

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.5, tr = 10.883, c = 0.8 mm, E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.791, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-33 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

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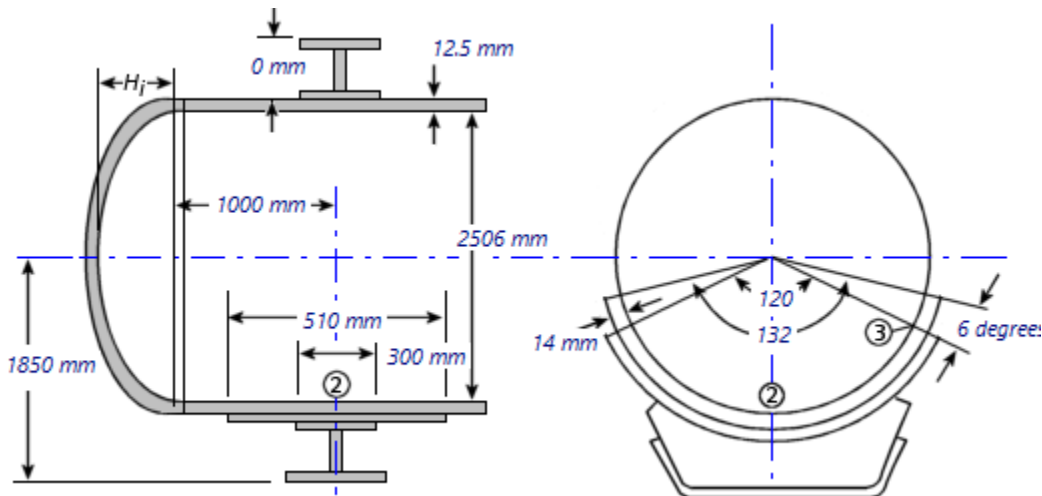
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1259.65	mm
Shell Thickness used in this Case	t	12.500	mm
Stiffened Vessel Length per 4.15.6	L	6261.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		91129.87	Kgf
Pressure used in Analysis	P	792.347	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	32.05	234.00
Long. Stress at Bottom of Midspan	53.26	234.00
Long. Stress at Top of Saddles	43.16	234.00
Long. Stress at Bottom of Saddles	42.15	234.00
Tangential Shear in Shell	13.14	138.58
Circ. Compressive Stress in Shell	5.22	234.00
Stiffener Circ. Stress at Shell	46.55	234.00



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Saddle Calcs: Test Case:

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Stiffener Circ. Stress at Tip	120.59	234.00
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0 (636.9/2 + 0) \cdot 1850.0/2216.1592$$

$$= 797.5 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max (\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max (362.43, 0) \cdot 1850.0/3300.0007$$

$$= 203.2 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max (\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 90332 + \max (203, 797, 0, 0)$$

$$= 91129.9 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (0.0 + \pi/4 (2.731 \cdot 1.2)^2)$$

$$= 7108.300 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	92332.50	Kgf
Transverse Shear Load	318.45	Kgf
Longitudinal Shear Load	362.43	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min (2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin (\theta/2), 2 \cdot R_m)$$

$$= \min (2 (2506.0/2 + 12.5 + 14.0) \sin (120.0/2), 2 \cdot 1259.65)$$

$$= 2216.159 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0365	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0300			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -91130 \cdot 1000.0 [1 - (1 - 1000.0/6261.0 + (1259.65^2 - 0.0^2) / (2 \cdot 1000.0 \cdot 6261.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 6261.0))]$$

$$= -3007.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 91130 \cdot 6261/4 ((1 + 2(1260^2 - 0^2)/(6261^2)) / (1 + (4 \cdot 0) / (3 \cdot 6261)) - 4 \cdot 1000/6261)$$

$$= 63059.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 792.347 \cdot 1259.65 / (2 \cdot 11.7) - 63059.9 / (\pi \cdot 1259.7^2 \cdot 11.7)$$

$$= 32.05 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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$$\begin{aligned}
 &= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 792.347 \cdot 1259.65 / (2 \cdot 11.7) + 63059.9 / (\pi \cdot 1259.7^2 \cdot 11.7) \\
 &= 53.26 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M1 / (\pi \cdot R_m^2 \cdot t) \\
 &= 792.347 \cdot 1259.65 / (2 \cdot 11.7) - 3007.7 / (\pi \cdot 1259.7^2 \cdot 11.7) \\
 &= 43.16 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 \cdot t) \\
 &= 792.347 \cdot 1259.65 / (2 \cdot 11.7) + 3007.7 / (\pi \cdot 1259.7^2 \cdot 11.7) \\
 &= 42.15 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L-2a) / (L + (4 \cdot h_m / 3)) \\
 &= 91130(6261.0 - 2 \cdot 1000.0) / (6261.0 + (4 \cdot 0.0 / 3)) \\
 &= 62019.5 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$\begin{aligned}
 &= T / (\pi \cdot R_m \cdot t) \\
 &= 62019.54 / (\pi \cdot 1259.65 \cdot 11.7) \\
 &= 13.14 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(R_m \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1259.65 \cdot 11.7)} \\
 &= 94.692 \text{ mm}
 \end{aligned}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k/A \\
 &= -0.76 \cdot 91130 \cdot 0.1 / 130.04 \\
 &= -5.22 \text{ N./mm}^2
 \end{aligned}$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$\begin{aligned}
 &= -K8 \cdot Q/A + K6 \cdot Q \cdot R_m \cdot c1/l \\
 &= -0.3405 \cdot 91130 / 130.04 + 0.0529 \cdot 91130 \cdot 1259.65 \cdot 112.87 / 9600.66 \\
 &= 46.55 \text{ N./mm}^2
 \end{aligned}$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$\begin{aligned}
 &= -K8 \cdot Q/A - K6 \cdot Q \cdot R_m \cdot c2/l \\
 &= -0.3405 \cdot 91130 / 130.04 - 0.0529 \cdot 91130 \cdot 1259.65 \cdot 156.83 / 9600.66 \\
 &= -120.59 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	600.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12 \\
 &= ((1.56 \cdot \sqrt{(1253.8 \cdot 11.7)} + 510.0)^3) / 12 \\
 &= 9.329 \text{ cm}^4
 \end{aligned}$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 9.329 + 81.776 (354.036 - 5.85)^2$$

$$= 99149.375 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	698.9	11.7	5.9	81.8	47839.2	0.991E+05
Wearplate	510.0	14.0	18.7	71.4	133518.0	0.803E+05
Web	20.0	525.5	288.5	105.1	3031609.5	0.287E+05
BasePlate	600.0	45.0	573.7	270.0	15489899.0	0.131E+06
Totals	...	...	...	528.3	18702866.0	0.339E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 18702866 / 528.276$$

$$= 354.036 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 91129.867$$

$$= 18546.9043 \text{ Kg}$$

$$\text{Tension Stress, } S_t = (F_h / A_s) = 4.0736 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1850.0 - 1253.8 \cdot \sin(120.0/2) / 1.0472$$

$$= 813.116 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 15081.0781 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C1 / I) = 15.4502 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(91130 + 1203)600.0) / (4 \cdot 2350.0 \cdot 173.33)}$$

$$= 31.628 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1253.0 + 12.5 + 14.0)$$

$$= 2216.159 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2216.1589 / (5 - 1)$$

$$= 554.040 \text{ mm}$$

Baseplate Pressure Area [Ap]:



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$$\begin{aligned}
 &= e \cdot Bpwid / 2 \\
 &= 554.0397 \cdot 600.0 / 2 \\
 &= 1662.119 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth) \\
 &= 92332.5 / (2350.0 \cdot 600.0) \\
 &= 6.548 \text{ Kg/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 1662.1 \cdot 6.55 \\
 &= 10884.229 \text{ Kg}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= Rib \text{ Area} + Web \text{ Area} \\
 &= 56.0 + 55.404 \\
 &= 111.404 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / Ar \\
 &= 10884.2 / 111.404 \\
 &= 9.581 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	300.0	...	60.0	...	0.450E+04

Rib dimension [D]:

$$\begin{aligned}
 &= Saddle \text{ Width} - Web \text{ Thickness} \\
 &= 300.0 - 20.0 \\
 &= 280.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0.0 / 111.404 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= Saddle \text{ Width} / 2 \\
 &= 300.0 / 2 \\
 &= 150.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) } \\
 &= \sqrt{ (4500.0 / 111.404) } \\
 &= 63.556 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk \\
 &= 1850.0 - \cos(120.0 / 2) (1253.0 + 12.5 + 14.0) - 45.0 \\
 &= 1165.250 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) } \\
 &= \sqrt{ (2 \cdot \pi^2 \cdot 0.19994E+09 / 260.0) } \\
 &= 123.211
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL / r \\
 &= 1 \cdot 1165.25 / 63.556
 \end{aligned}$$



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= 18.334

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$

= $362.4 / (2 \cdot 2350.0) \cdot 554.04 \cdot 1165.25 / 2$

= 24.892 Kg-m.

Compressive Allowable, $KL/r < Cc$ (18.3343 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$

= $(1 - (18.33)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (18.33) / (8 \cdot 123.21) - (18.33^3) / (8 \cdot 123.21^3))$

= 149.3 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $Sc/Sca + (Rm \cdot C1 / I) / Sba$

= $9.58 / 149.31 + (24.89 \cdot 150.0 / 449999992) / 173.33$

= 0.069

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	280.0	0.0	56.0	0.0	0.450E+04
Web	554.0	20.0	0.0	110.8	0.0	36.9
Totals	...	...	...	166.8	...	0.454E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= $0.0 / 166.808$

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= $300.0 / 2$

= 150.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= $1850.0 - \cos(1279.5/2) (45.0 + 0.0 + 0.0) - 0.0$

= 525.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{4535.6 / 166.808}$

= 52.145 mm

Slenderness ratio [KL/r]:

= KL/r

= $1 \cdot 525.5 / 52.145$

= 10.078

Unit Force [Force,u]:

= $Fl / (2 \cdot \text{Baseplate Length})$

= $362.428 / (2 \cdot 2350.0)$

= 0.077 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force $\cdot e \cdot L$

= $0.077 \cdot 554.04 \cdot 525.5$

= 22.452 Kg-m.

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

= Bending Moment / Section Modulus

= $22.452 / 302373.438$



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$$= 0.728 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($10.0778 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r)^2 / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (10.08)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (10.08) / (8 \cdot 123.21) - (10.08^3) / (8 \cdot 123.21^3))$$

$$= 152.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 9.63 / 152.67 + (22.45 \cdot 150.0 / 4535.598) / 173.33$$

$$= 0.067$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	91535.0 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	203.2 Kgf
Maximum Transverse Force	Ft	318.4 Kgf
Maximum Longitudinal Force	Fl	362.4 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 362.4 / (9.797 \cdot 8)$$

$$= 0.5 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 318.4 / (9.797 \cdot 8)$$

$$= 0.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 362.43 / (103.2 \cdot 8.0)$$

$$= 0.0431 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1850.0 \cdot 318.45 + 0.0$$

$$= 589.14 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q_0$$

$$= 589.14 / 91535.0$$

$$= 6.44 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



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Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 0.0431, 0.0]

= 0.0431 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1259.65	mm
Shell Thickness used in this Case	t	12.500	mm
Stiffened Vessel Length per 4.15.6	L	6261.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1850.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		39409.55	Kgf
Pressure used in Analysis	P	792.347	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	38.07	234.00
Long. Stress at Bottom of Midspan	47.24	234.00
Long. Stress at Top of Saddles	42.87	234.00
Long. Stress at Bottom of Saddles	42.44	234.00
Tangential Shear in Shell	5.68	138.58
Circ. Compressive Stress in Shell	2.26	234.00
Stiffener Circ. Stress at Shell	20.13	234.00
Stiffener Circ. Stress at Tip	52.15	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3.0(636.9/2 + 0) • 1850.0/2216.1592

= 797.5 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(362.43, 0) • 1850.0/3300.0007

= 203.2 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 38612 + max(203, 797, 0, 0)



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= 39409.5 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766.08(0.0 + $\pi/4$ (2.731 • 1.2)²)

= 7108.300 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 40612.18 Kgf

Transverse Shear Load 318.45 Kgf

Longitudinal Shear Load 362.43 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(2506.0/2 + 12.5 + 14.0)sin(120.0/2), 2•1259.65)

= 2216.159 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0365 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0300

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -39410•1000.0[1 - (1 - 1000.0/6261.0 + (1259.65²-0.0²)/

(2•1000.0•6261.0)/(1+(4•0.0)/(3•6261.0))]

= -1300.7 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)

= 39410•6261/4((1+2(1260²-0²)/(6261²))/(1+(4•0)/

(3•6261)) - 4•1000/6261)

= 27270.5 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= 792.347 • 1259.65/(2•11.7) - 27270.5/(π •1259.7²•11.7)

= 38.07 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 792.347 • 1259.65/(2 • 11.7) + 27270.5/(π • 1259.7² • 11.7)

= 47.24 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

= P • Rm/(2t) - M1/(π • Rm² • t)

= 792.347 • 1259.65/(2 • 11.7) - -1300.7/(π • 1259.7² • 11.7)

= 42.87 N./mm²

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

= P • Rm/(2t) + M1/(π •Rm²t)

= 792.347 • 1259.65/(2•11.7) + -1300.7/(π •1259.7²•11.7)

= 42.44 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 39410(6261.0 - 2 • 1000.0)/(6261.0 + (4 • 0.0/3))

= 26820.6 Kgf



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Shear Stress in the shell with a single ring (4.15.13) $[\tau_1]$:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 26820.65 / (\pi \cdot 1259.65 \cdot 11.7)$$

$$= 5.68 \text{ N./mm}^2$$

Decay Length (4.15.20) $[x_1, x_2]$:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1259.65 \cdot 11.7)}$$

$$= 94.692 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) $[\sigma_6^*]$:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.76 \cdot 39410 \cdot 0.1/130.04$$

$$= -2.26 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) $[\sigma_8^*]$:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/I$$

$$= -0.3405 \cdot 39410/130.04 + 0.0529 \cdot 39410 \cdot 1259.65 \cdot 112.87/9600.67$$

$$= 20.13 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) $[\sigma_9^*]$:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/I$$

$$= -0.3405 \cdot 39410/130.04 - 0.0529 \cdot 39410 \cdot 1259.65 \cdot 156.83/9600.67$$

$$= -52.15 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	600.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell $[ShellInertia]$:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + WearPlateWidth)^3)/12$$

$$= ((1.56 \cdot \sqrt{(1253.8 \cdot 11.7)} + 510.0)^3)/12$$

$$= 9.329 \text{ cm}^4$$

Resolved Inertia for the Shell $[I, shell]$:

$$= ShellInertia + A_{Shell} (C_1 - Y)^2$$

$$= 9.329 + 81.776 (354.036 - 5.85)^2$$

$$= 99149.375 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	698.9	11.7	5.9	81.8	47839.2	0.991E+05
Wearplate	510.0	14.0	18.7	71.4	133518.0	0.803E+05
Web	20.0	525.5	288.5	105.1	3031609.5	0.287E+05
BasePlate	600.0	45.0	573.7	270.0	15489899.0	0.131E+06
Totals	...	...	...	528.3	18702866.0	0.339E+06

Distance to Centroid $[C_1]$:

$$= AY / A$$

$$= 18702866/528.276$$



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Saddle Calcs: Test Case:

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$$= 354.036 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 39409.547$$

$$= 8020.6978 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.7616 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1850.0 - 1253.8 \cdot \sin(120.0/2) / 1.0472$$

$$= 813.116 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 6521.8843 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 6.6815 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3 \cdot (39410 + 1203) \cdot 600.0) / (4 \cdot 2350.0 \cdot 173.33)}$$

$$= 20.976 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) \cdot (1253.0 + 12.5 + 14.0)$$

$$= 2216.159 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2216.1589 / (5 - 1)$$

$$= 554.040 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 554.0397 \cdot 600.0 / 2$$

$$= 1662.119 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 40612.18 / (2350.0 \cdot 600.0)$$

$$= 2.880 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1662.1 \cdot 2.88$$

$$= 4787.396 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 56.0 + 55.404$$



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$$= 111.404 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 4787.4/111.404$$

$$= 4.214 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	300.0	...	60.0	...	0.450E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 20.0$$

$$= 280.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/111.404$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0/2$$

$$= 150.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4500.0/111.404)}$$

$$= 63.556 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1850.0 - \cos(120.0/2) (1253.0 + 12.5 + 14.0) - 45.0$$

$$= 1165.250 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1165.25/63.556$$

$$= 18.334$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 362.4 / (2 \cdot 2350.0) \cdot 554.04 \cdot 1165.25/2$$

$$= 24.892 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($18.3343 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.33)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (18.33) / (8 \cdot 123.21) - (18.33^3) / (8 \cdot 123.21^3))$$

$$= 149.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.21/149.31 + (24.89 \cdot 150.0/44999992) / 173.33$$

$$= 0.033$$



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Saddle Calcs: Test Case:

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Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	280.0	0.0	56.0	0.0	0.450E+04
Web	554.0	20.0	0.0	110.8	0.0	36.9
Totals	...	...	...	166.8	...	0.454E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/166.808$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0/2$$

$$= 150.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1850.0 - \cos(1279.5/2)(45.0 + 0.0 + 0.0) - 0.0$$

$$= 525.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4535.6/166.808)}$$

$$= 52.145 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 525.5/52.145$$

$$= 10.078$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 362.428 / (2 \cdot 2350.0)$$

$$= 0.077 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.077 \cdot 554.04 \cdot 525.5$$

$$= 22.452 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 22.452/302373.438$$

$$= 0.728 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($10.0778 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.08)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (10.08) / (8 \cdot 123.21) - (10.08)^3 / (8 \cdot 123.21^3))$$

$$= 152.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 4.23/152.67 + (22.45 \cdot 150.0/4535.598) / 173.33$$

$$= 0.032$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate

Nbolts

8

Total Number of Bolts in Tension/Baseplate

Nbt

4



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Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load QO (Weight)	QO	39814.7 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	203.2 Kgf
Maximum Transverse Force	Ft	318.4 Kgf
Maximum Longitudinal Force	Fl	362.4 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= Fl / (Bolt Area • Number of Bolts)

= 362.4 / (9.797 • 8)

= 0.5 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= Ft / (Bolt Area • Number of Bolts)

= 318.4 / (9.797 • 8)

= 0.4 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= Fl / (BoltShearAllowable • Nbolts)

= 362.43 / (103.2 • 8.0)

= 0.0431 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments

= 1850.0 • 318.45 + 0.0

= 589.14 Kg-m.

Eccentricity (e):

= Rmom / QO

= 589.14 / 39814.69

= 14.80 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 0.0431, 0.0]

= 0.0431 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: Tube inlet

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Input, Nozzle Desc: Tube inlet

From: 30

Pressure for Reinforcement Calculations	P	1000.000	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	101.13	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2500.00	mm
Design Length of Section	L	1500.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1236.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	886.4000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		240.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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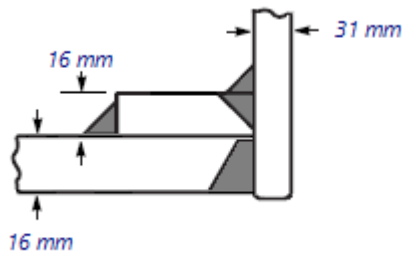
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Nozzle Calcs.: Tube inlet

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Reinforcement CALCULATION, Description: Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	13.562 in.
Actual Thickness Used in Calculation	1.219 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000.0 \cdot 1253.0 / (138 \cdot 1.0 - 0.6 \cdot 1000.0)$
 $= 9.1201 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000.0 \cdot 172.24 / (118 \cdot 1.0 - 0.6 \cdot 1000.0)$
 $= 1.4672 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.2099 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	688.9496 mm
Parallel to Vessel Wall, opening length	d	344.4748 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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Nozzle Calcs.: Tube inlet

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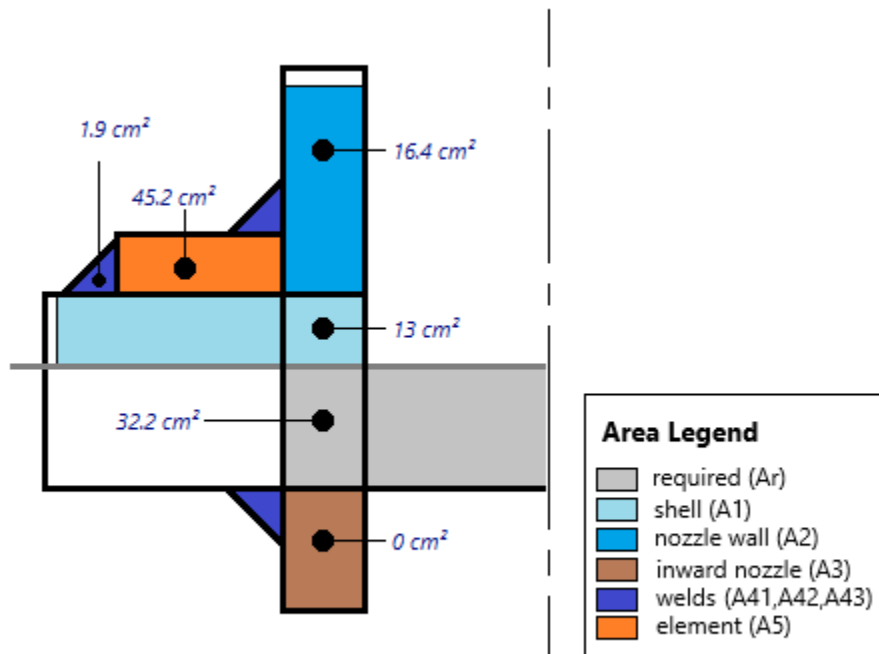


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	32.235	11.457	NA
Area in Shell	A1	13.017	21.865	NA
Area in Nozzle Wall	A2	16.393	16.536	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.924	1.924	NA
Area in Element	A5	45.208	45.208	NA
TOTAL AREA AVAILABLE	Atot	76.543	85.533	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 412.0264 16.0000 mm
Based on given Pad Diameter: 886.4000 0.3186 mm
Based on Shell or Nozzle Thickness: 412.0264 16.0000 mm

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (344.4748•9.1201•1.0+2•30.9626•9.1201•1.0•(1-0.86))
= 32.235 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 344.475(1.0 • 13.0 - 1.0 • 9.12) - 2 • 30.963
(1.0 • 13.0 - 1.0 • 9.1201) • (1 - 0.855)



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Nozzle Calcs.: Tube inlet

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$$= 13.017 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 32.5) \cdot (30.96 - 1.47) \cdot 0.8551$$

$$= 16.393 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 15.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 1.924 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (688.9496 - 406.4) \cdot 16.0 \cdot 1.0$$

$$= 45.208 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.4672 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 12.1201 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 12.1201 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb2 = 3.9187 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.331, \max(12.1201, 5.5)]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.4672, 8.3312)$$

$$= 8.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 30.963 = 27.092 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	11.7,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	283.3 N./mm ²	Passed
Occasional	:	2.6,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	7.2,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 27.092$, $t_r = 1.467$, $c = 0.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.054$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A 21°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 16.0$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.702$, Temp. Reduction = 17°C

Attachment governing, Attachment stress = Shell stress per Code



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Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 16.0$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.702$, Temp. Reduction = 17 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 16.0$, $tr = 9.12$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.702$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 16.0$, $tr = 9.12$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.702$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-31 °C
Gov. MDMT of the Reinforcement Pad	:	-31 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-31 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1000.00 / 1960.01 = 0.510$

Weld Size Calculations, Description: Tube inlet

Intermediate Calc. for nozzle/shell Welds	T_{min}	16.0000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$10.6050 = 0.7 * W_o$ mm
Pad Weld	$6.5000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) S_v)$
 $= \max(0, (32.2348 - 13.0172 + 2 \cdot 30.9626 \cdot 0.8551 \cdot (1.0 \cdot 13.0 - 9.1201)) 138)$
 $= 29933.09 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:



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$$= (A2 + A5 + A4 - (W_i - C_{an} / 707)^2 \cdot fr2) \cdot Sv$$

$$= (16.3935 + 45.2079 + 1.9239 - 0.0 \cdot 0.86) \cdot 138$$

$$= 89390.08 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (16.3935 + 0.0 + 1.9239 + (6.8836)) \cdot 138$$

$$= 35461.67 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (16.3935 + 0.0 + 1.9239 + 45.2079 + (6.8836)) \cdot 138$$

$$= 99076.34 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 406.4 \cdot 15.0 \cdot 0.49 \cdot 118$$

$$= 56456. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 886.4 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 115205. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 187.7187) \cdot (30.9626 - 0.0) \cdot 0.7 \cdot 118$$

$$= 153794. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 406.4 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 106357. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 406.4 \cdot (16.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 86415. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (115205 + 153794) = 268998 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (56456 + 106357 + 86415 + 0) = 249228 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (115205 + 86415 + 0) = 201620 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 268998 Kgf, must exceed W = 29933 Kgf	or W1 = 89390 Kgf
Path 2-2 = 249228 Kgf, must exceed W = 29933 Kgf	or W2 = 35461 Kgf
Path 3-3 = 201619 Kgf, must exceed W = 29933 Kgf	or W3 = 99076 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1185 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 16.6267 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 382.6267 mm

Input Echo, WRC297 Item 1, Description: Tube inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2500.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	344.475	mm
Nozzle Thickness	30.9626	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	886.400	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	1000.00	KPa.
Radial Load	P	-3059.35	Kgf
Circumferential Shear	Vc	2039.57	Kgf
Longitudinal Shear	Vl	2039.57	Kgf
Circumferential Moment	Mc	2283.42	Kg-m.
Longitudinal Moment	Ml	2283.42	Kg-m.
Torsional Moment	Mt	3262.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 406.4 + 2 \cdot 1.65 \cdot \sqrt{1267.5(16.0 - 3.0)} \\
 &= 830.004 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2535.000	mm
Nozzle Outside Diameter	(d) =	406.400	mm
Vessel Thickness used	(T) =	29.000	mm



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Nozzle Calcs.: Tube inlet

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Nozzle Thickness used (t) = 30.963 mm
 T / t = 0.937
 d / t = 13.126

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.499$$

Nr/P = 0.091
 Mr/P = 0.148
 M0/P = 0.050
 N0/P = 0.114
 MrD/Mc = 0.268
 NrDL/Mc = 0.127
 M0d/Mc = 0.097
 N0DL/Mc = 0.062
 MrD/Ml = 0.149
 NrDL/Ml = 0.084
 M0D/Ml = 0.053
 N0DL/Ml = 0.147

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	42	43	42	43
Outplane Stress Summary	26	48	87	25

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-58	58	58	-58
Inplane Pressure Stress	21	21	21	21
Inplane Stress Summary	-7	45	120	-60

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	36	52	120	86
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-38	38	38	-38
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21
Outplane Stress Summary	-6	48	78	-19

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-105	105	105	-105
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	42	43	42	43
Inplane Stress Summary	-35	112	191	-81

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	36	112	191	82

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	0	1	19	20



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-36	36	36	-36
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-20	14	67	-45

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	23	16	67	66

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	6	7	14	15

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-70	70	70	-70
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-54	48	100	-78

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside



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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3

Shear Stress Summary	2	2	4	4
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	60	48	100	94

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{1259.5(16.0 - 3.0)}$$

$$= 828.665 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2519.000 mm
Nozzle Outside Diameter	(d) =	886.400 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	270.963 mm
T / t	=	0.048
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.898$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.047
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.022
MrD/Mc	=	0.240
NrDL/Mc	=	0.104
M0d/Mc	=	0.072
N0DL/Mc	=	0.045
MrD/Ml	=	0.048
NrDL/Ml	=	0.052
M0D/Ml	=	0.015
N0DL/Ml	=	0.023

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-13	13	13	-13
Normal Pressure Stress	95	96	95	96
Outplane Stress Summary	103	90	136	70

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	75	-75	75	-75
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	47	47	47	47

Inplane Stress Summary	80	15	183	-55
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	3	3	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	103	90	183	125
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-64	64	64	-64
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	47	47	47	47

Outplane Stress Summary	1	89	143	-25
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	75	-75	75	-75
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-215	215	215	-215
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	95	96	95	96

Inplane Stress Summary	-50	228	410	-170
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
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Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	51	228	410	170

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	42	43	42	43	42	43	42	43
Circ. Pl (SUS)	-5	-5	13	13	-5	-5	11	11
Circ. Q (SUS)	-10	10	30	-30	-74	74	136	-136
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-2	-2	8	8	0	0	8	8
Long. Q (SUS)	-27	27	89	-89	-28	28	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	42.0	43.0	42.0	43.0	42.0	43.0	42.0	43.0
Pm+Pl (SUS)	37.1	38.1	55.0	56.0	37.1	38.1	53.0	54.0
Pm+Pl+Q (Total)	36.4	52.1	118.3	86.2	37.3	112.1	189.2	82.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	43.00	138.00	Passed
Pm+Pl (SUS)	56.04	207.00	Passed
Pm+Pl+Q (TOTAL)	189.22	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	95	96	95	96	95	96	95	96
Circ. Pl (SUS)	0	0	6	6	-7	-7	23	23
Circ. Q (SUS)	7	-7	33	-33	-140	140	290	-290
Long. Pm (SUS)	47	47	47	47	47	47	47	47
Long. Pl (SUS)	1	1	15	15	-3	-3	9	9
Long. Q (SUS)	32	-32	118	-118	-44	44	84	-84



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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	95.0	96.0	95.0	96.0	95.0	96.0	95.0	96.0
Pm+P1 (SUS)	95.0	96.0	101.0	102.0	88.0	89.0	118.0	119.0
Pm+P1+Q (Total)	102.2	89.1	180.0	125.0	52.0	229.0	408.0	171.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	96.00	138.00	Passed
Pm+P1 (SUS)	119.02	207.00	Passed
Pm+P1+Q (TOTAL)	408.01	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. P1 (SUS)	-5	-5	13	13	0	0	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. P1 (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-17	17	55	-55	-51	51	89	-89
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+P1 (SUS)	5.4	6.3	19.1	20.1	10.2	11.2	14.2	15.2
Pm+P1+Q (Total)	23.4	14.3	65.1	65.1	61.1	47.1	99.2	94.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed
Pm+P1 (SUS)	20.10	177.00	Passed
Pm+P1+Q (TOTAL)	99.19	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: Tube outlet

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Input, Nozzle Desc: Tube outlet

From: 30

Pressure for Reinforcement Calculations	P	1024.505	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	101.13	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2500.00	mm
Design Length of Section	L	1500.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1236.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-182 F22	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	886.4000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		240.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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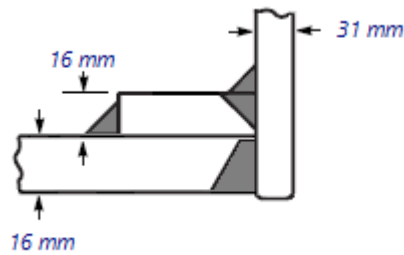
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Nozzle Calcs.: Tube outlet

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Reinforcement CALCULATION, Description: Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	13.562 in.
Actual Thickness Used in Calculation	1.219 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1024.51 \cdot 1253.0 / (138 \cdot 1.0 - 0.6 \cdot 1024.51)$
 $= 9.3445 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1024.51 \cdot 172.24 / (118 \cdot 1.0 - 0.6 \cdot 1024.51)$
 $= 1.5034 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.2099 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	688.9496 mm
Parallel to Vessel Wall, opening length	d	344.4748 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

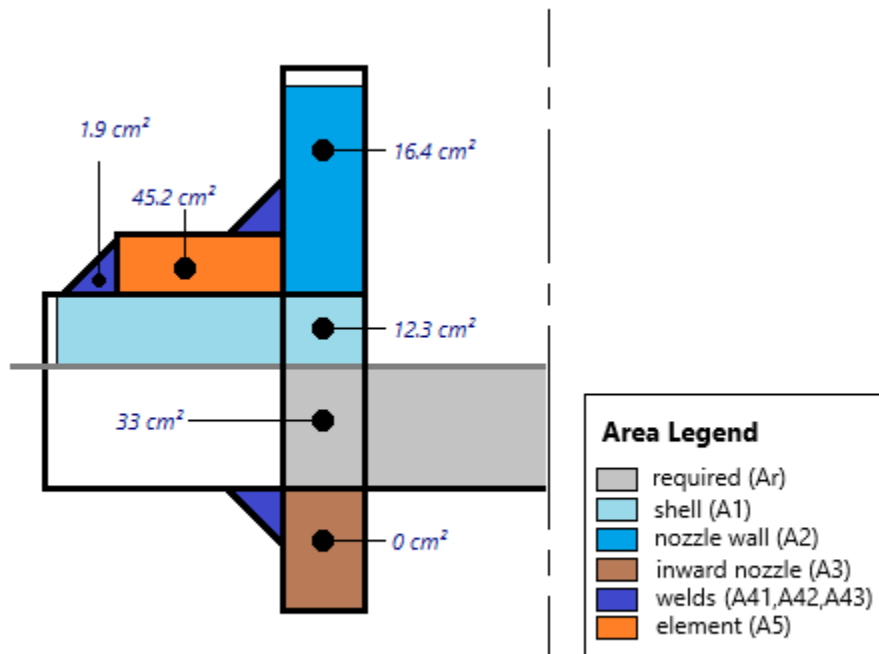


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	33.028	11.457	NA
Area in Shell	A1	12.264	21.865	NA
Area in Nozzle Wall	A2	16.373	16.536	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.924	1.924	NA
Area in Element	A5	45.208	45.208	NA
TOTAL AREA AVAILABLE	Atot	75.769	85.533	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 421.8181 16.0000 mm
Based on given Pad Diameter: 886.4000 0.8731 mm
Based on Shell or Nozzle Thickness: 421.8181 16.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (344.4748 \cdot 9.3445 \cdot 1.0 + 2 \cdot 30.9626 \cdot 9.3445 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 33.028 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 344.475(1.0 \cdot 13.0 - 1.0 \cdot 9.345) - 2 \cdot 30.963$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 9.3445) \cdot (1 - 0.855)$$



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$$= 12.264 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot f_{r2}$$

$$= (2 \cdot 32.5) \cdot (30.96 - 1.5) \cdot 0.8551$$

$$= 16.373 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot f_{r3} + (W_i - c_{an}/0.707)^2 \cdot f_{r2} + W_p^2 \cdot f_{r4}$$

$$= 15.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 1.924 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (688.9496 - 406.4) \cdot 16.0 \cdot 1.0$$

$$= 45.208 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 1.5034 \text{ mm}$

Wall Thickness per UG16(b), $t_{r16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $t_{rb1} = 12.3445 \text{ mm}$

Wall Thickness $t_{b1} = \max(t_{rb1}, t_{r16b}) = 12.3445 \text{ mm}$

Wall Thickness, shell/head, external pressure $t_{rb2} = 3.9187 \text{ mm}$

Wall Thickness $t_{b2} = \max(t_{rb2}, t_{r16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $t_{b3} = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[8.331, \max(12.3445, 5.5)]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.5034, 8.3312)$$

$$= 8.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 30.963 = 27.092 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	11.7,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	283.3 N./mm ²	Passed
Occasional	:	2.6,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	7.2,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 27.092$, $t_r = 1.503$, $c = 0.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.055$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B 1°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 16.0$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.719$, Temp. Reduction = 16°C

Attachment governing, Attachment stress = Shell stress per Code



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Nozzle Calcs.: Tube outlet Nozl: 10 10:13am Sep 9, 2026

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.719$, Temp. Reduction = 16 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 16.0, tr = 9.345, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.719$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 9.345, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.719$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-30 °C
Gov. MDMT of the Reinforcement Pad	:	-30 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-47 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1024.51 / 1960.01 = 0.523$

Weld Size Calculations, Description: Tube outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$10.6050 = 0.7 * Wo \text{ mm}$
Pad Weld	$6.5000 = 0.5 * TminPad$	$8.4840 = 0.7 * Wp \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 * tn * fr1 * (E1 * t - tr)) Sv)$
 $= \max(0, (33.0282 - 12.2641 + 2 * 30.9626 * 0.8551 * (1.0 * 13.0 - 9.3445)) 138)$
 $= 31942.13 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:



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$$= (A2 + A5 + A4 - (W_i - C_{an} / 707)^2 \cdot fr2) \cdot Sv$$

$$= (16.3734 + 45.2079 + 1.9239 - 0.0 \cdot 0.86) \cdot 138$$

$$= 89361.81 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (16.3734 + 0.0 + 1.9239 + (6.8836)) \cdot 138$$

$$= 35433.40 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (16.3734 + 0.0 + 1.9239 + 45.2079 + (6.8836)) \cdot 138$$

$$= 99048.07 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 406.4 \cdot 15.0 \cdot 0.49 \cdot 118$$

$$= 56456. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 886.4 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 115205. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 187.7187) \cdot (30.9626 - 0.0) \cdot 0.7 \cdot 118$$

$$= 153794. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 406.4 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 106357. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 406.4 \cdot (16.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 86415. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (115205 + 153794) = 268998 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (56456 + 106357 + 86415 + 0) = 249228 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (115205 + 86415 + 0) = 201620 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 268998 Kgf, must exceed W = 31942 Kgf	or W1 = 89361 Kgf
Path 2-2 = 249228 Kgf, must exceed W = 31942 Kgf	or W2 = 35433 Kgf
Path 3-3 = 201619 Kgf, must exceed W = 31942 Kgf	or W3 = 99048 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1185 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 16.6267 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 382.6267 mm

Input Echo, WRC297 Item 1, Description: Tube outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2500.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	344.475	mm
Nozzle Thickness	30.9626	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	886.400	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	1024.51	KPa.
Radial Load	P	-3059.35	Kgf
Circumferential Shear	Vc	2039.57	Kgf
Longitudinal Shear	Vl	2039.57	Kgf
Circumferential Moment	Mc	2283.43	Kg-m.
Longitudinal Moment	Ml	2283.43	Kg-m.
Torsional Moment	Mt	3262.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 406.4 + 2 \cdot 1.65 \cdot \sqrt{1267.5(16.0 - 3.0)} \\
 &= 830.004 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2535.000	mm
Nozzle Outside Diameter	(d) =	406.400	mm
Vessel Thickness used	(T) =	29.000	mm



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Nozzle Thickness used (t) = 30.963 mm
 T / t = 0.937
 d / t = 13.126

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.499$$

Nr/P = 0.091
 Mr/P = 0.148
 M0/P = 0.050
 N0/P = 0.114
 MrD/Mc = 0.268
 NrDL/Mc = 0.127
 M0d/Mc = 0.097
 N0DL/Mc = 0.062
 MrD/Ml = 0.149
 NrDL/Ml = 0.084
 M0D/Ml = 0.053
 N0DL/Ml = 0.147

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	43	44	43	44
Outplane Stress Summary	27	49	88	27

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-58	58	58	-58
Inplane Pressure Stress	21	21	21	21
Inplane Stress Summary	-7	46	120	-59

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	36	53	121	87
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-38	38	38	-38
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21
Outplane Stress Summary	-5	49	78	-18

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-105	105	105	-105
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	43	44	43	44
Inplane Stress Summary	-34	113	192	-80

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	35	113	192	81

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	0	1	19	21



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-36	36	36	-36
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-20	14	67	-45

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	23	16	67	66

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	6	7	14	15

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-70	70	70	-70
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-54	48	100	-78

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside



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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3

Shear Stress Summary	2	2	4	4
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	60	48	101	94

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{1259.5(16.0 - 3.0)}$$

$$= 828.665 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2519.000 mm
Nozzle Outside Diameter	(d) =	886.400 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	270.963 mm
T / t	=	0.048
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.898$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.047
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.022
MrD/Mc	=	0.240
NrDL/Mc	=	0.104
M0d/Mc	=	0.072
N0DL/Mc	=	0.045
MrD/Ml	=	0.048
NrDL/Ml	=	0.052
M0D/Ml	=	0.015
N0DL/Ml	=	0.023

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	98	99	98	99
Outplane Stress Summary	105	92	139	73

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



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Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	75	-75	75	-75
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	49	49	49	49

Inplane Stress Summary	82	17	184	-53
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	3	3	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	106	92	184	127
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-64	64	64	-64
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	49	49	49	49

Outplane Stress Summary	2	90	144	-24
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	75	-75	75	-75
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-215	215	215	-215
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	98	99	98	99

Inplane Stress Summary	-48	231	412	-167
------------------------	-----	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
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Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	50	231	412	167

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	43	44	43	44	43	44	43	44
Circ. Pl (SUS)	-5	-5	13	13	-5	-5	11	11
Circ. Q (SUS)	-10	10	30	-30	-74	74	136	-136
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-2	-2	8	8	0	0	8	8
Long. Q (SUS)	-27	27	89	-89	-28	28	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	43.0	44.0	43.0	44.0	43.0	44.0	43.0	44.0
Pm+Pl (SUS)	38.1	39.0	56.0	57.0	38.1	39.1	54.0	55.0
Pm+Pl+Q (Total)	37.4	52.7	118.3	87.2	36.3	113.1	190.2	81.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	44.00	138.00	Passed
Pm+Pl (SUS)	57.04	207.00	Passed
Pm+Pl+Q (TOTAL)	190.22	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	98	99	98	99	98	99	98	99
Circ. Pl (SUS)	0	0	6	6	-7	-7	23	23
Circ. Q (SUS)	7	-7	33	-33	-140	140	290	-290
Long. Pm (SUS)	49	49	49	49	49	49	49	49
Long. Pl (SUS)	1	1	15	15	-3	-3	9	9
Long. Q (SUS)	32	-32	118	-118	-44	44	84	-84



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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	98.0	99.0	98.0	99.0	98.0	99.0	98.0	99.0
Pm+P1 (SUS)	98.0	99.0	104.0	105.0	91.0	92.0	121.0	122.0
Pm+P1+Q (Total)	105.2	92.1	182.0	126.0	51.0	232.0	411.0	168.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	99.00	138.00	Passed
Pm+P1 (SUS)	122.02	207.00	Passed
Pm+P1+Q (TOTAL)	411.01	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. P1 (SUS)	-5	-5	13	13	0	0	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. P1 (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-17	17	55	-55	-51	51	89	-89
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+P1 (SUS)	5.4	6.3	19.1	20.1	10.2	11.2	14.2	15.2
Pm+P1+Q (Total)	23.4	14.3	65.1	65.1	61.1	47.1	99.2	94.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed
Pm+P1 (SUS)	20.10	177.00	Passed
Pm+P1+Q (TOTAL)	99.19	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell outlet

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Input, Nozzle Desc: Shell outlet

From: 70

Pressure for Reinforcement Calculations	P	624.564	KPa.
Temperature for Internal Pressure	Temp	230	°C
Parent Material		SA-516	70
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2506.00	mm
Shell Finished (Minimum) Thickness	t	24.5000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5453.23	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266	2
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.80	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		42.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm
Flange Material		SA-266	2
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	101.6140	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	0.8000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.5000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.



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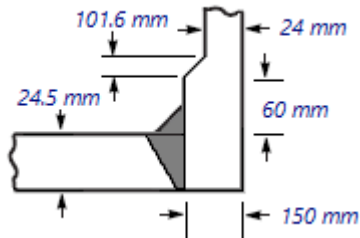
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Nozzle Calcs.: Shell outlet

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Reinforcement CALCULATION, Description: Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	42.000 in.
Actual Thickness Used in Calculation	0.945 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 624.56 \cdot 1253.8 / (138 \cdot 1.0 - 0.6 \cdot 624.56)$
 $= 5.6904 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 624.56 \cdot 534.2 / (137 \cdot 1.0 - 0.6 \cdot 624.56)$
 $= 2.4458 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 126.0 / 0.5773$
 $= 218.2385 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	2136.8000 mm
Parallel to Vessel Wall, opening length	d	1068.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.2500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 136.8 / 138.0)$
 $= 0.991$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 136.8 / 138.0)$
 $= 0.991$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.991, 1.0)$
 $= 0.991$



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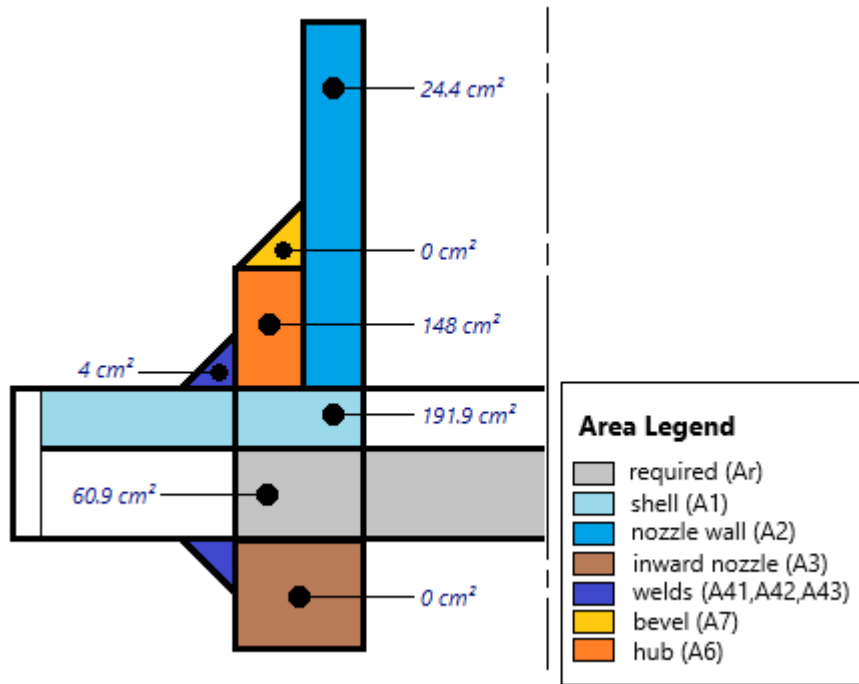
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Nozzle Calcs.: Shell outlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	60.943	NA	NA
Area in Shell	A1	191.948	NA	NA
Area in Nozzle Wall	A2	24.380	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	3.965	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	148.013	NA	NA
TOTAL AREA AVAILABLE	Atot	368.306	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (1068.4 \cdot 5.6904 \cdot 1.0 + 2 \cdot 149.2 \cdot 5.6904 \cdot 1.0 \cdot (1 - 0.99))$$

$$= 60.943 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 1068.4(1.0 \cdot 23.7 - 1.0 \cdot 5.69) - 2 \cdot 149.2$$

$$(1.0 \cdot 23.7 - 1.0 \cdot 5.6904) \cdot (1 - 0.991)$$

$$= 191.948 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 59.25)(23.2 - 2.45) \cdot 0.9913$$

$$= 24.380 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 20.0^2 \cdot 0.9913 + (0.0)^2 \cdot 0.9913$$

$$= 3.965 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(59.3, 350.0, 60.0)) \cdot (150.0 - 24.0) \cdot 0.991$$

$$= 148.013 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.2458 mm
Wall Thickness per UG16(b),	tr16b = 3.3000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 6.4904 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 6.4904 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.3000 mm
Wall Thickness per table UG-45	tb3 = 9.1312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.131, \max(6.4904, 3.3)]$$

$$= 6.4904 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.2458, 6.4904)$$

$$= 6.4904 \text{ mm}$$

Available Nozzle Neck Thickness = 24.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 16.3,	Allowable	: 136.8 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 327.2 N./mm ²	Passed
Occasional	: 7.0,	Allowable	: 181.9 N./mm ²	Passed
Shear	: 7.2,	Allowable	: 95.8 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 24.0, tr = 2.424, c = 0.8 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.105, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 24.5, tr = 5.69, c = 0.8 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.24, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 624.56/1960.01 = 0.319

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 = (Hub Thickness - Neck Thickness) / tan(30)
 = 126.0/0.5773
 = 218.2385 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1602.6000 mm
 Parallel to Vessel Wall, opening length d 801.3000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.2500 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	40.631	40.631
Area Available in Shell	A1	95.740	95.740
Area Available in Nozzle Wall	A2	24.380	24.380
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	3.965	3.965
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	148.013	148.013
TOTAL AREA AVAILABLE	Atot	272.098	272.098

The area available without a pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

The flange is considered in the moment of inertia calculations.

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

	Area	Distance	Inertia	d	I+Ad²
Nozzle	86.6984	186.8500	10089.6387	27.197	10730.9355
Hub	139.6168	59.2862	2008.1526	100.367	16072.3926
Shell	89.8704	11.8500	42.0661	147.803	19674.8418
Flange	81.9861	463.8267	830.9437	304.174	76685.8281
Sum	398.1717				123163.99

Membrane stress per Appendix 1-7 (1 or 2) Sm 29.65 N./mm²
 Allowable Membrane Stress S 138.00 N./mm²
 Bending stress at the inside of the Nozzle Sb 10.07 N./mm²
 Combined Membrane plus Bending Stress Sm + Sb 39.73 N./mm²
 Allowable Membrane + Bending Stress 1.5 * S 207.00 N./mm²
 M.A.W.P for the Appendix 1-7 Stress calc 2906.53 KPa.

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 31.605 cm²
 Area from Shell = 41.047 cm²
 Total Area, As = 208.273 cm²

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:



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$$= P [R (R_n + t_n + \sqrt{(r_m \cdot t)}) + R_n (t + \sqrt{(R_{nm} \cdot t_n)})] / A_s$$

$$= 624.564 [1253.8 (534.2 + 23.2 + 173.193) + 534.2 (136.228)] / 208.273$$

$$= 29.653 \text{ N./mm}^2$$

The area A_s was computed per figure 1-7-1 for the S_m calculation.

Moment [M]:

$$= (R_n^3 / 6 + R \cdot R_n \cdot e) P$$

$$= (534.2^3 / 6 + 1253.8 \cdot 534.2 \cdot 147.803) / 624.564$$

$$= 7923.408 \text{ Kg-m.}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= (M (e + t / 2)) / I$$

$$= (7923.408 (147.803 + 23.7 / 2)) / 123163.992$$

$$= 10.072 \text{ N./mm}^2$$

Weld Size Calculations, Description: Shell outlet

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * W_o mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (60.9434 - 191.9479 + 2 \cdot 149.2 \cdot 0.9913 \cdot (1.0 \cdot 23.7 - 5.6904)) / 138)$$

$$= \max(0, -109379.39) \text{ Kg}$$

For hub type nozzles, A_2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an} / 707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (172.3926 + 3.9652 - 0.0 \cdot 0.99) \cdot 138$$

$$= 248163.11 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (172.3926 + 0.0 + 3.9652 + (70.1062)) \cdot 138$$

$$= 346813.59 \text{ Kg}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (172.3926 + 0.0 + 3.9652 + 0.0 + (70.1062)) \cdot 138$$

$$= 346813.59 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi / 2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416 / 2.0) \cdot 1366.7999 \cdot 20.0 \cdot 0.49 \cdot 137$$

$$= 293497. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 608.8) \cdot (150.0 - 0.8) \cdot 0.7 \cdot 137$$

$$= 2786400. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:



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$$\begin{aligned}
 &= (\pi/2) \cdot D_o \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 1366.7999 \cdot (12.5 - 0.0) \cdot 0.74 \cdot 138 \\
 &= 279453. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (293497 + 2786400) = 3079897 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (293497 + 0 + 279453 + 0) = 572950 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (293497 + 279453 + 0) = 572950 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 3079896 Kgf, must exceed W = 0 Kgf	or W1 = 248163 Kgf
Path 2-2 = 572949 Kgf, must exceed W = 0 Kgf	or W2 = 346813 Kgf
Path 3-3 = 572949 Kgf, must exceed W = 0 Kgf	or W3 = 346813 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2128 KPa.

The Drop for this Nozzle is : 202.7745 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 577.2745 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 831.54 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1068.40 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: Shell outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.8000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2506.000	mm
Shell Thickness	24.5000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.8000	mm
Nozzle Diameter	1066.800	mm
Nozzle Thickness	150.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 136.80	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



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Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	624.56	KPa.
Radial Load	P	-8030.81	Kgf
Circumferential Shear	Vc	5353.89	Kgf
Longitudinal Shear	Vl	5353.89	Kgf
Circumferential Moment	Mc	15734.27	Kg-m.
Longitudinal Moment	Ml	15734.27	Kg-m.
Torsional Moment	Mt	22477.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.535) is > 0.5
check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (9.112) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1366.8 + 2 \cdot 1.65 \cdot \sqrt{1265.65(24.5 - 0.8)}$$

$$= 1938.338 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2531.300 mm
Nozzle Outside Diameter	(d) =	1366.800 mm
Vessel Thickness used	(T) =	23.700 mm
Nozzle Thickness used	(t) =	149.200 mm
T / t	=	0.159
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.580$$

Nr/P	=	0.042
Mr/P	=	0.069
M0/P	=	0.021
N0/P	=	0.040
MrD/Mc	=	0.243
NrDL/Mc	=	0.089
M0d/Mc	=	0.072
N0DL/Mc	=	0.070
MrD/Ml	=	0.051
NrDL/Ml	=	0.040
M0D/Ml	=	0.016
N0DL/Ml	=	0.035

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-7	-7	7	7
Outplane Bending (Ml)	-19	19	19	-19
Normal Pressure Stress	32	33	32	33



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Outplane Stress Summary	29	33	82	8
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-61	61	61	-61
Inplane Pressure Stress	16	16	16	16

Inplane Stress Summary	10	17	149	-89
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	30	34	149	98

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-86	86	86	-86
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	-60	76	139	-67
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	-17	-17	17	17
Inplane Bending (Mc)	-293	293	293	-293
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32	33	32	33

Inplane Stress Summary	-214	256	407	-293
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		3	3	3	3
Shear Stress Summary		2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		214	256	407	294

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the hoop direction)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)		5	5	5	5
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		-7	-7	7	7
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		2	3	2	3
Hoop Stress Summary		0	1	15	15

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)		0	0	0	0
Axial Bending (P)		1	-1	1	-1
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		0	0	0	0
Axial Bending (ML)		0	0	0	0
Axial Pressure Stress		1	1	1	1
Axial Stress Summary		0	0	4	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress)		Outside	Inside	Outside	Inside
Shear due to (Vc)		0	0	0	0
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		0	0	0	0
Shear Stress Summary		0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Outside	Inside	Outside	Inside
Two * Max Shear Stress		1	1	15	15

Nozzle Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3
Hoop Stress Summary	-6	-5	22	22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-6	6	6	-6
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-4	5	9	-4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	6	10	22	27

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	33	32	33	32	33	32	33
Circ. Pl (SUS)	-2	-2	12	12	-12	-12	22	22
Circ. Q (SUS)	-2	2	36	-36	-235	235	351	-351
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-2	-2	12	12	-9	-9	19	19
Long. Q (SUS)	-3	3	119	-119	-69	69	103	-103
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	32.0	33.0	32.0	33.0	32.0	33.0	32.0	33.0



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Pm+Pl (SUS)	30.1	31.1	44.1	45.1	20.1	21.1	54.1	55.0
Pm+Pl+Q (Total)	28.9	33.9	147.1	100.1	215.0	256.0	405.1	296.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	55.05	207.00	Passed
Pm+Pl+Q (TOTAL)	405.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	3	2	3	2	3	2	3
Circ. Pl (SUS)	-2	-2	12	12	-9	-9	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	1	-1	1	-1	-5	5	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Pm+Pl (SUS)	1.0	1.0	14.0	15.0	8.0	7.0	21.0	22.0
Pm+Pl+Q (Total)	2.0	1.0	14.0	15.0	7.0	12.0	21.0	28.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	136.80	Passed
Pm+Pl (SUS)	22.00	205.20	Passed
Pm+Pl+Q (TOTAL)	28.00	410.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell inlet

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Input, Nozzle Desc: Shell inlet

From: 70

Pressure for Reinforcement Calculations	P	600.000	KPa.
Temperature for Internal Pressure	Temp	230	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2506.00	mm
Shell Finished (Minimum) Thickness	t	24.5000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5415.96	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.80	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		42.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm
Flange Material		SA-266 2	
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	72.1687	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	0.8000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.5000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.



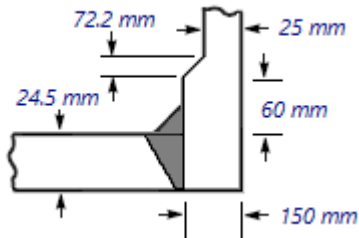
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Nozzle Calcs.: Shell inlet

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Reinforcement CALCULATION, Description: Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	42.000 in.
Actual Thickness Used in Calculation	0.984 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 600.0 \cdot 1253.8 / (138 \cdot 1.0 - 0.6 \cdot 600.0)$
 $= 5.4660 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 600.0 \cdot 534.2 / (137 \cdot 1.0 - 0.6 \cdot 600.0)$
 $= 2.3493 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 125.0 / 0.5773$
 $= 216.5064 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	2136.8000 mm
Parallel to Vessel Wall, opening length	d	1068.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.2500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 136.8 / 138.0)$
 $= 0.991$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 136.8 / 138.0)$
 $= 0.991$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.991, 1.0)$
 $= 0.991$



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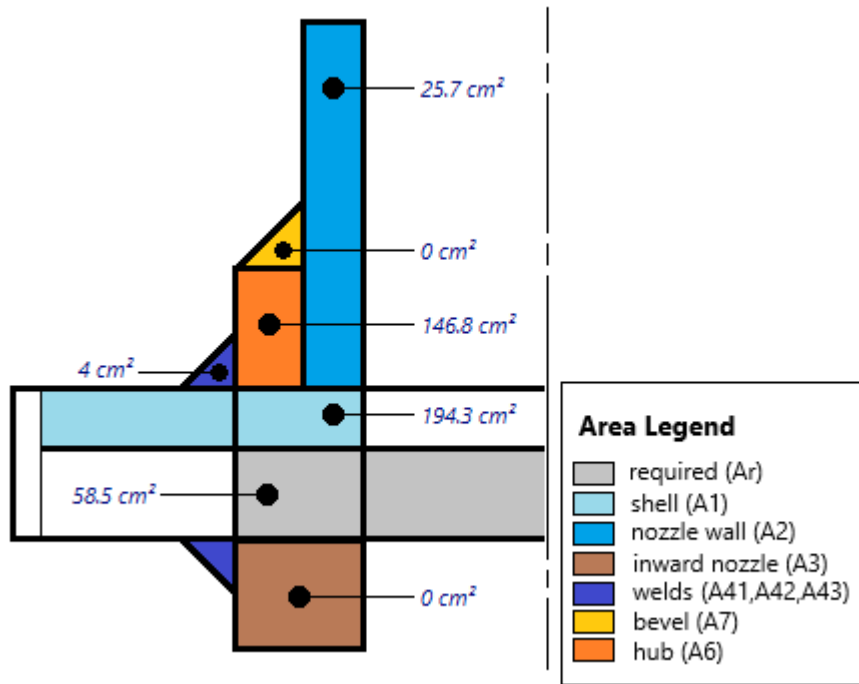
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Nozzle Calcs.: Shell inlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	58.540	NA	NA
Area in Shell	A1	194.339	NA	NA
Area in Nozzle Wall	A2	25.668	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	3.965	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	146.838	NA	NA
TOTAL AREA AVAILABLE	Atot	370.811	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (1068.4 \cdot 5.466 \cdot 1.0 + 2 \cdot 149.2 \cdot 5.466 \cdot 1.0 \cdot (1 - 0.99))$$

$$= 58.540 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 1068.4(1.0 \cdot 23.7 - 1.0 \cdot 5.466) - 2 \cdot 149.2$$

$$(1.0 \cdot 23.7 - 1.0 \cdot 5.466) \cdot (1 - 0.991)$$

$$= 194.339 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 59.25)(24.2 - 2.35) \cdot 0.9913$$

$$= 25.668 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 20.0^2 \cdot 0.9913 + (0.0)^2 \cdot 0.9913$$

$$= 3.965 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(59.3, 350.0, 60.0)) \cdot (150.0 - 25.0) \cdot 0.991$$

$$= 146.838 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.1493 mm
Wall Thickness per UG16(b),	tr16b = 3.3000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 6.2660 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 6.2660 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.3000 mm
Wall Thickness per table UG-45	tb3 = 9.1312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.131, \max(6.266, 3.3)]$$

$$= 6.2660 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.1493, 6.266)$$

$$= 6.2660 \text{ mm}$$

Available Nozzle Neck Thickness = 25.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 15.4,	Allowable	: 136.8 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 328.1 N./mm ²	Passed
Occasional	: 6.5,	Allowable	: 181.9 N./mm ²	Passed
Shear	: 6.9,	Allowable	: 95.8 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 25.0, tr = 2.329, c = 0.8 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.096, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 24.5, tr = 5.466, c = 0.8 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.231, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 600.00/1960.01 = 0.306

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 125.0/0.5773
= 216.5064 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1602.6000 mm
Parallel to Vessel Wall, opening length d 801.3000 mm
Normal to Vessel Wall (Thickness Limit), no pad Tlnp 59.2500 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		39.029	39.029
Area Available in Shell A1		96.933	96.933
Area Available in Nozzle Wall A2		25.668	25.668
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		3.965	3.965
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		146.838	146.838
TOTAL AREA AVAILABLE Atot		273.404	273.404

The area available without a pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

The flange is considered in the moment of inertia calculations.

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

	Area	Distance	Inertia	d	I+Ad²
Nozzle	90.4354	186.8500	10524.5371	23.790	11036.3779
Hub	120.1054	50.3007	1178.5485	112.759	16449.4609
Shell	89.8704	11.8500	42.0661	151.210	20590.3750
Flange	83.0999	463.6713	844.8311	300.612	75939.8594
Sum	383.5111				124016.08

Membrane stress per Appendix 1-7 (1 or 2) Sm 30.55 N./mm²
Allowable Membrane Stress S 138.00 N./mm²
Bending stress at the inside of the Nozzle Sb 9.99 N./mm²
Combined Membrane plus Bending Stress Sm + Sb 40.54 N./mm²
Allowable Membrane + Bending Stress 1.5 * S 207.00 N./mm²
M.A.W.P for the Appendix 1-7 Stress calc 2710.26 KPa.

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 33.561 cm²
Area from Shell = 41.047 cm²
Total Area, As = 194.713 cm²

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:



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$$= P \left[R \left(R_n + t_n + \sqrt{(r_m \cdot t)} \right) + R_n \left(t + \sqrt{(R_{nm} \cdot t_n)} \right) \right] / A_s$$

$$= 600.0 \left[1253.8 \left(534.2 + 24.2 + 173.193 \right) + 534.2 \left(138.68 \right) \right] / 194.713$$

$$= 30.55 \text{ N./mm}^2$$

The area A_s was computed per figure 1-7-1 for the S_m calculation.

Moment [M]:

$$= (R_n^3/6 + R \cdot R_n \cdot e) P$$

$$= (534.2^3/6 + 1253.8 \cdot 534.2 \cdot 151.21) 600.0$$

$$= 7751.406 \text{ Kg-m.}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= (M (e + t/2)) / I$$

$$= (7751.406 (151.21 + 23.7/2)) / 124016.078$$

$$= 9.995 \text{ N./mm}^2$$

Weld Size Calculations, Description: Shell inlet

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * W _o mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) S_v)$$

$$= \max(0, (58.5403 - 194.3395 + 2 \cdot 149.2 \cdot 0.9913 \cdot (1.0 \cdot 23.7 - 5.466)) 138)$$

$$= \max(0, -115192.28) \text{ Kg}$$

For hub type nozzles, A_2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an}/707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (172.5059 + 3.9652 - 0.0 \cdot 0.99) \cdot 138$$

$$= 248322.52 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (172.5059 + 0.0 + 3.9652 + (70.1062)) \cdot 138$$

$$= 346973.00 \text{ Kg}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (172.5059 + 0.0 + 3.9652 + 0.0 + (70.1062)) \cdot 138$$

$$= 346973.00 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 1366.7999 \cdot 20.0 \cdot 0.49 \cdot 137$$

$$= 293497. \text{ Kg}$$

Shear, Nozzle Wall [S_{nw}]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 608.8) \cdot (150.0 - 0.8) \cdot 0.7 \cdot 137$$

$$= 2786400. \text{ Kg}$$

Tension, Shell Groove Weld [T_{ngw}]:



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$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 1366.7999 \cdot (12.5 - 0.0) \cdot 0.74 \cdot 138 \\
 &= 279453. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (293497 + 2786400) = 3079897 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (293497 + 0 + 279453 + 0) = 572950 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (293497 + 279453 + 0) = 572950 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 3079896 Kgf, must exceed W = 0 Kgf	or W1 = 248322 Kgf
Path 2-2 = 572949 Kgf, must exceed W = 0 Kgf	or W2 = 346973 Kgf
Path 3-3 = 572949 Kgf, must exceed W = 0 Kgf	or W3 = 346973 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2152 KPa.

The Drop for this Nozzle is : 202.7745 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 577.2745 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1068.40 mm

Warning:
Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: Shell inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.8000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2506.000	mm
Shell Thickness	24.5000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.8000	mm
Nozzle Diameter	1066.800	mm
Nozzle Thickness	150.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 136.80	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



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Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	600.00	KPa.
Radial Load	P	-8030.81	Kgf
Circumferential Shear	Vc	5353.89	Kgf
Longitudinal Shear	Vl	5353.89	Kgf
Circumferential Moment	Mc	15734.27	Kg-m.
Longitudinal Moment	Ml	15734.27	Kg-m.
Torsional Moment	Mt	22477.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.535) is > 0.5
check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (9.112) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1366.8 + 2 \cdot 1.65 \cdot \sqrt{1265.65(24.5 - 0.8)}$$

$$= 1938.338 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2531.300 mm
Nozzle Outside Diameter	(d) =	1366.800 mm
Vessel Thickness used	(T) =	23.700 mm
Nozzle Thickness used	(t) =	149.200 mm
T / t	=	0.159
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.580$$

Nr/P	=	0.042
Mr/P	=	0.069
M0/P	=	0.021
N0/P	=	0.040
MrD/Mc	=	0.243
NrDL/Mc	=	0.089
M0d/Mc	=	0.072
N0DL/Mc	=	0.070
MrD/Ml	=	0.051
NrDL/Ml	=	0.040
M0D/Ml	=	0.016
N0DL/Ml	=	0.035

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-7	-7	7	7
Outplane Bending (Ml)	-19	19	19	-19
Normal Pressure Stress	31	32	31	32



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Outplane Stress Summary	28	32	81	7
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-61	61	61	-61
Inplane Pressure Stress	15	15	15	15

Inplane Stress Summary	10	17	149	-89
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	29	33	149	97
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-86	86	86	-86
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	15	15	15	15

Outplane Stress Summary	-61	75	139	-68
-------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	-17	-17	17	17
Inplane Bending (Mc)	-293	293	293	-293
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	31	32	31	32

Inplane Stress Summary	-215	255	406	-295
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		3	3	3	3
Shear Stress Summary		2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		215	255	406	295

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the hoop direction)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)		5	5	5	5
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		-7	-7	7	7
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		2	2	2	2
Hoop Stress Summary		0	1	14	15

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)		0	0	0	0
Axial Bending (P)		1	-1	1	-1
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		0	0	0	0
Axial Bending (ML)		0	0	0	0
Axial Pressure Stress		1	1	1	1
Axial Stress Summary		0	0	4	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress)		Outside	Inside	Outside	Inside
Shear due to (Vc)		0	0	0	0
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		0	0	0	0
Shear Stress Summary		0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Outside	Inside	Outside	Inside
Two * Max Shear Stress		1	1	14	15

Nozzle Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	-6	-5	21	22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-6	6	6	-6
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-4	5	9	-4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	6	10	22	27

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	32	31	32	31	32	31	32
Circ. Pl (SUS)	-2	-2	12	12	-12	-12	22	22
Circ. Q (SUS)	-2	2	36	-36	-235	235	351	-351
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-2	-2	12	12	-9	-9	19	19
Long. Q (SUS)	-3	3	119	-119	-69	69	103	-103
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	31.0	32.0	31.0	32.0	31.0	32.0	31.0	32.0



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Pm+Pl (SUS)	29.1	30.1	43.1	44.1	19.1	20.1	53.1	54.0
Pm+Pl+Q (Total)	27.9	32.9	146.1	100.1	216.0	255.0	404.1	297.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	138.00	Passed
Pm+Pl (SUS)	54.05	207.00	Passed
Pm+Pl+Q (TOTAL)	404.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	-2	-2	12	12	-9	-9	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	1	-1	1	-1	-5	5	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	1.0	1.0	14.0	14.0	8.0	8.0	21.0	21.0
Pm+Pl+Q (Total)	2.0	0.0	14.0	14.0	7.0	13.0	21.0	27.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	136.80	Passed
Pm+Pl (SUS)	21.00	205.20	Passed
Pm+Pl+Q (TOTAL)	27.00	410.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell-1

Shell Maximum Design Pressure	Psd,max	600.00 KPa.
Shell Maximum Operating Pressure	Psox,max	600.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	14.0000 mm
Shell Internal Corrosion Allowance	cas	0.8000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2506.000 mm
Shell Temperature for Internal Pressure	Ts	230.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2500.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4424
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6100.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5850.00 mm
Design Temperature of the Tubes		82.00 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		187.00 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	437.34 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1124.300 mm
Largest Center-to-Center Tube Distance	U1	41.2750 mm



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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	38519.9 cm ²
Perimeter of Tube Layout	Cp	7243.0 mm

Tubesheet Data:

Tubesheet TYPE: Stationary, Gasketed both Sides, Conf. d
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	82.00 °C
Tubesheet Material		SA-182 F60
Tubesheet Material UNS Number		S32205
Tubesheet Allowable Stress at Temperature	S	187.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	125.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	2578.000 mm
Dimension G for the Channel Side	Gc	2553.285 mm
Area of the Untubed Lanes	AL	595.3 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	2900.000 mm
Flange Inside Diameter	B	2510.000 mm
Flange Face Outside Diameter	Fod	2581.000 mm
Flange Face Inside Diameter	Fid	2510.000 mm
Gasket Outside Diameter	Go	2570.000 mm
Gasket Inside Diameter	Gi	2526.000 mm
Small end Hub thk.	g0	12.5000 mm
Large end Hub thk.	g1	19.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	2535.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:

Diameter of Bolt Circle	C	2750.000 mm
Nominal Bolt Diameter	dB	57.1500 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	36
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Alternate Flange Operating Bolt Load, Wm1	584935.19 Kg
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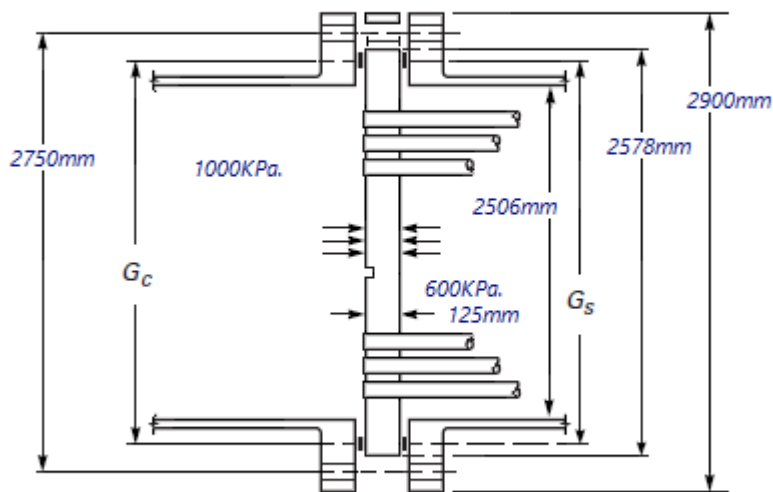
Alternate Flange Seating Bolt Load, Wm2 560459.62 Kgf
Alternate Flange Design Bolt Load, W 1394340.25 Kgf

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with None
Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	230.0 °C	0.19046E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1
Tubes - TM-1 Material Group H



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Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19529E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20179E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	82.0 °C	0.19529E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1
TubeSheet - TM-1 Material Group H

Elastic Mod. at Design Temperature	82.0 °C	0.19529E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1000.0 \cdot (25.4/2 - 0)) / (187.0 \cdot 1.0 + 0.4 \cdot 1000.0)$$

$$= 0.0678 + 0.0000 = 0.0678 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-5 at 82.00 °C
Elastic Modulus for Material 190996000.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5850.00	12.04	50.0000	0.0075908	182.68
EMAWP = (4*B)/(3*(D/T)) = (4 *182.675)/(3 *12.0379) = 20232.123 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4125	25.40	5850.00	61.57	50.0000	0.0002901	27.71
EMAWP = (4*B)/(3*(D/T)) = (4 *27.7098)/(3 *61.5728) = 600.0093 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4125 mm
Allowable Internal Pressure at Corroded thickness	33277.66 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	20232.12 KPa.
Required External Design Pressure	600.00 KPa.
Required Thickness due to Shell Side pressure	0.4125 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	N = (Goc-Gic) / 2	22.000 mm
Basic Gasket Width,	b0 = N / 2	11.000 mm
Effective Gasket Width,	b = sqrt(b0) * 2.5	8.355 mm
Gasket Reaction Diameter,	G = Go-2.0*b	2553.291 mm

Results for ASME Stationary Tubesheet Calculations for Configuration d:

Results for Tubesheet Calculations Original Thickness:



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ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$= l_{tx} / h$ (modified for corrosion if present)

$= 0/125.0$

$= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:

$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$

$= \max(25.4 - 2 \cdot 2.11 (0.19529E+09/.19529E+09) \cdot$
 $(186/186) \cdot (0); 25.4 - 2 \cdot 2.11)$

$= 25.4000$ mm

Equivalent outer tube limit circle diameter $[Do]$:

$= 2 \cdot r_o + dt$

$= 2 \cdot 1124.3 + 25.4$

$= 2274.000$ mm

Determine the basic ligament efficiency for shear $[\mu]$:

$= (p - dt) / p$

$= (31.75 - 25.4) / 31.75$

$= 0.200$

Equivalent outer tube limit radius $[ao]$:

$= Do / 2$

$= 2274.0 / 2$

$= 1137.0000$ mm

Effective tube pitch $[p^*]$:

$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$

$= 31.75 / \sqrt{ (1 - 4 \cdot \min(595.31 \cdot 100.0 ; 4 \cdot 2274.0 \cdot 31.75) / (\pi \cdot 2274.0^2)) }$

$= 31.9853$ mm

Effective ligament efficiency for bending $[\mu^*]$:

$= (p^* - d^*) / p^*$

$= (31.9853 - 25.4) / 31.9853$

$= 0.2059$

Ratio $[ps]$:

$= as / ao$

$= 1276.6425 / 1137.0$

$= 1.1228$

Ratio $[pc]$:

$= ac / ao$

$= 1276.6425 / 1137.0$

$= 1.1228$

Compute parameter $[xt]$:

$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$

$= 1 - 4424 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1137.0))^2$

$= 0.6162$

Determine parameter $[xs]$:

$= 1 - Nt(dt / (2 \cdot ao))^2$

$= 1 - 4424 (25.4 / (2 \cdot 1137.0))^2$

$= 0.4480$

Determine the value $[h'g]$:

$= \max((hg - CATC); 0)$ (For pressure only cases)

$= \max((0 - 0); 0)$

$= 0.000$ mm

ASME Section VIII-2 Part 4.18.9.4 Step 2:



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Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 3.937008$; $\mu^* = 0.205885$; $E^* = 31699788$ KPa.

$E^*/E = 0.162318$; $\nu^* = 0.450106$

Note: As h/p (3.937) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:

$$= \left(\left(24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.45^2)4424 \cdot 0.19529E+09 \cdot 2.11 \right. \right.$$

$$\left. \left(25.4 - 2.11 \right) 1137.0^2 \right) / (31699788 \cdot 5850.0 \cdot 125.0^3) \right)^{1/4}$$

$$= 7.3385$$

Values from Table 4.18.3

$$Z_d = 0.003738 ; Z_v = 0.018464 ; Z_m = 0.193274$$

$$Z_a = 0.457704E+03 ; Z_w = 0.018464$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio $[K]$:

$$= A / Do$$

$$= 2578.0/2274.0$$

$$= 1.1337$$

Compute coefficient $[F]$:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_C + E \cdot \ln(K))$$

$$= (1 - 0.45) / 31699788 \cdot (0 + 0 +$$

$$0.19529E+09 \cdot \ln(1.13))$$

$$= 0.4251$$

Compute parameter $[\phi]$:

$$= (1 + \nu^*)F$$

$$= (1 + 0.4501) \cdot 0.4251$$

$$= 0.6164$$

Compute parameter $[Q1]$:

$$= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.1228 - 1 - 0.6164 \cdot 0.0185) / (1 + 0.6164 \cdot 0.1933)$$

$$= 0.099573$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor $[\gamma_b]$:

$$= (G_c - G_s) / Do \text{ (config d)}$$

$$= (2553.2849 - 2553.2849) / 2274.0 = 0$$

Calculate parameter $[\omega_s]$:

$$= ps \cdot ks \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.1228 \cdot 0 \cdot 0 \cdot 0 (1 + 125.0 \cdot 0)$$

$$= 0.0000 \text{ mm}^2$$

Calculate parameter $[\omega^*s]$:

$$= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega_s$$

$$= 1137.0^2(1.123^2 - 1) \cdot (1.123 - 1) / 4 - 0$$

$$= 10348.7168 \text{ mm}^2$$

Parameter $[\omega_c]$:

$$= pc \cdot kc[\delta_C (Cm\theta + \beta_C + Cq\Delta \cdot h \cdot \beta_C^2) -$$



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$$\begin{aligned} & \zeta c(C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2) \\ & = 1.123 \cdot 0 [0 (0 + 0 \cdot 0 \cdot 125.0) - \\ & \quad 0 (0/125.0 + 0 \cdot 0/2)] \\ & = 0.0000 \text{ mm}^2 \end{aligned}$$

Parameter [$\omega \cdot c$]:

$$\begin{aligned} & = a o^2 [(p c^2 + 1) \cdot (p c - 1) / 4 - (p s - 1) / 2] - \omega c \\ & = 1137.2 [(1.12282^2 + 1) \cdot (1.12282 - 1) / 4 - (1.12282 - 1) / 2] - 0 \\ & = 10348.7236 \text{ mm}^2 \end{aligned}$$

Pressure [$P \cdot S$]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [$P \cdot C$]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 600.0 - 1000.0 = -400.0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q_2]:

$$\begin{aligned} & = [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + \\ & \quad W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\ & = [((10348.717 \cdot 600.0 - 10348.724 \cdot 1000.0) - \\ & \quad (0 \cdot 0 - 0 \cdot 0)) \cdot 0. + \\ & \quad 1394340 \cdot 0 / (2 \cdot \pi)] / (1 + 0.6164 \cdot 0.19327) \\ & = -377.1912 \text{ Kg} \end{aligned}$$

Calculate factor [Q_3]:

$$\begin{aligned} & = Q_1 + 2 \cdot Q_2 / (P_e \cdot a o^2) \\ & = 0.1 + 2 \cdot -377.191 / (-400.0 \cdot 1137.0^2) \\ & = 0.11388 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.077858

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} & = (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a o / (H - h'g))^2 \cdot P_e \\ & = (1.5 \cdot 0.0779 / 0.2059) \cdot (2 \cdot 1137.0 / (125.0 - 0))^2 \cdot -400.0 \\ & = -75.0964 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 187.0 = 280.49 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned} & = (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a o / (h - h'g))^2 \cdot P_e \\ & = (1.5 \cdot 0.0721 / 0.2059) \cdot (2 \cdot 1137.0 / (62.23 - 0))^2 \cdot -400.0 \\ & = 280.4949 \text{ N./mm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} & = h + C_{ats} + C_{atc} \\ & = 62.2304 + 0 + 0 \\ & = 62.2304 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [τ_{limit}]:

$$\begin{aligned} & = (2 \cdot \mu \cdot h / a o) \cdot \min(0.8S; 0.533S_y) \\ & = (2 \cdot 0.2 \cdot 125.0 / 1137.0) \cdot 149.597 \\ & = 6.5786 \text{ N./mm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}



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Tubesheet Shear Stress - final thickness [tf]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (38519.9 \cdot -400.0) / (0.2 \cdot 7.111 \cdot 7243.0)$$

$$= 149.5973 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 7.1105 + 0 + 0$$

$$= 7.1105 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$= \max(H_{reqB}, H_{reqS})$$

$$= \max(62.2304; 7.1105)$$

$$= 62.2304 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = -1.2009, F_{tmax} = 8.4688$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((600.0 \cdot 0.448 - 1000.0 \cdot 0.6162) - (-400.0) \cdot -1.201) /$$

$$(0.6162 - 0.448)$$

$$= -4.9224 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((600.0 \cdot 0.448 - 1000.0 \cdot 0.6162) - (-400.0) \cdot 8.469) /$$

$$(0.6162 - 0.448)$$

$$= 18.0788 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 18.079 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 186.9966 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.004 + 0.114 \cdot 0.018)7.338^4); 1.25)$$

$$= 1.2500 \text{ (Should be } \leq 2)$$

Determine the factor [rt]:

$$= (\sqrt{d_t^2 + (d_t - 2 \cdot t_t)^2}) / 4$$

$$= (\sqrt{25.4^2 + (25.4 - 2 \cdot 2.11)^2}) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{yt})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19529E+09 / 437)}$$

$$= 93.8884$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 0 \cdot 0 / 8.268$$

$$= 0.0000$$

Buckling Allowable Stress [Stb]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (437 / 1.25) \cdot (1 - 0 / (2 \cdot 93.888))$$

$$= 186.997 \text{ N./mm}^2 \text{ (Never greater than } S_{ot})$$



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Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 18.08 \cdot 1.5438$$

$$= 284.61 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 186.997$$

$$= 2943.791 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 187.0$$

$$= 2880.8000 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 186.997 / 186.997$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 187.0$$

$$= 1864.0472 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1.0 \cdot 1.0} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 2943.79 is $\geq$ tube strength 2943.79 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Computations completed for ASME tubesheet configuration: d

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	75.1 $\leq$	280.5 N./mm ²	Pass
Tubesheet Shear Stress	8.5 $\leq$	149.6 N./mm ²	Pass
Maximum Stress in the Tubes	18.1 $\leq$	187.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-4.9 $\leq$	-187.0 N./mm ²	Pass
Maximum Force on any one Tube	284.6 $\leq$	2943.8 Kgf	Pass



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Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	62.230	125.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	100.652	4.222	188	280	21	150 Ps+Pt-Th D1 Ok
D2uc	76.875	2.533	113	280	13	150 Ps+Pt-Th D2 Ok
D3uc	62.230	1.689	75	280	9	150 Ps+Pt-Th D3 Ok
D1c	116.669	4.222	242	280	24	150 Ps+Pt-Th-c D1 Ok
D2c	92.876	2.533	145	280	15	150 Ps+Pt-Th-c D2 Ok
D3c	78.231	1.689	97	280	10	150 Ps+Pt-Th-c D3 Ok
Max:	116.6689	4.222 mm	0.862	0.163	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+)-Th]:
With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
With or Without Corrosion Allowance

[D1, D2, D3]:
Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	Ten	---- Tube Stresses			Tube Loads		Shell Stress		Channel Stress		Pass
		Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	47	186	-11	-186	735	2943	...	...	...	...	Ok
D2uc	29	186	-29	-186	451	2943	...	...	...	...	Ok
D3uc	18	186	-5	-186	285	2943	...	...	...	...	Ok
D1c	55	186	-12	-186	862	2943	...	...	...	...	Ok
D2c	33	186	-33	-186	527	2943	...	...	...	...	Ok
D3c	21	186	-5	-186	335	2943	...	...	...	...	Ok
Max RATIO		0.293		0.179		0.293		...		...	

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
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Tubesheet thickness :	116.669	125.000	Ok
Tube thickness :	0.413	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	1160.35	1.000	1160.35	1.000
Tubesheet Shear Stress	6131.75	1.000	6131.75	1.000
Tube Tensile Stress	3414.94	1.000	3353.69	1.000
Tube Compressive Stress	15694.75	1.000	3353.69	1.000
Tube-Tubesheet Joint load	3414.94	1.000	3353.69	1.000
Tube Pressure Stress	33277.65	1.000	20232.11	1.000
Minimum MAWP	1160.35		1160.35	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	1494.04	1.000	1494.04	1.000
Tubesheet Shear Stress	7031.83	1.000	7031.83	1.000
Tube Tensile Stress	4004.44	1.000	3920.48	1.000
Tube Compressive Stress	17304.96	1.000	3920.48	1.000
Tube-Tubesheet Joint load	4004.44	1.000	3920.48	1.000
Tube Pressure Stress	33277.65	1.000	22169.73	1.000
Minimum MAPnc	1494.04		1494.04	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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Input Echo, Tubesheet Number 1, Description: Floating tubesh

Shell Data:

Main Shell Description: Shell-1

Shell Maximum Design Pressure	Psd,max	600.00 KPa.
Shell Maximum Operating Pressure	Psox,max	600.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	14.0000 mm
Shell Internal Corrosion Allowance	cas	0.8000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2506.000 mm
Shell Temperature for Internal Pressure	Ts	230.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2500.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4424
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6100.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5850.00 mm
Design Temperature of the Tubes		82.00 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		187.00 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	437.34 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1124.300 mm
Largest Center-to-Center Tube Distance	U1	41.2750 mm



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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	38519.9 cm ²
Perimeter of Tube Layout	Cp	7243.0 mm

Tubesheet Data:

Tubesheet TYPE: Floating, Tubesheet Gasketed, Conf. B
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	82.00 °C
Tubesheet Material		SA-182 F60
Tubesheet Material UNS Number		S32205
Tubesheet Allowable Stress at Temperature	S	187.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	125.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2482.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		No
Area of the Untubed Lanes	AL	595.3 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	2482.000 mm
Flange Inside Diameter	B	2288.000 mm
Flange Face Outside Diameter	Fod	2363.000 mm
Flange Face Inside Diameter	Fid	2288.000 mm
Gasket Outside Diameter	Go	2360.000 mm
Gasket Inside Diameter	Gi	2320.000 mm
Gasket Factor,	m	4.25
Gasket Design Seating Stress	y	69.64 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	4.0000 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2405.000 mm
Nominal Bolt Diameter	dB	34.9250 mm
Thread Series		UNC Thread Series
Number of Bolts	n	44
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	109.0000 mm
Is Bolt Load Transferred to the Tubesheet		Yes

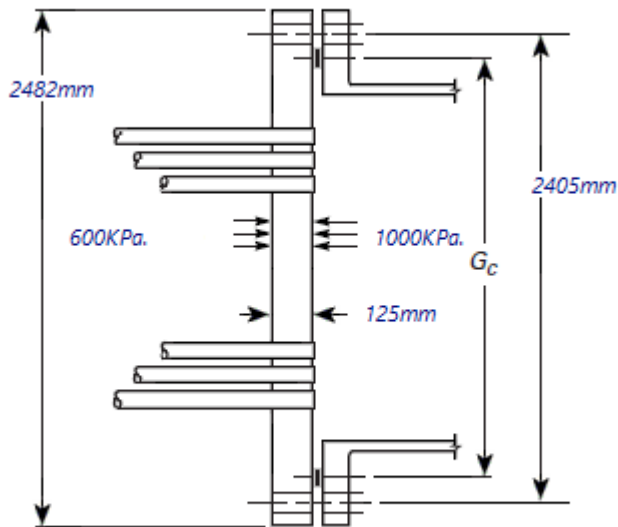
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No
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Tubesheet Gasketed, Extended as a Flange



Configuration B:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	230.0 °C	0.19046E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1
Tubes - TM-1 Material Group H

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19529E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20179E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	82.0 °C	0.19529E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1
TubeSheet - TM-1 Material Group H



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Elastic Mod. at Design Temperature	82.0 °C	0.19529E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(1000.0 \cdot (25.4/2 - 0)) / (187.0 \cdot 1.0 + 0.4 \cdot 1000.0)$

= $0.0678 + 0.0000 = 0.0678$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-5	at	82.00 °C
Elastic Modulus for Material			190996000.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5850.00	12.04	50.0000	0.0075908	182.68

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 182.675) / (3 \cdot 12.0379) = 20232.123$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4125	25.40	5850.00	61.57	50.0000	0.0002901	27.71

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 27.7098) / (3 \cdot 61.5728) = 600.0093$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4125 mm
Allowable Internal Pressure at Corroded thickness	33277.66 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	20232.12 KPa.
Required External Design Pressure	600.00 KPa.
Required Thickness due to Shell Side pressure	0.4125 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	20.000 mm
Basic Gasket Width,	$b_0 = N / 2$	10.000 mm
Effective Gasket Width,	$b = \sqrt{b_0} \cdot 2.5$	7.966 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 \cdot b$	2344.068 mm

Results for ASME Floating Tubesheet Calculations for Configuration B:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio $[\rho]$:

= l_{tx} / h (modified for corrosion if present)

= 0/125.0

= 0.0000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:



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$$\begin{aligned}
 &= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt) \\
 &= \max(25.4 - 2 \cdot 2.11 (0.19529E+09/.19529E+09) \cdot \\
 &\quad (186/186) \cdot (0); 25.4 - 2 \cdot 2.11) \\
 &= 25.4000 \text{ mm}
 \end{aligned}$$

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot ro + dt \\
 &= 2 \cdot 1124.3 + 25.4 \\
 &= 2274.000 \text{ mm}
 \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned}
 &= (p - dt) / p \\
 &= (31.75 - 25.4) / 31.75 \\
 &= 0.200
 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned}
 &= Do / 2 \\
 &= 2274.0 / 2 \\
 &= 1137.0000 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned}
 &= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) } \\
 &= 31.75 / \sqrt{ 1 - 4 \cdot \min(595.31 \cdot 100.0 ; 4 \cdot 2274.0 \cdot 31.75) / (\pi \cdot 2274.0^2) } \\
 &= 31.9853 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ *]:

$$\begin{aligned}
 &= (p* - d*) / p* \\
 &= (31.9853 - 25.4) / 31.9853 \\
 &= 0.2059
 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned}
 &= as / ao \\
 &= 1172.0342 / 1137.0 \\
 &= 1.0308
 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned}
 &= ac / ao \\
 &= 1172.0342 / 1137.0 \\
 &= 1.0308
 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned}
 &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\
 &= 1 - 4424 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1137.0))^2 \\
 &= 0.6162
 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned}
 &= 1 - Nt(dt / (2 \cdot ao))^2 \\
 &= 1 - 4424 (25.4 / (2 \cdot 1137.0))^2 \\
 &= 0.4480
 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned}
 &= \max((hg - CATC); 0) \text{ (For pressure only cases)} \\
 &= \max((0 - 0); 0) \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:



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ASME Fl-TS Calc: DESIGN CASE

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$h/p = 3.937008$; $\mu^* = 0.205885$; $E^* = 31699788$ KPa.
 $E^*/E = 0.162318$; $\nu^* = 0.450106$

Note: As h/p (3.937) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= \left(\left(24(1 - \nu^*)Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.45^2)4424 \cdot 0.19529E+09 \cdot 2.11 \right. \right. \\ \left. \left. (25.4 - 2.11)1137.0^2 \right) / (31699788 \cdot 5850.0 \cdot 125.0^3) \right)^{1/4}$$

$$= 7.3385$$

Values from Table 4.18.3

$Z_d = 0.003738$; $Z_v = 0.018464$; $Z_m = 0.193274$
 $Z_a = 0.457704E+03$; $Z_w = 0.018464$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 2482.0/2274.0$$

$$= 1.0915$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.45) / 31699788 \cdot (0 + 0 +$$

$$0.19529E+09 \cdot \ln(1.09))$$

$$= 0.2965$$

Compute parameter [ϕ]:

$$= (1 + \nu^*)F$$

$$= (1 + 0.4501) 0.2965$$

$$= 0.4300$$

Compute parameter [Q1]:

$$= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.0308 - 1 - 0.43 \cdot 0.0185) / (1 + 0.43 \cdot 0.1933)$$

$$= 0.021119$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [γ_b]:

$$= (G_c - C) / Do \text{ (config b)}$$

$$= (2344.0684 - 2405.0) / 2274.0 = -0.02679$$

Calculate parameter [ω_s]:

$$= ps \cdot ks \cdot \beta_s \cdot \delta_s(1 + h \cdot \beta_s)$$

$$= 1.0308 \cdot 0 \cdot 0 \cdot 0(1 + 125.0 \cdot 0)$$

$$= 0.0000 \text{ mm}^2$$

Calculate parameter [ω^*s]:

$$= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega_s$$

$$= 1137.0^2(1.031^2 - 1) \cdot (1.031 - 1) / 4 - 0$$

$$= 623.1480 \text{ mm}^2$$

Parameter [ω_c]:

$$= pc \cdot kc[\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2))$$

$$= 1.031 \cdot 0 [0(0 + 0 \cdot 0 \cdot 125.0) -$$

$$0(0/125.0 + 0 \cdot 0/2))$$

$$= 0.0000 \text{ mm}^2$$

Parameter [ω^*c]:

$$= ao^2[(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega_c$$

$$= 1137.2^2[(1.03081^2 + 1) \cdot (1.03081 - 1) / 4 - (1.03081 - 1) / 2] - 0$$

$$= 623.1490 \text{ mm}^2$$



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ASME Fl-TS Calc: DESIGN CASE

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Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$P_e = P_s - P_t$

= 600.0 - 1000.0 = -400.0 KPa.

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((\omega_s \cdot P_s - \omega_c \cdot P_t) - (\omega_s \cdot P_s - \omega_c \cdot P_c)) + W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((623.148 \cdot 600.0 - 623.149 \cdot 1000.0) - (0 \cdot 0 - 0 \cdot 0)) \cdot 0. + 524750 \cdot -0.027 / (2 \cdot \pi)] / (1 + 0.42997 \cdot 0.19327)$$

= -2089.5923 Kgf

Calculate factor [Q3]:

$$= Q_1 + 2 \cdot Q_2 / (P_e \cdot a_o^2)$$

$$= 0.021 + 2 \cdot -2089.592 / (-400.0 \cdot 1137.0^2)$$

= 0.10037

Fm Value from Table 4.18.3 = 0.072033

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.072 / 0.2059) \cdot (2 \cdot 1137.0 / (125.0 - 0))^2 \cdot -400.0$$

= -69.4778 N./mm²

Allowable Tubesheet Bending Stress [σ allowed]:

= 1.5 · S = 1.5 · 187.0 = 280.49 N./mm²

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.0637 / 0.2059) \cdot (2 \cdot 1137.0 / (58.487 - 0))^2 \cdot -400.0$$

= -280.4844 N./mm²

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

= h + Cats + Catc
= 58.4867 + 0 + 0
= 58.4867 mm

H reqd. = HReqB = 58.4867 mm

Computations completed for ASME tubesheet configuration: B

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	69.5 <=	280.5 N./mm ²	Pass

Floating Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 6 Load Cases:

Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Case Type	Pass/Fail



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ASME Fl-TS Calc: DESIGN CASE

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D1uc	77.726	44.879	127	280	...	...	Ps+Pt-Th	D1	Ok
D2uc	45.888	...	59	280	...	...	Ps+Pt-Th	D2	Ok
D3uc	58.487	...	69	280	...	...	Ps+Pt-Th	D3	Ok
D1c	77.726	44.879	127	280	...	...	Ps+Pt-Th-c	D1	Ok
D2c	45.888	...	59	280	...	...	Ps+Pt-Th-c	D2	Ok
D3c	58.487	...	69	280	...	...	Ps+Pt-Th-c	D3	Ok

Max: 77.7258 44.879 mm 0.452 ... (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	77.726	125.000	Ok
Tubesheet thickness flanged extension :	44.879	109.000	Ok
Tube thickness :	0.413	2.110	Ok
Tube-tubesheet fillet weld leg :	0.000	2.110	Ok
Tube-tubesheet groove weld leg :	0.000	2.110	Ok

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	2249.57	1.000	2860.62	1.000
Tube Pressure Stress	33277.65	1.000	20232.11	1.000
Tubesheet Extension Stress	1068.90	...	No Calc	No Calc
Minimum MAWP	1068.90		2860.62	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	2249.57	1.000	2860.62	1.000
Tube Pressure Stress	33277.65	1.000	22169.73	1.000
Tubesheet Extension Stress	1068.90	...	No Calc	No Calc
Minimum MAPnc	1068.90		2860.62	



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ASME Fl-TS Calc: DESIGN CASE

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(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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Floating Head: Operating

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Input Echo, Floating Head Item 1, Description: Operating

Floating Head Type		D	
Tube Side (Internal) Design Pressure	Pts	1000.00	KPa.
Shell Side (External) Design Pressure	Pss	600.00	KPa.
Design Temperature for Spherical Head	Temp	230.00	°C
Head Material		SA-240	
Head Material UNS Number		S32205	
Head Allowable Stress at Temperature	Sho	171.60	N./mm ²
Head Allowable Stress at Ambient	Sha	187.00	N./mm ²
Crown Radius for Spherical Head	L	2113.200	mm
Head Thickness	Th	25.000	mm
Tube Side (Internal) Corrosion Allowance	Cats	0.0000	mm
Shell Side (External) Corrosion Allowance	Cass	0.0000	mm
Flange Material		SA-182 F60	
Flange Material UNS Number		S32205	
Flange Allowable Stress at Temperature	Sfo	171.60	N./mm ²
Flange Allowable Stress at Ambient	Sfa	187.00	N./mm ²
Flange Outside Diameter	Fod	2482.000	mm
Flange Inside Diameter	Fid	2288.000	mm
Flange Thickness	Tf	195.0000	mm
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2405.000	mm
Nominal Bolt Diameter	dB	34.9250	mm
Type of Threads	UNC Thread Series		
Number of Bolts		44	
Full face Gasket Flange Option	Program Selects		
Flange Face Outside Diameter	Ffod	2363.000	mm
Flange Face Inside Diameter	Ffid	2288.000	mm
Gasket Outside Diameter	Go	2360.000	mm
Gasket Inside Diameter	Gi	2320.000	mm
Gasket Factor,	m	4.2500	
Gasket Design Seating Stress	y	69.64	N./mm ²
Flange Facing Sketch	Code Sketch 1a		
Column for Gasket Seating	Code Column I		
Gasket Thickness		4.0000	mm
Width of raised face		0.0000	mm
Dist. from Head Centerline to Flange Centroid hr		-14.9751	mm
The Flange is not Slotted.			
This unit does not have a Backing Ring.			
Use Full Bolt Load in Flange Design		Yes	

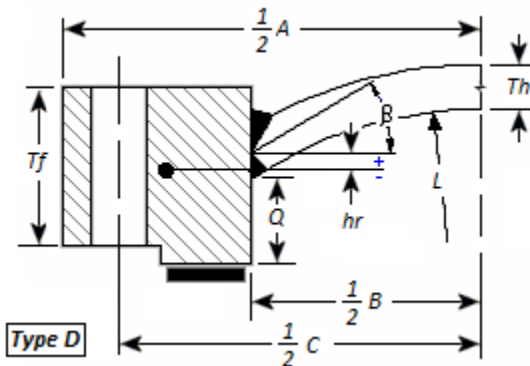


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Floating Head: Operating

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Internal Pressure results for Spherical Heads:

ASME Code, Section VIII, Division 1, 2025, App. 1-6

Thickness Due to Internal Pressure[thr]:

$$= 5PL / 6Sho$$

$$= (5 \cdot 1000.0 \cdot 2113.2) / (6 \cdot 171)$$

$$= 10.2629 \text{ mm}$$

Maximum Allowable Working Pressure at Given Thickness[Pa]:

$$= 6Sho \cdot (Th - Cass - Cats) / 5L$$

$$= (6 \cdot 171 \cdot 25.0) / (5 \cdot 2113.2)$$

$$= 2435.95 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold[Pnc]:

$$= 6Sha / 5L$$

$$= (6 \cdot 186 \cdot 25.0) / (5 \cdot 2113.2)$$

$$= 2654.54 \text{ KPa.}$$

Actual Stress at given Pressure and Thickness[Sact]:

$$= 5PL / 6(Th - Cass - Cats)$$

$$= (5 \cdot 1000.0 \cdot 2113.2) / (6 \cdot 25.0)$$

$$= 70. \text{ N./mm}^2$$

Spherical head Percent Elongation $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$: 0.882 %

External Pressure results, Spherical Head

ASME Code, Section VIII Division 1, 2025

External Pressure Chart HA-5 at 230.00 °C
Elastic Modulus for Material 180805456.00 KPa.

Results for Maximum Allowable External Pressure:

Corroded Thickness of Shell	TCA	25.0000	mm
Outside Crown Radius of Shell	OD/2	2138.2000	mm
Crown Radius / Thickness Ratio	$(OD/2)/T$	85.5280	
Geometry Factor, A	f(D/T, LD)	0.0014615	
Materials Factor, B	f(A, Chart)	108.2847	N./mm ²
Maximum Allowable Working Pressure		1266.00	KPa.
$EMAWP = B / ((OD/T)/2) = 108.2847 / (171.056/2.0) = 1266.0$			

Results for Req'd Thickness for Ext. Pressure (Tca):

Corroded Thickness of Head	TCA	15.5809	mm
Outside Diameter of Shell	ODCA	4276.400	mm
Diameter / Thickness Ratio	(D/T)	274.4641	
Geometry Factor, A	f(DT, LD)	0.0009109	
Materials Factor, B	f(A, Chart)	82.3495	N./mm ²
Maximum Allowable Working Pressure		600.04	KPa.
$EMAWP = B / ((D/T)/2) = 82.3495 / (274.4641/2) = 600.0402$			



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Floating Head: Operating

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Intermediate Calculations for Flanged portion:

ASME Code, Section VIII, Div. 1, 2025 App. 2

Gasket Contact Width, $N = (Goc - Gic) / 2$ 20.000 mm
 Basic Gasket Width, $b0 = N / 2$ 10.000 mm
 Effective Gasket Width, $b = \text{sqrt}(b0) * 2.5$ 7.966 mm
 Gasket Reaction Diameter, $G = Go - 2.0 * b$ 2344.068 mm

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$= 2a + 6t / (m + 0.5)$
 $= 2 * 34.925 + 6 * 195.0 / (4.25 + 0.5)$
 $= 316.166 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:

$= C * \sin(\pi / n)$
 $= 2405.0 * \sin(\pi / 44)$
 $= 171.571 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$= \max(\sqrt{Bs / (2a + t)}, 1)$
 $= \max(\sqrt{171.571 / (2 * 34.925 + 195.0)}, 1)$
 $= 1.0000$

Bolting Information for UNC Thread Series (Non Mandatory):

Total Area of Bolts 299.199 cm²

	Minimum	Actual	Maximum
Radial Distance between Bolts and Edge:	34.925	38.500	
Circ. Spacing between the Bolts:	77.800	171.571	316.166

Results for the Internal Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	440075.3	Kgf
Contact Load on Gasket Surfaces	Hp	50846.7	Kgf
Hydrostatic End Load at Flange ID	Hd	419274.6	Kgf
Pressure Force on Flange Face	Ht	20800.7	Kgf
Radial Component of Head Membrane Force	Hr	654573.6	Kgf
Operating Bolt Load:	Wm1	490922.0	Kgf
Gasket Seating Bolt Load	Wm2	416557.2	Kgf
Required Bolt Area	Am	279.911	cm ²

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:

$= Ab * Sb / (y * PI * (Go + Gi))$
 $= 299.199 * 172.0 / (69.64 * 3.14 * (2360.0 + 2320.0))$
 $= 5.026 \text{ mm}$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

= 38.100 mm [Note: Exceeds actual gasket width, 20.000]

Flange Design Bolt Load (Seating)	W	524750.2	Kgf
Gasket Seating Force (Operating)	Hg	50846.7	Kgf
Distance to Gasket Load Reaction	hg	30.4659	mm
Distance to Face Pressure Reaction	ht	44.4829	mm
Distance to End Pressure Reaction	hd	58.5000	mm

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	419275.	58.5000	1.0000
				24528. Kg-m.



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Floating Head: Operating

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Face Pressure,	Mt	20801.	44.4829	1.0000	925. Kg-m.
Gasket Load,	Mg	50847.	30.4659	1.0000	1549. Kg-m.
Floating Hd. Load,	Mh	654574.	-14.9751	1.0000	9802. Kg-m.
Gasket Seating,	Ma	524750.	30.4659	1.0000	15987. Kg-m.

Total Moment for Operation (Internal Pressure) 36804.965 Kg-m.

Total Moment for Gasket Seating (Int. Pressure) 15987.292 Kg-m.

Results for the External Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	264045.2	Kgf
Hydrostatic End Load at Flange ID	Hd	251564.7	Kgf
Pressure Force on Flange Face	Ht	12480.4	Kgf
Radial Component of Head Membrane Force	Hr	392744.2	Kgf
Gasket Seating Bolt Load	Wm2	416557.2	Kgf
Required Bolt Area	Am	237.510	cm ²

Flange Design Bolt Load (Seating)	W	524750.2	Kgf
Distance to Gasket Load Reaction	hg	30.4659	mm
Distance to Face Pressure Reaction	ht	44.4829	mm
Distance to End Pressure Reaction	hd	58.5000	mm

Summary of Moments for External Pressure:

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	251565.	28.0341	1.0000	7053. Kg-m.
Face Pressure,	Mt	12480.	14.0171	1.0000	175. Kg-m.
Floating Hd. Load,	Mh	392744.	-14.9751	1.0000	5882. Kg-m.
Gasket Seating,	Ma	524750.	30.4659	1.0000	15987. Kg-m.

Total Moment for Operation (External Pressure) 13108.983 Kg-m.

Total Moment for Gasket Seating (Ext. Pressure) 15987.292 Kg-m.

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 1000.0 \cdot 2288.0 \cdot \sqrt{(4 \cdot 2113.2 \cdot 2113.2 - 2288.0 \cdot 2288.0) / (8 \cdot 171.598(2482.0 - 2288.0))}$$

$$= 30.5306 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 36804(2482.0 + 2288.0) / (171.598 \cdot 2288.0)(2482.0 - 2288.0)$$

$$= 22603.3984 \text{ mm}^2$$

Required thickness for Main Flange, Internal Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 30.531 + \sqrt{(30.531 \cdot 30.531 + 22603.398)}$$

$$= 183.9435 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 15987(2482.0 + 2288.0) / (186.997 \cdot 2288.0)(2482.0 - 2288.0)$$

$$= 9009.9404 \text{ mm}^2$$

Required thickness for Main Flange, Internal Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 0 + \sqrt{(0 \cdot 0 + 9009.94)}$$

$$= 94.9207 \text{ mm}$$

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 600.0 \cdot 2288.0 \cdot \sqrt{(4 \cdot 2113.2 \cdot 2113.2 - 2288.0 \cdot 2288.0) / (8 \cdot 171.598(2482.0 - 2288.0))}$$



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FileName : 196-C_Channel MOC=CS_new

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Floating Head: Operating

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= 18.3183 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $13108(2482.0+2288.0)/(171.598 \cdot 2288.0)(2482.0-2288.0)$

= 8050.7495 mm²

Required thickness for Main Flange, External Operating Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $18.318 + \sqrt{(18.318 \cdot 18.318 + 8050.75)}$

= 109.8951 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $15987(2482.0+2288.0)/(186.997 \cdot 2288.0)(2482.0-2288.0)$

= 9009.9404 mm²

Required thickness for Main Flange, External Bolt-up Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $0 + \sqrt{(0 \cdot 0 + 9009.94)}$

= 94.9207 mm

Summary of Required Thicknesses(mm):

	Head	Flange
Tubeside (Internal) Pressure	10.2629	183.9435
Shellside (External) Pressure	15.5809	109.8951
Tubeside Gasket Seating Load		94.9207
Shellside Gasket Seating Load		94.9207
Maximum + Corrosion Allowance	15.5809	183.9435
Actual Thickness as Given	25.0000	195.0000

Floating Head MAWP and MAPnc (KPa.):

Stress Condition	Tubeside		Shellside	
	MAWP	MAPnc	MAWP	MAPnc
Flange Stress	1100.93	1199.75	1499.84	1634.70
Floating head Stress	2435.85	2654.44	1265.90	1534.35
Bolt Stress	1068.81	1068.81	...	...
Minimum MAWP & MAPnc	1068.81	1068.81	1265.90	1534.35

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: